OBSERVATION-RELATIVE MONITORABILITY OF BOUNDED TEMPORAL IMPLICATION

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2012 ACM CCS: Theory of computation → Logic and verification; Verification by model checking; Modal and temporal logics; Software and its engineering → Formal software verification.

Key words and phrases: Linear Temporal Logic, logical implication, runtime verification, bounded eventuality, observation model, calibration, safety properties, temporal implication, monitorability, observation theory.

Preprint submitted to
Logical Methods in Computer Science

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Abstract. A material conditional "P implies Q" has a timeless, definite truth value, but knowing whether it holds does not: every physical or computational check of the conditional is an act of observation, carried out at particular moments and at finite resolution, with each event separated from its observation by delay, drift, and jitter. These distortions are unavoidable, and they produce a paradox of observed implication: an internally sound monitor can contradict the timeless truth value in both directions, reporting a violation that is only an observation artifact (a spurious falsifier), or reporting satisfaction while the implication is vacuous or absent (a phantom trigger) — not because the instrument is broken, but because observation has a class. We make this precise for the temporal implication G(p -> F[<=k] q) rendered from a static conditional, and establish a single headline result: the monitorability of the bounded implication is joint in the pair (formula, observation model) — the formula being the temporal rule itself, the observation model the sampling regime through which its signals are read (when the monitor looks, and the delay, drift, jitter, and finite resolution between an event and its observation) — and is not a property of the formula alone. A change of observation model alone — a shift of sampling phase, a coarser resolution — can move the property across the safety / cosafety / liveness / coliveness classification, so what a monitor may attribute depends on the observation regime, not on the formula. The reference-class apparatus and its causal-soundness conditions form the semantic layer over which this single monitorability result is stated: the six antecedent-validated signatures are the reference on which the forward reading is well-posed, and every other observed class is a causally-graded departure — with coliveness derived from the clopen window skeleton, not imposed. The remaining results support this thesis: the space of possible observations partitions into a taxonomy of eight observation classes, complete by construction and machine-checked; a monotone ladder of declared observation theories grades which classes become True-attributable, each rung admitting one previously ignored timing effect; and internally sound observers that disagree hold different rungs rather than contradicting each other. A universal implication is refuted by a single counterexample yet never finitely confirmed, a Popperian falsify-not-confirm asymmetry. In consequence, a monitor deployed without a declared observation model has no well-defined soundness guarantee. A companion engineering paper [BBI26] constructs the optimal, self-tuning monitor that operates under this relativity.

1. Introduction

1.1. **Context and Motivation.** The material conditional $P \Rightarrow Q$ is one of the most fundamental connectives in classical logic. Its truth table assigns a definite Boolean value to every combination of P and Q—crucially, the conditional is *false* only when P is true and Q is false. This purely extensional characterisation is *timeless*: no notion of precedence, delay, or observability is required.

In practice, however, every empirical or computational evaluation of an implication is performed by an *agent* (a runtime monitor, a scientific experimenter) that must:

(i) determine the truth values of P and Q at specific instants;
(ii) account for the propagation delay between the event that makes P (or Q) true and the instant at which this truth becomes visible to the monitor;
(iii) tolerate possible imprecision (jitter) in the sampling mechanism.

Karl Popper's doctrine of falsifiability [Pop59] demands that a scientific hypothesis of the form "P implies Q" be exposed to the risk of refutation; yet the refutation itself is an observational act that is subject to the temporal constraints listed above. A naive identification of the static conditional with its temporal counterpart can produce both *false*

positives (the implication appears violated when it is in fact satisfied, due to observation delay) and *false negatives* (a genuine violation is masked by sampling jitter).

Thesis. Three questions about a monitored implication must be separated, and are answered by three different mechanisms. *What is observed* is fixed by the observation model: the monitoring class—safety, cosafety, liveness, or coliveness—is not fixed by the logical formula alone but jointly determined by the formula and the observation model, and the observation space partitions geometrically into eight classes. *What can be read* is fixed by the tuning space: tuning the observation kernel induces *discrete regime transitions*—the same formula migrates between LTL fragments depending on the observer's temporal resolution—up to a universal-readability result, guarded by a non-circularity boundary. *What may be attributed True* is fixed by the declared observation theory: a one-way observational-soundness bridge, graded by a monotone ladder of physical premises, connects assumed ideal regularities to what a finite-resolution monitor can verify. The paper's headline is this pair: a taxonomy of eight observation classes placed against the established verdict-availability landscape, and, orthogonal to it, a monotone ladder whose rungs each admit one previously unmodelled timing effect under a declared premise and thereby grade which classes become True-attributable — an axis absent from the monitorability literature, which classifies by which verdicts are reachable rather than by which premises license attribution. Section 1.3 states the apparent paradox that this separation dissolves.

Headline question. This paper answers one question: is the monitorability of the bounded temporal implication $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ a property of the formula alone, or jointly of the *(formula, observation model)* pair? — a question of what can be *observed* and *classified*. Here the *formula* is the temporal rule $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ itself, and the *observation model* is the sampling regime through which its signals are read: when and how often the monitor looks, and the delay, drift, jitter, and finite resolution (Definition 3.2) that separate an event from its observation. The bound k belongs to the formula; the threshold is normalised (Remark 3.6). The formula is fixed and abstract; the observation model is the tunable sampling layer, and it is what makes the same implication observable as one class or another. The complementary control question — whether that observation class can be *steered* — is taken up in the monitor-engineering paper [BBI26].

1.2. **Static vs. Temporal Implication.** We draw a sharp distinction between:

- **Static implication** ($P \Rightarrow Q$): a Boolean function evaluated on a single, perfectly synchronised snapshot of P and Q.
- **Temporal implication** ($\mathbf{G}(P \to \mathbf{F}_{\leq k} Q)$): an LTL formula that requires, at every time step, that if P holds then Q holds within a bounded window of k ticks.

The gap between these two notions is populated by parameters that we collectively call the *observation model*: clock skew S, drift D, jitter J, and propagation delay L.

Why the static reading is not truly atemporal. The static conditional is evaluated on a *single, perfectly synchronised snapshot* of P and Q. That snapshot is not the absence of time but a *degenerate limit* of it: reading both P and Q on one valuation presupposes that the evaluation itself has zero duration, $\Delta t_{\text{read}} \to 0$, so that neither signal can change, race, or pass through an undefined intermediate state while it is being read. Atomicity is therefore a *temporal* hypothesis — a zero-width instant — smuggled in under the appearance of timelessness, not the elimination of time. Read immanently, this is the demand that the excluded middle hold

at a point, $\mathbf{G}(P \vee \neg P)$: a *safety* property (it denotes Σ^ω, the degenerate $\mathsf{S} \cap \mathsf{L}$ tautology cell of [Bra26a] — equivalently $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$, since Σ^ω is trivially cosafety) whose per-tick verdict is well-defined only if each tick is an atomic, race-free instant.

The classical equivalence $P \Rightarrow Q \equiv \neg P \vee Q$ inherits the same concealed premise: it holds *on the nose* only in the already-collapsed model where reading is instantaneous. Restore a finite reading interval and it no longer holds verbatim — "if P" becomes "if and when P is activated", and the disjunct is not falsified but *deferred*: the excluded middle is not violated, only delayed by the response horizon.

Every live, non-atomic system violates the zero-width assumption: between the onset of P and the settling of Q there is always a gap z with $0 < z \leq \Delta$. The difference between the two readings is thus not gap versus no-gap but *hidden* versus *declared*. The static connective *hides* the gap and demands $\Delta t_{\text{read}} \to 0$ as an unreachable ideal, treating time as noise to be eliminated; the bounded temporal reading *declares* it structurally as $0 < z \leq \Delta$, treating time as constitutive. Minimal causal reading needs precisely what the static connective omits and the temporal one supplies: a temporal order on the two onsets, a finite response delay, and one bit of persistent state to remember that P fired while Q is awaited — the memory-bearing crossing primitive of Definition 3.5, as against the memoryless Boolean.

The two-read race. A monitor cannot sample P and Q at the same instant: it reads P at t_1 and Q at $t_2 = t_1 + \delta$ with $\delta > 0$. The Boolean it computes, $\neg P(t_1) \vee Q(t_2)$, coincides with $P \Rightarrow Q$ only when both signals are constant on $[t_1, t_2]$ — the very atomicity the static reading presupposes. With $\delta > 0$ the two reads can straddle a change: P holds at t_1 but Q has not yet answered at t_2, so the monitor returns *violated*; yet Q rises at some $t_3 \in (t_2,\ t_1{+}k]$, a response within the admissible window but after the single sample. The verdict is then an artifact of *when* the monitor looked, not of whether the implication holds — the concrete mechanism behind the gap $0 < z \leq \Delta$. The sound object is therefore not the instantaneous conditional but the bounded temporal reading that gives the consequent a window, $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$, which lies in the safety fragment and is finitely monitorable (Section 1.3; Theorems 5.3 and 7.8 and Remark 5.5) — the paper's "dissolve, not solve". That window is legitimate only when calibrated: its evidence must overlap the admissible response interval (Definition 8.4), a finite-latency bounded precedence must size it (Section 4.4, condition (CC1)), and it must cohere with the governing physical theory (Section 11).

The conclusion is not that static implication is *false* but that it is not genuinely *atemporal*: it necessitates a concealed temporal condition, and the honest move is to make that condition explicit. That move is exactly the bounded temporal reading $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$, which is why the resolution of the paradox (Section 1.3) *dissolves* it rather than solving it — the retyping is forced by the static connective's own hidden premise, not adopted for convenience. This is a motivating argument, not a formal impossibility result: it is offered as the paper's reasoned justification for retyping the connective (HEURISTIC).

1.3. **The Paradox of Observed Implication.** The distinction above generates the tension that this paper treats as its central object. It can be stated as an apparent paradox in three acts.

Act 1 — logic. The material conditional $P \Rightarrow Q$ has a definite, timeless truth value (Table 1): given the values of P and Q, nothing further is needed, and nothing about time can change it. Yet this timeless definition already generates the classical *paradoxes of material implication* [Lew12]: a false antecedent makes the conditional vacuously true, so arbitrary

implications become provable whenever the antecedent fails — the truth value is definite, but its *epistemic worth* is not.

TABLE 1. Truth table of the material conditional $P \Rightarrow Q$. The value is definite and *timeless*, yet the two false-antecedent rows are *vacuously true*: their verdict $\top$ carries no epistemic worth — the classical paradoxes of material implication [Lew12]. This is exactly the tension the paper resolves once the conditional is evaluated by a temporal, *observational* monitor on the tick grid: the truth value stays definite, but its *observed* verdict has a class. The rightmost column records what an internally sound monitor can still get wrong — delay and jitter fracture precisely the two contentful (true-antecedent) rows (Section 1.3), a distinction made operational by the eight-class taxonomy of Section 7.

P	Q	$P \Rightarrow Q$	**Epistemic worth**	**What an observational monitor can still get wrong**
$\top$	$\top$	$\top$	contentful	delay can defer the confirmation past the window: a *spurious falsifier* (false alarm)
$\top$	$\bot$	$\bot$	contentful	jitter can mask the missing confirmation: a *masked violation* (false negative)
$\bot$	$\top$	$\top$	*none* (vacuous†)	— (antecedent never fires; nothing to observe)
$\bot$	$\bot$	$\top$	*none* (vacuous†)	— (antecedent never fires; nothing to observe)

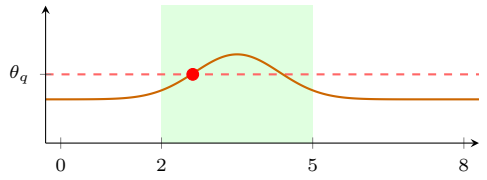
† *Vacuously true*: the value is definite ($\top$) but carries no epistemic worth. Rows three and four ($P = \bot$) are exactly the vacuous rows revisited in operational form in Section 6.

Act 2 — monitors. Every physical evaluation of that conditional is temporal, and there exist traces on which an *internally sound* monitor reports *Violated* while the ground-truth implication holds (a delay- or glitch-induced *spurious falsifier*), and traces on which it reports *Satisfied* while the implication is vacuous or absent (phantom triggers; the $P \wedge Q$ fallacy of Proposition 4.5). The verdict of a sound instrument can contradict the timeless truth value in both directions — not because the instrument is broken, but because *observation has a class*: the same formula is safety, cosafety, or liveness depending on the observation model (Section 7). The classical vacuity annoyance of Act 1 returns here in sharpened, operational form: a monitor that never observes the antecedent cannot distinguish vacuous truth from its own miscalibration (Section 6).

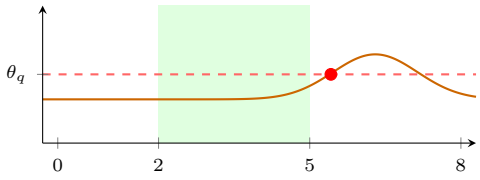
Act 3 — physical theories. Two observers of the *same* observed trace may legitimately disagree: under synchrony premises a silent window is an unresolvable *Quaestio*, while under declared relativistic premises the same trace is True-attributable — its late evidence is the dilated image of an in-bound confirmation (Proposition 11.2). Verdicts are indexed by the declared physics of observation (Section 11).

The paradox is only apparent, and the thesis above is its resolution: once *what is observed*, *what can be read*, and *what may be attributed True* are separated — each with its

own formal machinery — the three acts answer three different questions, and no two sound answers ever contradict each other *about the same question*. Figure 1 shows the smallest instance of Act 2: one world, two verdicts.



(A) Ground truth: the Q-event falls inside the window — the implication holds.



(B) Observed through channel delay: the same event arrives late — a sound monitor reports *Violated*.

FIGURE 1. The smallest instance of the paradox (Act 2). Both panels show the consequent signal s_q against its threshold θ_q, with the observation window $[x, x{+}k]$ (green) opened by the antecedent at $x{=}2$. In the world (left) the Q-crossing lands in-window; through channel delay (right) the *same* event is observed late, and an internally sound monitor reports *Violated*. Neither the world nor the monitor is wrong — they answer different questions.

Resolution strategy: dissolve, not solve. The three acts share one grammatical error rather than three separate ones: they read a temporal, causal relation through a static connective. We therefore do not adjudicate the classical paradoxes of material implication [Lew12] on their own atemporal, truth-functional terms — there a false antecedent stays vacuously true, and no redistribution of truth values removes it. We *dissolve* the paradox instead, by changing the type of logic — a change *forced*, not chosen, by the static connective's own hidden atomicity premise (Section 1.2): the object of study is no longer the static conditional $P \Rightarrow Q$ but its bounded temporal reading $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$, in which precedence, delay, and a finite response horizon are first-class. So read, the paradox is not a theorem to be defeated but an artefact of the connective; it evaporates once the relation is typed temporally, and the three-way separation above is how that temporal reading is then instrumented. On the temporal reading we impose a *causal model* with two separable order premises. Observation *transport* — how an emitter's event is carried to the monitor — is a Lamport happened-before / consistency-model question [Lam78], formalised on the verdict side as the theory ladder of Section 11. The *physical-causation window* — the onset of the antecedent preceding the onset of the consequent within a fixed latency horizon — is a distinct *bounded-latency* premise on the observed process, *not* a happened-before edge: independent emitters P and Q share no communication and hence no happened-before relation, and Lamport's order asserts only *potential* causality. This window premise is what the causal-coherence conditions formalise (Section 4.4): condition (CC1) (window plausibility, $\tau_{\min} \leq k\,\Delta t \leq \tau_{\max}$) together with condition (CC4) (tick–time correspondence) fixes the admissible latency window, and Theorem 4.3 establishes that this coherence is necessary for reliable verification. The causal model is in turn anchored on a *window of validity* — the k-bounded eventuality $\mathbf{F}_{\leq k}$ (Definition 5.1) and its band partition (Section 3.10), both defined here; its engineering *formalisation* (runtime kernel and tuning) is developed in the companion paper [BBI26]. The arc is thus a single line: a static conditional is paradoxical; we retype it as a bounded temporal relation; on that relation we set a bounded-latency

causal window, transported through a declared consistency model; and that window lives on the bounded window of validity defined here.

Why the horizon is bounded. The window is finite by necessity, not convenience. With a plain unbounded eventuality the implication reads $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$, a response property that lies in the *liveness* fragment: it has no finite bad prefix — a consequent still outstanding can always arrive later — so no finite observation can ever falsify it, and a runtime monitor can never emit a rejecting ($\bot$) verdict on it. Bounding the eventuality by k migrates the trace property into the *safety* fragment: for $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ a consequent still missing k ticks after its antecedent *is* a finite bad prefix, and that is exactly what makes a real rejecting ($\bot$) verdict reachable. Boundedness is therefore what moves the property from liveness into the safety fragment, and that migration — not the value of k — is the precondition for finite-horizon monitorability (established formally in Lemma 7.9 and Table 8).

1.4. **Gap in the Literature.** Extensive work exists on LTL model checking [Pnu77, CGP99], runtime verification [LS09, BDD+18], timed logics [AD94, Ras05], and STL/MTL monitoring of continuous signals [MN04, DM10, DDG+17]. However, to the best of our knowledge, no prior work has: (a) shown that the separation between ideal continuous-time semantics and discrete observed semantics induces a deterministic reclassification of the monitored property across LTL fragments, formalised via an explicit factored projection pipeline; (b) geometrically classified the observation space that arises when a *single* propositional implication is monitored through such a pipeline, with a machine-verified refining enumeration; or (c) classified those scenarios by LTL fragment with instrument-parametrised observability conditions.

Division of labour. The observation primitives — touch and crossing on the tick grid, with their capability calculus — are those of [Bra26b]; the four-family safety/cosafety/liveness/coliveness classification is that of [Bra26a]; the present paper develops the theory of observation-relative monitorability and its causal reading, while the engineering control layer — selection, self-tuning, and physical bounding of that observation class — is developed in [BBI26].

1.5. **Contributions and Outline.** *Primary contribution.* The paper's single headline result is *observation-relative monitorability* (Theorem 7.11, Section 7.5): the monitorability fragment — safety, cosafety, liveness, or coliveness — of the bounded implication $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ is fixed by the pair *(formula, observation model)*, not by the formula alone, and a change of observation model alone can move the property across that classification. Its soundness backbone is the causal-coherence necessity theorem (Theorem 4.3); its supporting apparatus is the eight-class reference taxonomy and the departure axes catalogued below. A companion engineering paper [BBI26] carries the complementary half of the diptych: the optimal, self-tuning monitor that operates under this relativity.

What is genuinely new. The novelty is not a new impossibility or hardness result — Theorem 7.11 is, as its own proof records, a consolidation of standard finite-observation facts. What is new is the *joint* quantification of monitorability over the pair *(formula, observation model)*: which family — safety, cosafety, liveness, or coliveness — a property occupies is *undefined at deployment until the observation model is declared*, and a change of observation model alone migrates the property across families. Fragment membership is thus a property

of the (formula, model) pair, not of the formula in isolation. We advance this as a framing, and claim its value as framing — a way of asking where a property lives before asking whether it can be monitored — rather than as a new complexity or hardness bound.

The supporting contributions C1–C9 that follow fall into the three layers of the thesis — what is observed (C1–C2), what can be read (C3, C7–C8), and what may be attributed True (C4, C6, C9) — grounded throughout in instrumentation practice (C5). Each builds, classifies, or grounds the observation models over which the headline result quantifies.

What is observed.

C1. *Dual semantics with factored observation pipeline.* Ideal traces τ_{ideal} are separated from observed traces τ_{obs} by the factored projection $\Pi_{\Delta t,\boldsymbol{\theta},k} = \Pi_k \circ \mathcal{K}_{\boldsymbol{\theta}} \circ S_{\Delta t}$ (Sections 3 and 10).

C2. *Geometric classification of the observation space.* It partitions into eight classes (Table 4), read as crossing/touch patterns on the tick grid (Definition 3.5); complete by construction, its finer signature enumeration is machine-verified (Section 6), and it separates *observational* from *logical* vacuity into provably different verdict cells.

What can be read.

C3. *Tuning as regime migration, up to universal readability.* Tuning $\mathcal{K}_{\boldsymbol{\theta}}$ moves the observed trace between *confirmation sets* (Definition 8.7) without altering Φ_k, potentially across fragment boundaries (Sections 8.3 to 8.4), culminating in an *arbitrary-readability* result (universal readability in at most three tuning moves — the $\leq$3-move construction is developed in the engineering companion [BBI26]; here we state the requirement and its role in the readability thesis): every class is reachable from every other in at most three monitor-internal operations, subject to per-signal feasibility, while the $\mathcal{L}_1^{\text{tun}}$ boundary [Bra26b] keeps readability from becoming truth-manufacture.

C7–C8. *Monitor selection and self-tuning (in brief).* A multi-axis criterion selects the monitor type with the best cost–coverage trade-off; a PLL/DLL loop adapts the kernel at runtime (Section 9), with *local* convergence proven and *global* convergence conjectured (witnessed on a finite prefix, Appendix B). The selection criterion and the self-tuning loop are constructed in the companion engineering paper [BBI26]; here we state the requirement and prove local convergence at its proven scope, deferring the full-loop and global treatment to that paper.

What may be attributed True.

C4. *Bridge proposition (observational soundness).* A unidirectional result connecting ideal causal regularity ($\Psi_{\text{lag},\Delta}$) to satisfaction of Φ_k under explicit regime assumptions (Proposition 10.7); the converse does not hold (Remark 10.8).

C6. *Calibration theorems and the theory ladder.* Each class maps to the appropriate safety–cosafety–liveness fragment (Section 7), closing with a *theory ladder of verdict justification* (Section 11) — synchrony, relativistic, signal-regularity, open. This orthogonal axis grades which classes become True-attributable as each rung declares a premise accounting for one previously unmodelled timing effect — distinct from cataloguing which properties are monitorable.

C9. *Causal validity of signatures (supporting apparatus).* Which observed-implication signatures may be read as causally valid is fixed by the declared physics (Section 4) — the reference-class layer over which the headline result Theorem 7.11 quantifies: the six antecedent-validated signatures are the reference on which the forward reading is well-posed, and every other observed class is a causally-graded departure, with coliveness

derived from the clopen window skeleton, not imposed. Conditions CC1–CC4 anchor the timing model to physical process dynamics, with five falsification pathologies (F1–F5) covering the observation space at signature level (Section 4.4; the class-level crosswalk is deferred to [BBI26]).

Grounded in instrumentation (C5). An explicit mapping ties the four-stage kernel to real measurement chains (detector → ADC → conditioning → decision → gating), grounded in Nyquist–Shannon, signal-detection theory, and PLL/CDR architectures (Sections 4.4 and 8.4).

1.6. **Research Questions.** The paper answers five research questions; each is resolved by material already developed in the body, and none is left open.

RQ1. Is monitorability of a bounded temporal implication a property of the formula alone, or of the pair *(formula, observation model)*? Answered by the observation-relative monitorability theorem (Theorem 7.11, Section 7.5).

RQ2. Can a sound monitor change which verdict it reaches without altering the formula, and what bounds that freedom? Answered by the reading layer — kernel, interval refinement, and tuning up to universal readability (Section 8; universal readability in at most three tuning moves, engineering companion [BBI26]) — and the ideal–observed bridge (Section 10).

RQ3. Under what physical premises may a reached verdict be attributed *True*, rather than manufactured by the observation channel? Answered by the causal-coherence conditions and necessity theorem (Section 4.4, Theorem 4.3).

RQ4. How do successive premises grade which classes become True-attributable? Answered by the theory ladder of verdict justification (Section 11).

RQ5. Where do the bounded-implication patterns sit in the safety/cosafety/liveness/coliveness classification, and how does observational vacuity differ from logical vacuity? Answered by the fragment classification (Section 5, Table 8), its Peled–Havelund placement (Section 7, Proposition 7.1), and the vacuity analysis (Section 6, Remark 7.3).

1.7. **Scope and Non-Goals.** To prevent mis-reading we state what this paper is *not*:

(1) **Not a new monitoring algorithm.** We do not propose a new STL/MTL monitor. The four-stage kernel $\mathcal{K}_{\boldsymbol{\theta}}$ is an analytic model of existing measurement chains; the algorithmic toolbox (Section 8.4) catalogues known signal-processing primitives, not new ones.

(2) **Not causal inference.** We do not perform interventionist causal inference (Pearl, Granger): the ideal predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ formalise *observational regularities consistent with causal mechanisms* under stated regime assumptions, not causal direction, and the Bridge Proposition is strictly one-way (see C4 and Remark 10.8).

(3) **Not a security or indistinguishability argument.** The evaluation-agent model (Definition 10.3) draws on the structural pattern of specification refinement; no cryptographic, adversarial, or computational-indistinguishability claim is made (Section 10).

(4) **Not an extension of Nyquist/SDT/PLL theory.** We invoke Nyquist sampling, signal detection theory, and PLL timing recovery as *existing* analytical tools; the contribution is the formal link between these tools and the monitor's kernel stages, not a novel result in any of those fields.

(5) **Single formula scope.** All combinatorial results are specific to the single bounded-eventuality formula $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$. Multi-property or multi-formula extensions are future work (Section 16).

1.8. **Paper Organisation.** After related work (Section 2), the body follows the three questions of the thesis in order. *What is observed*: Section 3 (observation model and eight-class geometry), Section 5 (LTL patterns), Section 6 (observational vacuity), Section 7 (fragment geometry and calibration). *What can be read*: Section 8 (kernel, interval refinement, tuning theory up to universal readability), realised in brief by Section 9 as monitor selection and a self-tuning loop. *What may be attributed True*: Section 10 (the ideal–observed bridge), Section 4.4 (physical premises), and Section 11 (the theory ladder resolving the third act). Sections 12 and 12.4 let the three answers come apart on concrete traces; Sections 13, 15 and 16 conclude, and the appendices hold the Python artefacts, the extended interval semantics, and the Schmitt-trigger witness (the full engineering development, including the worked gallery, is in a related paper [BBI26]). The stakes: without an explicit observation model the question "does the monitor operate in the safety, cosafety, or liveness fragment?" has no well-defined answer for a deployed system — RV frameworks that fix a fragment assignment implicitly assume observation parameters known and stable, an assumption that routinely fails in distributed, non-synchronised, or drifting environments.

Reader's guide (what depends on what). The paper's technical results — observation-relative monitorability (Theorem 7.11), the safety/cosafety/liveness/coliveness classification (Table 8, Proposition 7.1), and the causal-coherence necessity theorem (Theorem 4.3) — are established *independently* of the interpretive material of this paper. In particular they do not logically depend on the motivating reductio of Section 1.2 (that snapshot atomicity is a temporal hypothesis smuggled in under the appearance of timelessness) or on the theory ladder of verdict justification (Section 11); those sections supply motivation and interpretation, not premises of any theorem. A reader concerned only with the formal core may take Theorems 4.3 and 7.11 and the fragment classification on their own terms; the philosophical framing can be read, or skipped, without affecting their proofs.

2. Related Work

This section positions our contribution relative to four strands of prior research and summarises the comparison in Table 2.

2.1. **Runtime Verification and Monitoring of Temporal Properties.** Runtime verification (RV) monitors a running system against a formal specification and raises verdicts online. Leucker and Schallhart [LS09] survey three-valued and four-valued semantics for finite-prefix monitoring; Bauer et al. [BLS11] define the three-valued semantics LTL_3, with a timed variant, together with incremental automata-based monitor construction. Havelund and Roşu [HR01] propose rewriting-based monitors, while Bartocci et al. [BDD+18] and Sánchez et al. [SSA+19] survey real-time and distributed monitoring challenges. Bonakdarpour et al. [BNF13] develop *time-triggered* runtime verification, in which the monitor samples the program state at fixed intervals rather than at event boundaries — the closest RV analogue of the tick-grid observation model studied here — and characterise the sampling

periods under which verdicts remain sound. Cimatti et al. [CTT22] develop *assumption-based* runtime verification, in which a monitor's soundness is stated relative to an explicitly declared environment assumption rather than in the absolute; the present paper's declared observation theory and its monotone ladder are a graded, physically-instantiated instance of this stance, where each ladder rung is a strictly stronger channel assumption under which more observation classes become True-attributable. The closest prior is the assumption-relative monitorability of Henzinger and Saraç [HS20], in which an assumption A restricts the admissible trace space; the present observation model is a *specific* instantiation of that stance — the timing 4-tuple (L, S, D, J) of Definition 3.2 together with a normalised threshold — whose distinct contribution is the resulting migration of the property across the safety/cosafety/liveness/coliveness four-family classification, an organisation that assumption-relative monitorability does not itself supply. Falcone et al. [FKRT21] classify sixty RV tools along a taxonomy whose dimensions (trace model, sampling discipline, verdict domain) correspond to axes that the present paper treats as first-class semantic parameters rather than tool-implementation choices. The canonical Safety–Progress account of what can be verified and enforced at runtime is due to Falcone, Fernandez and Mounier [FFM12], whose verify-and-enforce taxonomy underlies the verdict-availability reading used here. Francalanza et al. [FPS17] survey runtime verification for decentralised and distributed systems. A distinct distributed-computing line asks *how many* verdict values (opinions) a set of fault-prone distributed monitors must be able to emit in order to decide a predicate [FRT14, BFR$^+$16]; the ladder developed here is orthogonal to that question — it asks *which* declared transport premises license a True-attribution for a *single* monitor — but both quantify the gap between raw observation and sound conclusion. Peled and Havelund [PH19] refine the safety/liveness dichotomy of Alpern–Schneider into a tripartite verdict-axis partition (always / sometimes / never finitely refutable, and symmetrically for satisfiability), yielding nine classes of ω-regular properties differentiated by which verdicts a runtime monitor can produce; their follow-up survey [HP23] consolidates this classification and its monitoring algorithms. Diekert and Leucker [DL14] anchor the same body of results in the Cantor topology, characterising the monitorable properties as those whose boundary is nowhere dense. Aceto, Achilleos, Francalanza, Ingólfsdóttir and Lehtinen [AAF$^+$19] chart the modern monitorability hierarchy across branching and linear time; our classification instead indexes by the *(formula, observation-model)* pair and its induced safety/cosafety/liveness/coliveness migration, orthogonal to their branching-versus-linear monitorable-fragment hierarchy.

Our classification uses the four families *safety, cosafety, liveness, and coliveness*, read against the Peled–Havelund verdict-axis partition (the four corner cells $AFR \cap AFS$, $AFR \cap NFS$, $NFR \cap AFS$, $NFR \cap NFS$ in their Fig. 1, plus the safety-only and cosafety-only edge cells). Coliveness K enters precisely as the one-sided cell $\mathsf{S} \cap \mathsf{K}$ that the implication patterns of Table 8 occupy — finitely refutable, never finitely confirmable. The full four-family intersection algebra of safety, cosafety, liveness, and coliveness is systematised in a related paper [Bra26a], and its constructive derivation for the implication patterns — coliveness obtained as the universal-$\mathbf{G}$ lift of the clopen window skeleton — is given here in Section 4.5; the deciding prefix-filtration monitor that realises it is the operational object of the engineering companion [BBI26].

For *continuous-time* signals, Signal Temporal Logic (STL) [MN04] extends the bounded real-time logic $\mathrm{MITL}_{[a,b]}$, a fragment of metric temporal logic [Koy90], with numerical predicates over real-valued signals. Donzé and Maler [DM10] give STL a quantitative, space- and

time-robust satisfaction semantics, and Deshmukh et al. [DDG+17] develop robust online STL monitoring over sampled signals. Timed-pattern matching over continuous streams is studied by Ulus et al. [UFAM16] using derivatives of timed regular expressions. Two neighbours sit especially close to our bounded-future setting. Ho, Ouaknine and Worrell [HOW14] give a trace-length-independent online monitoring procedure for MTL in dense time — the closest online-monitoring analogue of our bounded-future $\mathbf{F}_{\leq k}$ implication; and Alur, Feder and Henzinger [AFH96], by relaxing punctuality to MITL, identify the decidable home of the bounded eventuality $\mathbf{F}_{\leq k}$ used throughout this paper (a preliminary version appeared at PODC 1991). Both, however, evaluate an MTL/MITL formula against a *fixed* observation of the trace; our point is orthogonal — the observation *model* itself (latency, jitter, drift) migrates the property across the safety/cosafety/liveness/coliveness families, so monitorability is joint in the pair *(formula, observation model)*, a dependence their fixed-trace setting does not address.

How we differ. Existing STL/MTL monitors evaluate arbitrary formulas assuming the sampling pipeline is given and correct. Even for the simplest bounded-eventuality formula $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$, the interaction between sampling period, decision kernel, and window bound generates eight structurally distinct observation classes with distinct fragment assignments: the pipeline itself, not just its output, is a first-class object of analysis. Consequently an STL/MTL monitor correct under its assumed regime may silently become unsound when parameters drift into a different fragment — a failure mode our classification makes visible.

2.2. **Observation Models and Partial Observability.** Pnueli [Pnu77] introduces LTL for fully observable program traces. Alur and Dill [AD94] extend to timed automata; Raskin [Ras05] introduces hybrid automata and surveys their decidable subclasses, timed automata among them.

Partial observability and uncertainty in monitoring are studied by Stoller et al. [SBS+11], who combine runtime verification with state estimation, and by Basin et al. [BKZ15], who address failure-aware distributed monitoring. Ganguly et al. [GMB21] monitor distributed computations under *partial synchrony*, where a declared bound on clock skew determines which event orderings the monitor must treat as possible; their skew bound plays the same role as the cumulative offset Δ of our observation channel, and their enumeration of skew-consistent interleavings is the distributed counterpart of the ambiguity that our window classes render geometric. Beer et al. [BBDER01] and Kupferman and Vardi [KV03] formalise vacuity in model checking, and Kupferman and Vardi [KV01] characterise model checking of safety properties through bad prefixes.

How we differ. Our *observational vacuity* (Section 6) is caused not by formula structure but by the monitor's temporal inability to observe the antecedent — absent in classical model checking. The factored pipeline $S_{\Delta t} \to \mathcal{K}_{\boldsymbol{\theta}} \to \Pi_k$ (Definition 10.2) makes each information-loss stage explicit, separating how sampling ambiguity, kernel masking, and window truncation contribute to the ideal–observed gap.

2.3. **Signal-Processing and Detection-Theory Foundations.** The instrumentation grounding of the monitor model draws on well-established results that we invoke but do not claim as novel contributions:

- **Nyquist–Shannon sampling theorem** [Sha49]: sampling at rate a signal containing no frequencies above $f_{\max}$ is completely determined by samples spaced $1/(2f_{\max})$ apart; our same-bin ambiguity (Section 3.4) is a direct consequence of violating this bound.
- **Signal detection theory (SDT)** [GS66, Kay98]: the false-positive/false-negative trade-off is governed by the likelihood ratio and the Bayes risk $R(\theta) = c_{\mathrm{FP}}\,\alpha(\theta) + c_{\mathrm{FN}}\,\beta(\theta)$; our threshold-tuning analysis instantiates this for the monitor's edge-detection stage (developed in [BBI26]).
- **Phase-locked loops (PLL/CDR)** [Bes07, Gar86, MM76, Raz02]: the self-tuning loop (Section 9) is a direct adaptation of classical timing-recovery architectures.

How we differ. We do not extend SDT, Nyquist theory, or PLL design. Instead we show that the four-stage kernel $\mathcal{F}\circ\mathcal{A}\circ\mathcal{D}\circ\mathcal{M}$ is a formal abstraction of real measurement chains (detector $\to$ ADC $\to$ conditioning $\to$ decision $\to$ gating), grounding the monitor model in established instrumentation practice rather than an *ad hoc* construction.

2.4. **Safety–Liveness Classification and Bounded Model Checking.** Alpern and Schneider [AS87] show, constructively on Büchi automata, that every ω-regular property is the intersection of a safety and a liveness property. Manna and Pnueli [MP90, MP95] develop the safety–progress hierarchy of temporal properties. Clarke, Grumberg and Peled [CGP99] give the standard account of model checking; Biere et al. [BCCZ99] introduced bounded model checking, and Biere et al. [BHJ$^+$06] give linear-size encodings of bounded LTL model checking.

Manna and Pnueli's hierarchy [MP90] refines the trichotomy into six classes (safety, guarantee, obligation, recurrence, persistence, reactivity); in that finer key the unbounded pattern $\mathcal{L}_3$ is a *response* formula, hence a recurrence property, while the bounded patterns are safety. The trichotomy used here is the coarsest reading compatible with the four-family classification of a related paper [Bra26a].

How we differ. Our calibration theorems (Section 7) map each observation class to a position in the safety–cosafety–liveness spectrum *parameterised* by the observation-model parameters. Where Peled–Havelund [PH19] and Diekert–Leucker [DL14] classify by *verdict-availability*, we add an orthogonal observation-theory ladder grading True-attribution by which previously unmodelled timing effect a declared premise accounts for.

2.5. **Causality Positioning.** We do not perform interventionist causal inference (Pearl, Granger). The ideal predicates Ψ_{ov} and $\Psi_{\mathrm{lag},\Delta}$ (Definition 8.4) formalise *observational regularities consistent with causal mechanisms* under stated regime assumptions; the Bridge Proposition (Proposition 10.7) connects them to satisfaction of Φ_k, not to underlying causal structure. The converse fails (Remark 10.8): Φ_k satisfaction on τ_{obs} is necessary but not sufficient for inferring an ideal-level causal regularity.

TABLE 2. Comparison of monitoring frameworks.

Framework	Signal model	Sampling	Obs. kernel	Tuning loop	Pattern tax.	Guarantees
STL/MTL monitoring [MN04, DM10, DDG$^+$17]	Continuous real-valued	Assumed given	N/A	N/A	N/A (robustness degree)	Quantitative robustness
Timed automata monitoring [AD94, UFAM16]	Timed words	Clock regions	N/A	N/A	N/A	Decidability, complexity
Classical LTL RV [LS09, BLS11]	Discrete symbolic	Perfect	N/A	N/A	Monitorability classes	3/4-valued verdicts
RV with state estimation [SBS$^+$11, BKZ15]	Partially observable	Imperfect	State estimator	N/A	N/A	Probabilistic guarantees
This work	Ideal continuous $\to$ discrete observed	$S_{\Delta t}$ (explicit)	$\mathcal{K}_{\boldsymbol{\theta}}$ (4-stage)	PLL/CDR (self-tuning)	8 window classes	Obs. soundness (Proposition 10.7)

3. Formal Model

3.1. **Processes and Events.** We consider three processes operating on a shared discrete timeline $\mathbb{T} = \{0, 1, 2, \ldots\}$ measured in *ticks*.

Definition 3.1 (Emitter and Monitor Processes)**.** Let $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{M}$ be three processes:

- $\mathcal{P}$: the *antecedent emitter*, which produces the event e_P at some tick $t_P \in \mathbb{T}$.
- $\mathcal{Q}$: the *consequent emitter*, which produces the event e_Q at some tick $t_Q \in \mathbb{T}$.
- $\mathcal{M}$: the *monitor* (the observing process), which samples the truth values of P and Q at ticks t_M^P and t_M^Q.

Definition 3.2 (Observation Parameters)**.** The observation model is characterised by four non-negative integer parameters:

$$S \in \mathbb{N}_0 \qquad \text{(clock skew)}, \tag{3.1}$$

$$D \in \mathbb{N}_0 \qquad \text{(drift rate)}, \tag{3.2}$$

$$J \in \mathbb{N}_0 \qquad \text{(jitter half-width)}, \tag{3.3}$$

$$L \in \mathbb{N}_0 \qquad \text{(propagation delay)}. \tag{3.4}$$

Here S measures the clock skew between emitters $\mathcal{P}/\mathcal{Q}$ and monitor $\mathcal{M}$; D is the drift accumulation rate per tick; J is the sampling jitter half-width; and L is the propagation delay from emitter to monitor.

Definition 3.3 (Monitor Sampling Instants)**.** The monitor $\mathcal{M}$ observes P at tick

$$t_M^P = t_P + L + S + D \cdot t_P + \epsilon_P, \qquad |\epsilon_P| \le J, \tag{3.5}$$

and Q at tick

$$t_M^Q = t_Q + L + S + D \cdot t_Q + \epsilon_Q, \qquad |\epsilon_Q| \le J. \tag{3.6}$$

3.2. **Trace Model.**

Definition 3.4 (Trace)**.** A *trace* is an infinite sequence $\sigma = \sigma_0\,\sigma_1\,\sigma_2\,\cdots$ where each $\sigma_i \subseteq AP$ is a set of atomic propositions that hold at tick i. The set of atomic propositions is $AP = \{p,\ q,\ w_p,\ w_q\}$, where:

- $p \in \sigma_i$ iff $\mathcal{P}$ emits e_P at tick i;
- $q \in \sigma_i$ iff $\mathcal{Q}$ emits e_Q at tick i;
- $w_p \in \sigma_i$ iff $\mathcal{M}$ observes P=*true* at tick i;
- $w_q \in \sigma_i$ iff $\mathcal{M}$ observes Q=*true* at tick i.

Under the monodimensional observation alphabet (Definition 3.8), at most one of $\{w_p, w_q\}$ holds at any tick i.

3.3. **Notation Summary.** Table 3 collects the principal symbols used throughout the paper for quick reference.

3.4. **Observation as Crossing and Touch on the Tick Grid.** The band tokens of Σ are not primitive: each is produced by a threshold event on an underlying signal, sampled onto the tick grid. Following the dual-primitive account of threshold monitoring in a related paper [Bra26b], we read every observation as one of two modalities.

Definition 3.5 (Crossing and touch observation)**.** Let s_p, s_q be the underlying signals for P and Q, with thresholds θ_p, θ_q, sampled at tick period Δt. The monitor observes each channel through one of two modalities.

- *Crossing* (latched validation): an up-crossing $s_p(t^-) < \theta_p \le s_p(t)$ fires *validation* and latches the channel to $P^\top$; the channel *holds* $P^\top$ until a down-crossing $s_p(t^-) \ge \theta_p > s_p(t)$ fires *invalidation*, latching $P^\perp$. Crossing thus carries one bit of persistent state — the current validity level — the causal, memory-bearing primitive of [Bra26b].
- *Touch* (memoryless extremum): a sampled local extremum of s_p at the threshold fires a single tangency event with no latched level — the memoryless, window-bearing primitive of [Bra26b].

The symmetric definitions hold for Q with θ_q. Under the crossing modality the observation stream σ_{obs} is exactly the latched band signal: $w_p(t) = \top$ on the maximal interval from a validation to the next invalidation, and w_q analogously.

The restriction to these two modalities is not a simplification: by the ternary-rung closure theorem of a related paper [Bra26b], *every* three-valued threshold observable in its scope (instant-level, at most one-step lookback, firing rule a Boolean combination of the three value classes) is event-stream-equivalent, under threshold tuning, to either the crossing or the touch primitive, so the pair is exhaustive for the observation model of this paper.

TABLE 3. Principal notation and symbols.

Symbol	Description
$\mathcal{P},\ \mathcal{Q}$	Antecedent / consequent emitter processes
$\mathcal{M}$	Monitor (observer) process
$e_P,\ e_Q$	Events emitted by $\mathcal{P}$, $\mathcal{Q}$
$t_P,\ t_Q$	Emission ticks of e_P, e_Q
$t_M^P,\ t_M^Q$	Monitor sampling instants for P, Q
L	Propagation delay (emitter $\to$ monitor)
S	Clock skew
D	Drift accumulation rate per tick
J	Jitter half-width
k	Bounded-eventuality window bound
x	Opener tick (*bandx*): first tick $\mathcal{M}$ observes $P{=}\top$
Δ	Maximum observation error $L{+}S{+}D{+}J$
σ	Trace $\sigma_0\,\sigma_1\,\cdots \in (2^{AP})^\omega$
AP	$\{p,\ q,\ w_p,\ w_q\}$; at most one of w_p, w_q per tick
$n_0,\ n_1$	Early / late confirmation counts around x
e_n	Timing-error signal $(n_0 - n_1)/(n_0 + n_1 + \epsilon)$
$\mathbf{G},\ \mathbf{F},\ \mathbf{F}_{\leq k}$	LTL: always, eventually, bounded eventually
(A, B, C)	The eight observation presence cells (window classes)
$\hat{\varphi}$	Estimated phase offset (self-tuning loop)
$\mathcal{A}(w)$	Signature ambiguity of monitor type w
$\mathcal{S}(w)$	Opener sensitivity of monitor type w
$\mathcal{R}(w)$	Timing robustness of monitor type w
$\mathcal{C}(w)$	Causal incoherence penalty
τ_{detect}	Minimum causal detection latency
Δt	Tick duration (physical time per tick)
A, B, C	In-window crossing/touch observations (Definition 3.5)
Φ_0	Zero-delay LTL formula $\mathbf{G}(w_p \to w_q)$
Φ_k	Bounded-eventuality formula $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$
$M^-,\ M^+$	Early / late timing margins
θ	Decision threshold (TED)
$\mathcal{K}_{\boldsymbol{\theta}}$	Observation kernel (Definition 8.1)
$\mathcal{F}, \mathcal{A}, \mathcal{D}, \mathcal{M}$	Kernel stages: pre-filter, alignment, decision, gating
$\tau_{\min}$	Minimum pulse width (pre-filter)
$\boldsymbol{\theta}^\star$	Min-risk tunable state (Equation (8.4))
$\mathcal{K}_{\text{id}}$	Ideal (noiseless) observation map
$\alpha_q,\ \beta_q$	FP / FN rates, Q-channel (Equations (8.2) and (8.3))
Σ	Observation alphabet $\{\varnothing, P^\top, P^\perp, Q^\top, Q^\perp\}$
$\prec$	Token priority on Σ (Definition 3.12)
$\text{tok}(B)$	Priority-max token of band B (Definition 3.12)
$\pi_{8\to4}$	eight-tick $\to$ 4-band projection (Remark 3.13)
$\mathcal{L}_1$–$\mathcal{L}_6$	LTL implication patterns (Table 8)
(ρ, ω, π)	Monitor-type axes: repeat / opener / probe (Section 9)
T_1–T_9	Elementary tuning operations [BBI26]
$\mathcal{L}^{\text{tun}}_{1,2,3}$	Tuning-power hierarchy (strict) [Bra26b]
CC1–CC4	Causal-coherence conditions (Definition 4.2)
F1–F5	Falsification pathologies (Definition E.1)
$\mathcal{O}_{\text{sync}} \subset \{\mathcal{O}_{\text{rel}}, \mathcal{O}_{\text{sig}}\} \subset \mathcal{O}_{\text{open}}$	Observation-theory ladder (lattice; $\mathcal{O}_{\text{rel}}, \mathcal{O}_{\text{sig}}$ incomparable, Section 11)

Remark 3.6 (Threshold sensitivity and the single-θ convention). Each channel carries its own *sensitivity* knob: the threshold can be raised or lowered at will — the per-channel threshold-shift tuning (T_1), i.e. the moving-threshold sensitivity of a related paper [Bra26b]. Because the crossing and touch events of a channel depend only on the relation between its signal and its *own* threshold, the per-channel change of coordinates $s_p \mapsto s_p - \theta_p + \theta$ and $s_q \mapsto s_q - \theta_q + \theta$ maps both channels onto a common threshold θ without altering any event stream: without loss of generality one may draw a *single* semantic threshold. We adopt this single-θ representation wherever convenient (Table 4 and the worked visualisations of Appendix D), while Figure 2 retains distinct θ_p, θ_q to display the general two-threshold case. The normalisation is purely representational and spends none of the tuning budget: the two sensitivities remain independent degrees of freedom, and the joint two-channel tuning developed in the engineering companion [BBI26] may move them separately — or deliberately drive them to coincide, adopting the canonical single-threshold form as a tuning *target*.

The underlying model carries a threshold *per channel*: θ_P for the antecedent and θ_Q for the consequent, on generally different scales and in different units. The single θ drawn in the figures is the normalised exposition above — each channel rescaled onto a common level — and is legitimate because the eight classes are invariant under any per-channel monotone rescaling of a signal against its own threshold. Tuning, however, remains *per channel*: a consequent miscalibration is corrected by retuning θ_Q alone, leaving θ_P untouched. Each channel may itself be two-level, a hysteresis band $(\theta^{\text{low}}, \theta^{\text{high}})$ rather than a single level, as already visible in the encoding gallery.

Read this way, bounded-eventuality implication is a purely geometric statement on the tick grid: $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ holds iff, after every tick at which the crossing of θ_p latches $P^\top$, the crossing of θ_q latches $Q^\top$ within the next k ticks, i.e. $t_q \in [t_p, t_p{+}k]$. Figure 2 renders this reading: two signals and their thresholds in one diagram, the tick grid beneath, and the window $[x, x{+}k]$ in which the Q-crossing must land for the implication to be *produced*. Whether a given *theory* — a choice of modality, thresholds, tick period Δt, and bound k — produces satisfaction is read off spatially, as the question of whether the Q-crossing falls inside the shaded window; the definitive verdicts a theory can attain are read off from the cell it occupies in the safety/cosafety/liveness/coliveness classification of a related paper [Bra26a] — $\bot$ is attainable iff the property is not live, $\top$ iff it is not colive; monitorability is not a function of the cell in general, but on the three cells the implication patterns occupy ($\mathsf{S} \cap \mathsf{K}$, $\mathsf{C} \cap \mathsf{L}$, $\mathsf{L} \cap \mathsf{K}$) it is.

3.5. **The Observation Classes as Tick Diagrams.** *Presence axis (which events occur, order forgotten).* These eight window classes record, for a fixed antecedent opener, *which* of the three window events are present; the antecedent-validated reference class of Section 4.1 is the all-present cell $(1, 1, 1)$, and every other class here is a presence-departure from it.

The eight classes of the observation space are exactly the crossing/validation patterns of Definition 3.5: within the window $[x, x{+}k]$ each class is fixed by three in-window threshold crossings — A a Q-validation ($Q^\top$ up-crossing), B a Q-invalidation ($Q^\bot$ down-crossing), and C a P-retraction ($P^\bot$ down-crossing). Table 4 renders all eight as compact signal-vs-threshold panels, the antecedent signal s_p over the consequent signal s_q crossing the single semantic threshold θ (justified by the normalisation of Remark 3.6), so each (A, B, C) triple is read off directly from the in-window crossing geometry. Signal-level worked realisations of each class — with full token rows and per-class verdict discussion — are collected in

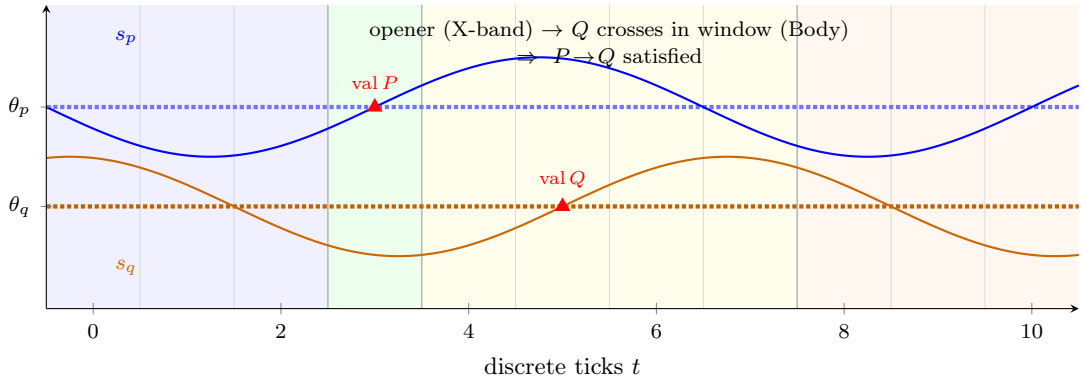


FIGURE 2. Geometric reading of bounded-eventuality implication on the tick grid, in a related threshold-monitor style [Bra26b]. The real signals s_p, s_q oscillate across their thresholds θ_p, θ_q; the *crossing* modality latches $P^{\top}$ at the up-crossing of θ_p (the green opener/X-band, tick x=3) and $Q^{\top}$ at the up-crossing of θ_q. The implication $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ is *produced* exactly when the Q-crossing lands in the window interior (the yellow Body band). A different theory (threshold, Δt, or bound k) slides the Q-crossing in or out of the window, changing the attainable verdict cell [Bra26a].

Appendix D. This gallery is a *monitor-side* observable: which in-window edges are present fixes which finite verdicts are *reachable*, with the causal reading deferred to the soundness rule (Theorem 4.3) and the tuning of the observation itself deferred to the companion [BBI26]. It classifies a window by *presence* — which of $\{Q^{\uparrow}, Q^{\downarrow}, P^{\downarrow}\}$ occur (2^3=8 cells), order-blind — and the benign/adverse split falls along the A axis: the four A=1 cells confirm the Q-crossing in-window, so the implication is *produced* — these four form the *confirming family*, a presence-reduction of the reference class in which the all-present $(1,1,1)$ of Section 4.1 degrades to the two-event $(1,1,0), (1,0,1)$ and the single-event $(1,0,0)$ as the retraction $P^{\downarrow}$ and the invalidation $Q^{\downarrow}$ drop out-of-window or go undetected while the onset $Q^{\uparrow}$ (A=1) is preserved (their tuning is developed in the companion [BBI26]) — while the four A=0 cells do not, with the silent cell $(0,0,0)$ the miscalibration case (s_q never crosses θ) and $(0,1,0)$ a *monitor-side refutation candidate*: a lone in-window $Q^{\perp}$ with A=0 presupposes an out-of-window $Q^{\top}$ onset that the order-blind, in-window cube cannot place, so by Proposition 4.4 it is a premature-falsification / retracted-confirmation candidate with an out-of-window onset, not a settled causal violation. The presence view thus *admits absence*: the A=0 rows are exactly the windows where the consequent never validates ($Q^{\uparrow}$ absent) — the violation, miscalibration, and open cases. The complementary *order* view — how the events that *are* present arrive — is developed for the antecedent-validated regime in Section 4.1; the two axes are orthogonal (presence × order), this gallery being the PRESENCE departure from that reference class. Crucially, only the presence view expresses a violation: a violation is the *absence* of a consequent onset ($Q^{\uparrow}$ absent) and so has no all-present four-event signature, tying to the crossing-CLOSED caveat of Section 4.1. The verdict-cell reading of the classes is developed in [Bra26a].

TABLE 4. The eight observation classes by in-window presence triple $(A, B, C) = (\exists\, Q^{\uparrow}, \exists\, Q^{\downarrow}, \exists\, P^{\downarrow})$, for a fixed antecedent-validated opener. The classification proper is the (A, B, C) cell; the class names are an informal, provisional gloss, not a definition. The eight window classes are the quotient of the 50000-entry eight-tick catalogue (k=3), invariants of the general observation model, whose worked signal-vs-threshold gallery is rendered in the engineering companion [BBI26]; the four-family $(\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K})$ placement is Table 8.

(A, B, C)	Informal gloss
$(1, 0, 0)$	static-like
$(1, 1, 0)$	bounded-temporal
$(1, 0, 1)$	bounded-temporal
$(1, 1, 1)$	bounded-temporal (the reference cell)
$(0, 0, 0)$	miscalibration (silent window)
$(0, 1, 0)$	monitor-refuted (out-of-window onset)
$(0, 0, 1)$	late retraction
$(0, 1, 1)$	complex / race

TABLE 5. Instantiations of the confirming ($Q^{\top}$-first) signatures. Each row is a physical or chemical realisation of one A=1 window class: the opener $P^{\top}$ (antecedent onset, the trigger crossing θ_P) is implicit, and the listed events are those falling in the in-band window $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ (k=3). These are the $Q^{\top}$-first branch of the confirming family, ordered by decreasing event count ($4 \to 3 \to 2$, counting the implicit opener): the shorter signatures are the longer ones seen through a narrower observation window, in which the later events fall out of band. Concretely, the same transient process yields $(1, 1, 1) \to (1, 1, 0), (1, 0, 1) \to (1, 0, 0)$ as the observation window is shortened. Developed realisations, with their threshold and window sensitivity, are collected in Appendix G.

Signature	(A, B, C)	System	P onset	Q onset	R
$Q^{\top}\, P^{\perp}\, Q^{\perp}$	$(1, 1, 1)$	resistively-heated sample	power on	T crosses then decays	–
$Q^{\top}\, Q^{\perp}\, P^{\perp}$	$(1, 1, 1)$	fluorophore photobleaching	laser on	fluoresce then bleach	–
$P^{\perp}\, Q^{\top}\, Q^{\perp}$	$(1, 1, 1)$	flash photolysis	flash pulse	transient rise/decay	[
$Q^{\top}\, Q^{\perp}$	$(1, 1, 0)$	sensory/receptor adaptation	sustained stimulus	response then adapts	–
$Q^{\top}\, P^{\perp}$	$(1, 0, 1)$	crystallisation from a seed	seed added	crystal latches	–
$P^{\perp}\, Q^{\top}$	$(1, 0, 1)$	phosphorescence	excitation flash	delayed persistent emission	–
$Q^{\top}$	$(1, 0, 0)$	irreversible product formation (sustained)	reagent held	product plateaus	–

3.6. **The Six Reference Signatures and Reinterpretation Under Context Extension.** The confirming realisations of Table 5 are windowed shadows of a small generating basis. Fix the opener $P^\top$ and let each channel bracket exactly once — the antecedent through $\{P^\top, P^\bot\}$ and the consequent through $\{Q^\top, Q^\bot\}$. The four events admit $3! = 6$ orderings with $P^\top$ first: the *six reference signatures* $\mathsf{R}_1, \ldots, \mathsf{R}_6$ of Table 6. All six carry the same presence triple $(A, B, C) = (1, 1, 1)$ — the all-present cell of Table 4 — so the presence view cannot tell them apart: their *order* is the only discriminant. They divide by whether the consequent affirms before it denies (R_1–R_3, confirming) or denies before it affirms (R_4–R_6, the invalidation-led family of Section 3.7).

TABLE 6. The six reference signatures: the orderings of the four bracketing events with the antecedent onset $P^\top$ fixed as opener. All share the presence triple $(A, B, C) = (1, 1, 1)$; the readings are the informal observed glosses of Table 4, not definitions.

ID	Signature	Consequent order	Reading
R_1	$P^\top Q^\top P^\bot Q^\bot$	affirm, then deny	clean forward (reference)
R_2	$P^\top P^\bot Q^\top Q^\bot$	affirm, then deny	retracted confirmation
R_3	$P^\top Q^\top Q^\bot P^\bot$	affirm, then deny	retracted confirmation
R_4	$P^\top Q^\bot P^\bot Q^\top$	deny, then affirm	spurious falsifier
R_5	$P^\top Q^\bot Q^\top P^\bot$	deny, then affirm	contested confirmation
R_6	$P^\top P^\bot Q^\bot Q^\top$	deny, then affirm	contested confirmation

Reduced signatures are subsequences. A monitor whose frame $[\texttt{pre}, \texttt{post}]$ is narrower than the process records only a *subsequence* of a reference signature: events outside the frame are unobserved. Every shorter window class of Table 4 is a reference signature seen through such a frame, and forgetting order collapses the six to the eight presence classes. A reduced signature therefore names not a single reference but the *set* of references of which it is a subsequence, with the missing events restorable on either side of the frame.

The two sides are not symmetric. Which side of the frame carries the missing events fixes how the reduced signature may be reread, and the two differ in kind.

- *After* `post` (late evidence): a missing consequent onset may still arrive beyond the window. The reading stays within the six — the antecedent remains the antecedent — and the outstanding obligation is coliveness: no finite frame confirms that the consequent will, or will not, yet answer.
- *Before* `pre` (prior activity): a consequent onset $Q^\top$ revealed before the trigger $P^\top$ reverses the roles. The consequent now precedes, and may explain, the antecedent; the reading leaves the six reference signatures for the consequent-first family — a causal-role inversion we treat as plausible rather than established.

This reversal is an artefact of the finite frame, not of the theory of time. It already arises under a single, absolute (Newtonian) time order — the earlier onset having merely fallen outside the window — so no relativistic or quantum account of time is needed to produce the confusion; abandoning a global time order would only compound it. The ambiguity is thus purely observational, a property of the frame rather than of physics. Thus $P^\top Q^\bot$,

the transient contradiction of the falsifying-premise case (Section 3.7), is a subsequence of $\mathsf{R}_4 = P^\top Q^\perp P^\perp Q^\top$ under a `post`-side restoration (antecedent P, transient falsification then recovery), and equally of $Q^\top P^\top Q^\perp P^\perp$ under a `pre`-side restoration in which an earlier $Q^\top$ makes the consequent the cause and inverts the implication. The antecedent–consequent swap is confined to the `pre` side: late evidence grades the attribution, prior evidence can overturn it.

3.7. **Invalidation-Led Signatures: Transient Contradiction of a Standing Consequent.** The invalidation-led signatures R_4–R_6 of Section 3.6 warrant a closer reading, for they are where observation and causation most readily part. They arise when the consequent already *holds* as the opener $P^\top$ fires and the antecedent triggers a transient *contradiction* $Q^\perp$ before the consequent re-confirms: a standing consequent can be disturbed only by first falling, so $Q^\perp$ precedes any returning $Q^\top$. Table 7 re-tabulates the three by the position of the contradiction gap $[Q^\perp, Q^\top]$ relative to the antecedent retraction $P^\perp$ — the only degree of freedom left once the presence triple is fixed at $(A, B, C) = (1, 1, 1)$.

TABLE 7. Invalidation-led signatures: an antecedent-triggered transient contradiction $Q^\perp$ of a standing consequent, followed by recovery $Q^\top$. The causal reading is an informal gloss, not a definition; it records how tightly the contradiction is bracketed by the antecedent's presence.

Signature	Contradiction gap vs. antecedent	Causal gloss
$P^\top Q^\perp P^\perp Q^\top$	gap contains the retraction $P^\perp$	antecedent sustains the contradiction
$P^\top Q^\perp Q^\top P^\perp$	gap lies inside the antecedent window	onset transient; consequent self-restores
$P^\top P^\perp Q^\perp Q^\top$	gap lies wholly after $P^\perp$	lagged; attribution to the antecedent is weak

The contradiction is confined to the gap $[Q^\perp, Q^\top]$. Whenever the consequent re-confirms, the recovery is itself a bounded-eventuality obligation (Definition 5.1): once contradicted, the consequent must re-confirm within a bounded horizon, and the signature is *temporary*. If no $Q^\top$ returns within the window, the contradiction is not observed to close and the recovery degrades to a coliveness obligation that no finite window can confirm — a *persistent* invalidation rather than a transient one.

3.8. **Signal Layers: Ground Truth, Measurement, and Observation.** The formal model involves three distinct signal layers, each required for a precise statement of what the monitor computes and can infer.

Definition 3.7 (Signal Layers)**.**

(1) **Ground-truth layer.** $(p(t), q(t))_{t\in\mathbb{T}}$ are the ideal Boolean signals produced by the emitter processes $\mathcal{P}$ and $\mathcal{Q}$. These are not directly accessible to $\mathcal{M}$.
(2) **Measurement layer.** $(\tilde{p}(t), \tilde{q}(t))$ are the signals arriving at the monitor's input after the observation channel (L, S, D, J) has introduced delay, jitter, and potential Boolean distortion (phantom triggers, missed events).

(3) **Observation layer.** $\sigma_{\text{obs}}(t) \in \Sigma$ is the single token emitted *after* the monitor's own processing: event detection, glitch suppression (filtering/hysteresis), and sampling-phase alignment. All formal definitions (Definition 3.4), features, and eight-tick signatures are stated exclusively in terms of σ_{obs}. The derived predicates w_p, w_q (Definition 3.8) project this token onto the Boolean propositions used in the LTL formulas of Section 5.

The eight window classes and the 50000-entry eight-tick catalogue that enumerates them (Appendix A-1) classify *observed* traces, not ground-truth behaviour: two physically identical ground-truth evolutions can land in different window classes if the observation channel (L, S, D, J) or the monitor's processing chain differs.

Definition 3.8 (Observation Alphabet)**.** At each tick t the monitor records a single observation token $\sigma_{\text{obs}}(t)$ drawn from a five-state monodimensional alphabet:

$$\Sigma \;=\; \{\, \varnothing,\; P^{\top},\; P^{\perp},\; Q^{\top},\; Q^{\perp} \,\}, \tag{3.7}$$

where the symbols denote respectively: no observation ($\varnothing$), antecedent confirmed ($P^{\top}$), antecedent retracted ($P^{\perp}$), consequent confirmed ($Q^{\top}$), and consequent contradicted ($Q^{\perp}$). At the opener tick x, the token is always $P^{\top}$ (by definition of trigger). At any subsequent tick $t > x$, the observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$ emits exactly one token; when both P and Q signals are present at the measured layer, the Q-channel event takes priority because it carries the diagnostic information for Φ_k (see Definition 8.1). For backward compatibility with the LTL formulas of Section 5, we define the derived predicates $w_p(t) \equiv [\sigma_{\text{obs}}(t) = P^{\top}]$ and $w_q(t) \equiv [\sigma_{\text{obs}}(t) = Q^{\top}]$. Feature A requires $\sigma_{\text{obs}}(t) = Q^{\top}$ at some tick; feature B requires $Q^{\perp}$; feature C requires $P^{\perp}$ at some tick $t > x$.

The monodimensional constraint (one symbol per tick) restricts the observation space but leaves the classification codomain unchanged: the eight classes are fixed by the crossing/-touch pattern of Definition 3.5 (Q-validation A, Q-invalidation B, P-retraction C), so the class count remains 8, quotiented from the 50000-entry eight-tick catalogue at k=3 (Remark A.1). Its algebraic reading as a parallel composition $E_P \parallel E_Q$ is developed in the supplementary material and a related paper [Bra26b].

3.9. **4-Band Window Model (Minimal Partition).** A minimal 4-band abstraction suffices for the semantics of bounded implication (though not for boundary-sensitive tuning).

Definition 3.9 (4-Band Temporal Partition)**.** Given opener tick x (Definition 3.10) and window bound k, the timeline is partitioned into four bands:

$$\begin{aligned} \texttt{pre}&: \; t < x && \text{(idle)} \\ \texttt{x}&: \; t = x && \text{(opener)} \\ \texttt{win}&: \; x{+}1 \le t \le x{+}k && \text{(window interior)} \\ \texttt{post}&: \; t > x{+}k && \text{(post-window)} \end{aligned} \tag{3.8}$$

$\mathbf{G}(w_p \to \mathbf{F}_{\le k} w_q)$ is satisfied when $w_q = \top$ appears in `win` for every trigger in `x`; every observation in these bands is a crossing or touch event (Definition 3.5), so the validity levels w_p, w_q are read directly off this partition. It is sufficient for the crossing/touch classification determining the eight window classes of Table 4 and backs the eight-class completeness check (concrete τ_1–τ_5 examples are in the supplementary material); the eight-tick refinement next adds boundary timing resolution without altering the eight-class quotient.

3.10. **The Eight-Tick Temporal Partition.** The 4-band partition of Section 3.9 is sufficient for the semantics of bounded implication, but provides no boundary resolution at the opener or at the closing tick. Boundary resolution is what separates an immediate confirmation from a last-admissible one — a distinction on which the diagnostic and tuning machinery of the second part of the paper depends (Sections 8 and 9). The band/tick observation model refined here is shared with the engineering companion [BBI26], which develops it in full generality as the observation layer for an arbitrary monitored property; the present section fixes only the instance the implication reading requires. We therefore refine the timeline into eight explicit ticks, resolving the window interior into one observation per tick:

Definition 3.10 (Opener Tick (bandx))**.** Let x denote the tick at which the monitor $\mathcal{M}$ first observes $P = \top$ and opens the bounded-eventuality window. Formally,

$$x \;=\; \min\{\, t \in \mathbb{T} : w_p \in \sigma_t \,\}. \tag{3.9}$$

Definition 3.11 (Eight-Tick Temporal Partition)**.** Around the opener tick x the timeline is partitioned into eight consecutive ticks, each carrying a single observation token from $\Sigma_{\text{obs}} = \{\varnothing, P^\top, P^\perp, Q^\top, Q^\perp\}$:

$$\begin{aligned}
\texttt{pre}&: \; t \le x{-}2 \quad \text{(out-of-band, free)}\\
\texttt{x-1}&: \; t = x{-}1 \quad \text{(out-of-band, free)}\\
\texttt{x}&: \; t = x \quad \text{(opener, pinned } P^\top\text{)}\\
\texttt{x+1}&: \; t = x{+}1 \quad \text{(in-band window)}\\
\texttt{m2}&: \; t = x{+}2 \quad \text{(in-band window)}\\
\texttt{cl}&: \; t = x{+}3 \quad \text{(in-band window, closing)}\\
\texttt{cl+1}&: \; t = x{+}4 \quad \text{(out-of-band, free)}\\
\texttt{post}&: \; t \ge x{+}5 \quad \text{(out-of-band, free)}
\end{aligned} \tag{3.10}$$

The in-band window thus spans the three consecutive ticks $x{+}1, x{+}2, x{+}3$ (the reference bounded-eventuality window), one observation per tick. The *in-band window* is the three ticks $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ — one observation each — which carry the three window events of a reference signature; the opener tick x is pinned to $P^\top$ (Definition 3.10), and the four out-of-band ticks $\texttt{pre}, \texttt{x-1}$ (before the opener) and $\texttt{cl+1}, \texttt{post}$ (after the closing tick) are each a free Σ_{obs} tick. The presence triple $(A, B, C) = (\exists\, Q^\top, \exists\, Q^\perp, \exists\, P^\perp)$ over the in-band window induces the eight window classes of Table 4.

Definition 3.12 (Band-Level Token Aggregation)**.** For any multi-tick band $B = \{t_1, \ldots, t_n\}$, the *canonical band token* is the priority-maximum observation over all ticks in B:

$$\text{tok}(B) \;=\; \max_{\prec}\{\, \sigma_{\text{obs}}(t) : t \in B \,\}, \tag{3.11}$$

where the total order on Σ is

$$\varnothing \;\prec\; P^\top \;\prec\; P^\perp \;\prec\; Q^\perp \;\prec\; Q^\top.$$

The ordering reflects diagnostic informativeness: Q-channel events ($Q^\top$, $Q^\perp$) outrank P-channel events because they carry the evidence for or against Φ_k; among Q events, $Q^\top$ (confirmation) outranks $Q^\perp$ (contradiction); $P^\perp$ (retraction) outranks $P^\top$ because it adds the $C{=}1$ feature. In the eight-tick partition (Definition 3.11) each tick already carries a single token, so tok is the identity on every tick; aggregation is needed only for intra-tick

same-bin collisions (where the order $\prec$ selects the emitted token) and for the coarser 4-band projection $\pi_{8\to4}$ (Remark 3.13), whose `win` band spans the three in-band ticks $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$.

The explicit opener tick `x` distinguishes the *opener* (where the trigger occurs) from `x+1` (the first opportunity for an immediate consequent response). This separation is essential for the signal-synchronisation analogy developed in Section 9: counts of confirming events in `x+1` versus
`x` directly define the timing-error signal used by the self-tuning loop.

Design rationale. The eight-tick partition keeps the opener and the first-response tick as separate ticks rather than merging them. The split into `x` and `x+1` is essential for three reasons: (i) it anchors all relative timing measurements at a unique opener tick; (ii) it maps the early/late counts n_0, n_1 of the timing-error detector (Section 9) one-to-one to distinct ticks, eliminating aggregation artefacts; and (iii) the dedicated closing tick `cl` at $t = x{+}k$ provides an unambiguous end-of-window marker aligned with k-bounded eventuality semantics.

Remark 3.13 (Projection to the 4-band model). The eight-tick partition is a refinement of the 4-band model (Definition 3.9). Define the projection $\pi_{8\to4}$ by combining the in-band window $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ into a single `win` band via the priority-max rule of Definition 3.12:

$$\mathrm{tok}(\texttt{win}) \;=\; \max_{\prec}\{\, \sigma_{\mathrm{obs}}^{x+1},\ \sigma_{\mathrm{obs}}^{m2},\ \sigma_{\mathrm{obs}}^{x+k} \,\},$$

while collapsing the out-of-band ticks `pre`, `x-1` to `pre`, mapping `x` $\mapsto$ `x`, and `cl+1`, `post` to `post`. This projection preserves the crossing/touch observations (A, B, C) of Definition 3.5—hence the eight window classes are identical under either partition. What the refinement adds is boundary timing sensitivity: the separate `x+1` and `cl` ticks enable the timing-error detector (Section 9) and the early/late counts that drive the self-tuning loop.

Concrete eight-tick trace examples. Each trace is a tuple

(`pre`, `x-1`, `x`, `x+1`, `m2`, `cl`, `cl+1`, `post`)

with $P^\top$ at `x` in all cases. Each tick contains a single observation token from Σ.

ID	eight-tick trace	Reading
τ_6	$(\varnothing, \varnothing, P^\top, Q^\top, \varnothing, \varnothing, \varnothing, \varnothing)$	confirmation at `x+1`
τ_7	$(\varnothing, \varnothing, P^\top, \varnothing, \varnothing, Q^\top, \varnothing, \varnothing)$	last-admissible at `cl`
τ_8	$(\varnothing, \varnothing, P^\top, \varnothing, \varnothing, \varnothing, \varnothing, Q^\top)$	late evidence in `post`
τ_9	$(Q^\top, \varnothing, P^\top, \varnothing, \varnothing, \varnothing, \varnothing, \varnothing)$	precedence anomaly (Q in `pre`)
τ_{10}	$(\varnothing, \varnothing, P^\top, Q^\perp, Q^\top, \varnothing, \varnothing, \varnothing)$	contradiction, then recovery ($(1, 1, 0)$)

Trace τ_6 versus τ_7 shows the boundary sensitivity gained by the eight-tick refinement: under $\pi_{8\to4}$, both map to the 4-band trace τ_1, yet their *timing* differs by $k{-}1$ ticks—a distinction that the early/late counts of the timing-error detector (developed in a related engineering paper [BBI26]) can detect and the self-tuning loop can correct.

3.11. **Observation Model and Environmental Generality.** The formal model imposes *no* causal or probabilistic constraints on the relationship between the emitter processes $\mathcal{P}$ and $\mathcal{Q}$. Four properties define the *maximal observation model*:

(1) **Independence.** Neither the emission of e_P nor the truth of P constrains the timing or truth value of Q, and vice versa.
(2) **Noise tolerance.** The channel (L, S, D, J) can distort timing and Boolean values alike: the monitor may observe $w_p = \top$ when $p = \bot$ (phantom trigger) or $w_q = \bot$ when $q = \top$ (missed confirmation).
(3) **Exogenous triggers.** $Q{=}\top$ may arise from causes unrelated to P (independent emitters, sensor cross-talk, environmental noise).
(4) **Multiple and repeated signals.** Over a multi-tick band the monitor may observe any token sequence from Σ; under the crossing modality the per-window validation count is capped by the number of touch events, i.e. local maxima at or above the threshold ($N_X \leq N_T$, up to one trailing-boundary correction when the window closes inside an above-threshold excursion [Bra26b]).

The eight observation classes of Table 4 enumerate every structurally distinct outcome under this model, so the partition is complete by construction: no environmental or causal assumption excludes a case. *Restricted environments* (deterministic latency, no exogenous triggers, no noise) are strict subsets reaching only a subset of the eight window classes; the full catalogue remains valid as an upper bound, with application-specific pruning deferred to Section 4.4. Two band-level constraints are absolute, by definition of the opener $x = \min\{t : w_p \in \sigma_t\}$: no $P^\top$ before the opener, and $\sigma_{\text{obs}}(x) = P^\top$ at it; all other bands, and the ground-truth propositions p, q, are unconstrained (the full per-band admissibility table is in the supplementary material).

Remark 3.14 (Forward-looking: beyond the monodimensional baseline). The monodimensional alphabet Σ is a *baseline* choice (compact 50000-entry eight-tick catalogue at $k{=}3$, deterministic P/Q priority rule). Two refinements are available for future work without altering the eight-class quotient: *micro-band spawning* (expanding a shared tick into two per-channel sub-ticks) and an *interval / phase monitor* (encoding edges, overlap, and phase via an augmented token set; the point-event/interval bridge is developed in Section 8.2). Neither is implemented here; the baseline suffices for the analyses of Sections 4.4 and 9.

4. Causal Validity of Observed-Implication Signatures

This section develops the reference-class and causal-soundness apparatus that the paper's headline result, Theorem 7.11 (Section 7.5), quantifies over; it is the semantic groundwork for that theorem, not an independent result.

With the eight-class tick geometry of Section 3 in place, we now ask which observed orderings of antecedent and consequent events may be *read* as causally valid instances of $P \to Q$. This section fixes which signatures are causally *valid* — a trace-side, semantic question: it settles which orderings the forward reading is well-posed on, and grades every other observed class as a causally-ordered departure. How to *transform* one observation class into another is the operational question, developed in the companion engineering paper [BBI26]; here we settle validity, not transformation.

4.1. **The Antecedent-Validated Reference Class.** The eight classes of Table 4 record, for a *fixed* opener, a static count of in-window crossings; they deliberately forget the *order* in which those crossings arrive. Ordering that count isolates a small canonical family, which we take as the reference class for the forward reading of $P \to Q$.

Four boundary events, one ordering. Fix a window opened by a single antecedent validation. Within the crossing modality of Definition 3.5 each channel both validates and retracts, giving four distinct boundary events:

$$P^{\uparrow} \text{ (antecedent validates)}, \quad P^{\downarrow} \text{ (antecedent retracts)},$$
$$Q^{\uparrow} \text{ (consequent validates)}, \quad Q^{\downarrow} \text{ (consequent invalidates)}.$$

A *signature* is a temporal ordering of these four events: the first — the *opener* — falls in band x, the remaining three in the observation window. These directional tokens are *not* the in-window feature triple (A, B, C) of Definition 3.5, where A already denotes a *consequent* validation; we track event *order* with $P^{\uparrow}, P^{\downarrow}, Q^{\uparrow}, Q^{\downarrow}$ precisely to avoid that symbol clash, and the antecedent-validating opener $P^{\uparrow}$ is the always-present opener x, not one of the counted triple.

There are $4! = 24$ orderings (PROVEN, elementary). The family splits by which channel owns the opener: the antecedent-first orderings number $2 \cdot 3! = 12$ (opener $\in \{P^{\uparrow}, P^{\downarrow}\}$), of which the $P^{\uparrow}$-opener alone contributes $3! = 6$. These six form the *antecedent-validated* reference class — the only regime in which the premise of $P \to Q$ is present when the window opens, so that the implication is genuinely put to the test.

The group-(a) forward anchor. Take first the three orderings in which the consequent *affirms* before it withdraws ($Q^{\uparrow}$ before $Q^{\downarrow}$) — group (a), the primary forward-reading anchor. Writing the opener $P^{\uparrow}$ once and then the three window events, these are

$$\sigma_1 = P^{\uparrow} \,\big|\, P^{\downarrow}\, Q^{\uparrow}\, Q^{\downarrow}, \qquad \sigma_2 = P^{\uparrow} \,\big|\, Q^{\uparrow}\, P^{\downarrow}\, Q^{\downarrow}, \qquad \sigma_3 = P^{\uparrow} \,\big|\, Q^{\uparrow}\, Q^{\downarrow}\, P^{\downarrow},$$

that is, $\sigma_1 = \neg A\, B\, \neg B$, $\sigma_2 = B\, \neg A\, \neg B$, $\sigma_3 = B\, \neg B\, \neg A$ in the antecedent/consequent validate–retract shorthand ($A \equiv P^{\uparrow}$, $\neg A \equiv P^{\downarrow}$, $B \equiv Q^{\uparrow}$, $\neg B \equiv Q^{\downarrow}$). On each of these the antecedent onset opens the window and a consequent onset answers inside it, so the forward reading "P then Q" applies directly; they are panel (a) of Figure 3. **Extending** to the symmetric case in which the consequent *denies* first ($Q^{\downarrow}$ before $Q^{\uparrow}$) adds group (b), panel (b), completing the six antecedent-validated signatures ($3! = 6$, PROVEN).

They are listed in Figure 3. These six signatures are the *reference class*: every other observed class is a *departure* from them along one independent axis, enumerated in the map below.

The departure map: every other class along one axis. Each observed class departs from the reference six along exactly one independent axis (the enumeration itself is framing, untagged).

- **Opener** (which channel owns band x): opener $\neq P^{\uparrow}$. A $P^{\downarrow}$ opener (six orderings) leaves the premise *not established* — the phantom-trigger departure (HEURISTIC); a $Q^{\uparrow}/Q^{\downarrow}$ opener (the remaining twelve) is *consequent-first*, an effect-before-cause causality inversion whose only reading *on the zero-skew synchrony rung* is retrocausal (and mundane — a skew reordering — once channel skew is admitted), hence outside the forward model (PLAUSIBLE). Both $\neg P^{\uparrow}$ families fall outside not only this reference class but the eight-class model of Table 4 itself, since every window class assumes the antecedent opener $P^{\uparrow}$; they form a separate *premise-not-established* class. The

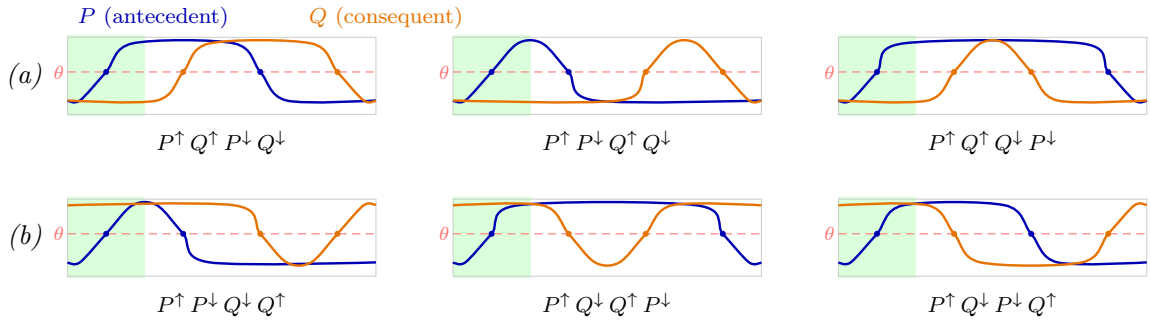


FIGURE 3. The six antecedent-validated signatures as signal-vs-threshold *waveforms*: the antecedent P (blue) and consequent Q (orange) signals cross the semantic threshold θ (red dashed) in the observed order (event dots on θ). Each panel opens with the antecedent validation $P^{\uparrow}$ in the shaded band x, then the three window events left to right. **(a)** the consequent affirms first ($Q^{\uparrow}$ before $Q^{\downarrow}$), so Q rises to a pulse; **(b)** it denies first ($Q^{\downarrow}$ before $Q^{\uparrow}$), so Q enters above θ and dips; the last panel of (b) is the spurious-falsifier order. The enumeration is PROVEN; these event orders are the reference class of Section 4.1.

forward-model exclusion of the consequent-first (effect-before-cause) inversion is made precise by the causal-coherence necessity of Theorem 4.3: a timing model that admits a consequent before its trigger cannot soundly verify $P \to Q$.

- **Presence** (which events occur, order forgotten): the eight window classes; the reference six are the all-present cell $(1, 1, 1)$ (Table 4).
- **Closure** (crossing-CLOSED vs. *open*): a statement may stay *open*, with no closing edge; then a violation of $P \to Q$ is the *absence* of a $Q^{\uparrow}$ onset, not a $Q^{\downarrow}$ event — the genuinely temporal, non-safety case (Section 8.2; [Bra26b]).
- **Out-of-band** (events in the *Pre* or *Post* bands, beyond band x and the window): a consequent onset $Q^{\uparrow}$ landing in the *Post* band is *late evidence*, and — the key unification (HEURISTIC) — this is exactly the theory ladder: each rung *accounts for* such a post-window onset under a declared premise (Section 11, Remark 11.5), attribution not manufacture. A *Pre*-band event is channel activity before the trigger (HEURISTIC).
- **Multiplicity** (more than one crossing per channel, repeated flicker): higher-order departures (HEURISTIC, future work) — noted, not developed here.

Order-refinement of the all-present cell $(1, 1, 1)$. All three counted events are present in each reference signature $(Q^{\uparrow}, Q^{\downarrow}, P^{\downarrow})$, so its static triple is (A, B, C)=$(1, 1, 1)$ — exactly $(1, 1, 1)$ (PROVEN, elementary). The six thus order-refine the single all-present cell, restoring the event order the static triple of Definition 3.5 forgets; they do *not* refine the sparser cells $(1, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$. Whether distinct $(1, 1, 1)$ signatures warrant distinct verdicts we leave open (CONJECTURED).

4.2. **Well-Posedness and the Soundness Rule.** The reference six are exactly the orderings on which the forward reading of $P \to Q$ is *well-posed*. We state the criterion as a soundness rule; it is a HEURISTIC schema — a discipline on how a reading is set up, not an impossibility theorem — whose formal necessity backbone is Theorem 4.3 below.

Remark 4.1 (Soundness rule for the forward reading (heuristic schema)). HEURISTIC. A reading of the observed implication $P \to Q$ on a signature is *well-posed and sound* only when

(i) its *trigger* is the antecedent onset $P^{\uparrow}$ opening band x — not a closing edge $P^{\downarrow}$, and not a consequent event;
(ii) its *response* is the consequent onset $Q^{\uparrow}$ — not the closing edge $Q^{\downarrow}$, which merely reports that Q had already validated; and
(iii) the onset lag lies in the admissible window $\tau_{\min} \le k\,\Delta t \le \tau_{\max}$ (condition (CC1)).

A genuine *violation* is then the *absence* of a $Q^{\uparrow}$ onset after the trigger, not the occurrence of a $Q^{\downarrow}$ event.

This rule is the informal content of the causal-coherence conditions CC1–CC4 (Definition 4.2), whose necessity is Theorem 4.3; we introduce no new primitive but read those conditions back onto the event order. Reading a closing edge ($P^{\downarrow}$ or $Q^{\downarrow}$) as a causal relatum is the *static collapse*: a $Q^{\downarrow}$ is a *point* feature reporting that a prior $Q^{\uparrow}$ has ended, whereas the temporal implication is about the onset $Q^{\uparrow}$ and its persistence — an *interval* feature (Section 8.2).

4.3. **The Physics-of-Causality Reading.** Reading a signature is assigning roles and following the timeline: the band-x event is the trigger, the consequent events are the response, and the second antecedent event modulates the story. Six archetypes exhaust the well-posed and boundary readings; the schema attaching an archetype to an ordering is HEURISTIC.

Archetype	**Signature**	**Reading**	**Status**
Clean forward	$P^{\uparrow}Q^{\uparrow}P^{\downarrow}Q^{\downarrow}$	antecedent asserts, consequent answers at once, both withdraw	reference
Retracted confirmation	$P^{\uparrow}P^{\downarrow}Q^{\uparrow}Q^{\downarrow}$	antecedent asserts then withdraws; consequent confirms then flickers off	reference
Contested confirmation	$P^{\uparrow}P^{\downarrow}Q^{\downarrow}Q^{\uparrow}$	consequent denies, then affirms	reference
Spurious falsifier	$P^{\uparrow}Q^{\downarrow}P^{\downarrow}Q^{\uparrow}$	consequent denies at once (looks like a violation), a late onset rescues	boundary
Phantom trigger	$P^{\downarrow}Q^{\uparrow}P^{\uparrow}Q^{\downarrow}$	consequent confirms while antecedent still absent ($Q^{\uparrow}$ before $P^{\uparrow}$)	departure
$\neg A \to A$ self-flip	$P^{\downarrow}P^{\uparrow}Q^{\uparrow}Q^{\downarrow}$	window opens on absence, antecedent then asserts	departure

The *phantom-trigger* and *spurious-falsifier* archetypes (Section 1.3) are now resolved to the event order that produces them; they mark where a naive reading becomes *ill-posed*:

- **Phantom trigger** ($Q^{\uparrow}$ before $P^{\uparrow}$): the consequent affirms while the antecedent is still absent, so there is no trigger to which the response can be attributed; reading it as "P caused Q" fails clause (i) of the soundness rule.
- **Spurious falsifier** ($P^{\uparrow}$ then $Q^{\downarrow}$, read as a violation): a $Q^{\downarrow}$ closing edge presupposes that Q had already validated, so the signature is a *retracted confirmation*, not a violation; mistaking the closing edge for the *absence* of $Q^{\uparrow}$ is the static collapse and fails clause (ii).

Beyond these, the *consequent-first* orderings ($Q^{\uparrow}$ or $Q^{\downarrow}$ owning band x) place an effect before its cause: on the zero-skew synchrony rung $\mathcal{O}_{\text{sync}}$ (Section 11), where observed order equals emission order, their only reading inverts the causal order and is admissible solely under exotic retrocausal physics — a non-linear-time model (a closed timelike structure) in which an effect may precede its cause. The retrocausal reading is therefore scoped to $\mathcal{O}_{\text{sync}}$, and two *mundane* readings dissolve it once that idealisation is dropped. First, an *uncontrolled experimental setting*: if the observation is not begun in a controlled environment at the trigger but opens mid-process, the first observed event may be a consequent invalidation $Q^{\downarrow}$ (or validation $Q^{\uparrow}$) rather than the antecedent onset $P^{\uparrow}$ — a windowing artefact of where observation began, not an acausal emission. Second, channel skew S can reorder two *independent* emitters' observed events — a skew-induced reordering. Under either reading the same consequent-first observation has an ordinary, non-retrocausal explanation, so we regard the forward-model exclusion of the consequent-first half as PLAUSIBLE. At bottom the exclusion is *definitional*: the opener band x is by construction opened by the first observed antecedent validation w_p ($P^{\uparrow}$), so a consequent-first ordering is simply not an opener — not a claim of physics. Its formal exclusion is the causality inversion of Theorem 4.3.

4.4. **Causal Coherence: the Necessity Backbone.** The soundness rule of Section 4.2 and the archetype reading of Section 4.3 are informal. We now give their formal backbone. The well-posed archetypes are exactly the orderings that satisfy the causal-coherence conditions CC1–CC4 below; the two ill-posed "curious" readings fail its clauses (i)/(ii) by reading a closing edge as a relatum; and the consequent-first orderings fail by causality inversion. In thesis terms, these conditions ground the *attribute-True* layer: they are the physical premises on which the first rung of the theory ladder (Section 11) will rest, and they make precise when the timing parameters $(k, \hat{\varphi}, M^{-}, M^{+}, \theta)$ — adopted by the monitor whose optimal selection and self-tuning are developed in the companion engineering paper [BBI26] — are compatible with the dynamics of the monitored process.

Read off the signal picture. Before the formal conditions, the causal-coherence requirements can be read directly off the sinusoid-versus-threshold picture of Figure 2, where the antecedent signal s_p crosses its threshold θ_p and the consequent signal s_q crosses θ_q. On that picture the conditions below are simply: the antecedent signal must cross θ_p *before* the consequent signal crosses θ_q (order); that consequent crossing must land *inside* the window ((CC1), timing); and the tick grid must be fine enough to resolve the fastest relevant signal scale ((CC4)). Causal coherence is thus a *bounded-latency* premise on the observed process, made visible on the crossing geometry — the same P-before-Q-within-k ordering that Theorem 4.3 requires for reliable verification, distinct from the happened-before order of the observation *transport* (Section 11).

4.4.1. *Principle of Causal Coherence.*

Definition 4.2 (Causal Coherence of a Timing Model)**.** Let $\mathcal{G}$ be a physical or computational process with characteristic time scales $\tau_1, \tau_2, \ldots$ governing the causal chain from antecedent P to consequent Q. A timing model $\mathcal{T} = (k, \hat{\varphi}, M^{-}, M^{+}, \theta, \Delta t)$ adopted by a monitor $\mathcal{M}$ is *causally coherent* with $\mathcal{G}$ if the following conditions hold:

(CC1) **Window plausibility.** The bounded-eventuality horizon satisfies $\tau_{\min} \leq k \cdot \Delta t \leq \tau_{\max}$, where $\tau_{\min}$ is the minimum physical latency from P to an observable Q, and

$\tau_{\max}$ is a physics-derived upper bound beyond which a co-occurrence of P and Q is no longer evidence of causation.

(CC2) **Phase admissibility.** The phase offset $\hat{\varphi}$ and the margins M^-, M^+ correspond to measurable delays in the observation chain: $|\hat{\varphi}| \leq L + S + D \cdot t_{\max} + J$, $M^- \leq L + J$, $M^+ \leq L + J + D \cdot k$.

(CC3) **Threshold detectability.** The decision threshold θ must lie strictly above the instrumental resolution (noise floor) of the monitor's sensor, ensuring that noise-induced fluctuations of the measured signal cannot cross θ and be registered as spurious events.

(CC4) **Tick–time correspondence.** The tick duration Δt must resolve the fastest relevant time scale: $\Delta t \leq \min(\tau_1, \tau_2, \ldots)$.

Condition (CC1) is the most consequential: it prevents both premature falsification (window too short) and vacuous confirmation (window too long). Conditions (CC2)–(CC4) ensure that the monitor's internal clock and measurement apparatus are physically realisable. The window $[\tau_{\min}, \tau_{\max}]$ is a physical-latency estimate — chemical-kinetics rate constants, a radar range-gate — so its tuning is structural, fixed by the process, rather than a free parameter (Appendix G).

Call the validation of $P \Rightarrow Q$ by a monitor $\mathcal{M}$ *reliable* over a trace family when (i) its verdicts are sound and complete with respect to the ground truth on that family (classification reliability), and (ii) its self-tuning loop converges (operational reliability).

Theorem 4.3 (Necessity of Causal Coherence for Implication Verification)**.** *If the timing model $\mathcal{T}$ is not causally coherent with the process $\mathcal{G}$ in the sense of Definition 4.2, then the validation of $P \Rightarrow Q$ by $\mathcal{M}$ is not reliable, irrespective of the observed eight-tick signatures: violations of the window or tick–time conditions systematically undermine reliable classification (clause (i)), and violations of the phase, margin, or threshold conditions systematically undermine the convergence of the self-tuning loop (clause (ii)); the clause-(ii) (convergence) half is* HEURISTIC *and operational — it rests on the convergence conditions of the engineering companion* [BBI26]*, not on a trace-level impossibility. The necessity is systematic in this sense; it is not the claim that every individual trace is forced to a wrong verdict.*

Proof. Cases 1–2 (window violations) and Case 5 (tick–time violation) exhibit failures of clause (i); Cases 3–4 (phase/margin and threshold violations) exhibit failures of clause (ii).

Case 1: $k \cdot \Delta t < \tau_{\min}$ ((CC1) violated, window too short). The causal response Q cannot appear within the observation window, so every trigger event is classified in a violation class (the four A=0 cells) even when the physical implication holds. The monitor produces systematic false negatives.

Case 2: $k \cdot \Delta t \gg \tau_{\max}$ ((CC1) violated, window too long). Within an excessively wide window, the probability of observing $Q = \top$ from a process *causally independent* of P approaches 1 (assuming Q has a positive base rate). The monitor therefore classifies most events in $(1, 0, 0)$ regardless of whether the causal link $P \to Q$ exists, producing a statistical inflation of false positives — a tendency whose probability approaches 1 as the window widens, not a per-trace forced verdict. This is the formal manifestation of the $P \wedge Q \not\Rightarrow (P \Rightarrow Q)$ fallacy (Section 4.4.2).

Case 3: Phase or margin violation ((CC2) violated). If $|\hat{\varphi}|$ exceeds the maximum physical delay, the opener tick x is placed outside the causal cone of P, so the `bandx` observation is causally disconnected from the antecedent.

Case 4: Threshold violation ((CC3) violated). If θ lies at or below the sensor resolution, noise fluctuations cross θ transiently and are registered as spurious events (the noise-induced false positive F5). These spurious crossings enter the confirmation counts from which the early–late timing error e_n is computed, independently of the true offset $\varphi^\star$; the self-tuning loop (Section 9, Appendix B) is then driven by noise rather than by the genuine timing error, and the error-signal and mean-field hypotheses on which its convergence rests are no longer secured: the loop tracks noise instead of converging to $\varphi^\star$ — an operational (convergence) failure, not a per-trace forced misclassification.

Case 5: Tick–time (Nyquist) violation ((CC4) violated, $\Delta t > \min_i \tau_i$). If the tick duration is coarser than the fastest relevant time scale, the sampling lattice under-resolves the P–Q transition: a validation and its retraction (or a transient Q-pulse of width $< \Delta t$) can fall between consecutive ticks and be aliased onto a single sample or missed entirely. The observed eight-tick signature is then not a faithful image of the emitter-side event pattern — two distinct ground-truth traces map to the same signature and to different verdicts — so the classification is not sound: a clause (i) failure, independent of window, phase, or threshold.

In every case the diagnostic output is unreliable; hence causal coherence (all of CC1–CC4) is necessary. □

A worked instantiation of CC1–CC4 on a chemical-kinetics measurement chain — including the derivation of $\tau_{\min}$ and $\tau_{\max}$ from rate constants and the resulting admissible window range — is provided in the supplementary material, together with the extension of the Stage-B optimality criterion by a causal penalty and the physically-guided variant of the self-tuning loop.

4.4.2. *Avoiding the $P \wedge Q \Rightarrow (P \to Q)$ Fallacy.* A recurring theme in this paper is that temporal co-occurrence of P and Q does not establish causal implication. Modelling the occurrences of Q as a Bernoulli process with per-tick base rate $\rho_Q \in (0, 1)$ independent of P, the probability of at least one Q in a window of $k{+}1$ ticks is $1 - (1 - \rho_Q)^{k+1} \to 1$ as $k \to \infty$: for a large enough window the monitor classifies almost every trigger event in $(1, 0, 0)$ even though P and Q are causally unrelated. This is the independent-Q inflation limit — quantified with its window floor k_0 in Proposition 4.5 — and the failure mode of Case 2 of Theorem 4.3. The remedy is condition (CC1): the upper bound $\tau_{\max}$ constrains k so that only causally plausible responses fall within the window. Equivalently (Section 5), the safety fragment $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ needs k small enough to exclude non-causal coincidences while the cosafety falsifier $\mathbf{F}(w_p \wedge \neg w_q)$ needs k large enough to exclude causal delays, and $[\tau_{\min}, \tau_{\max}]$ is the unique range satisfying both.

4.4.3. *A Preview of the Causal-Pathology Taxonomy.* The conditions CC1–CC4 of Definition 4.2 are abstract constraints. We now identify the *concrete* failure modes that arise when one or more conditions are violated, and map every failure mode onto the eight-tick signature space developed in Section 3.10.

The five concrete failure modes F1–F5 — premature falsification, causal inflation, parallel Q, probabilistic causation, and noise-induced false positive — together with their signature-level predicates and the fact that they cover 100% of the 50000 eight-tick signatures (so the CC conditions are non-vacuous: every signature falls in at least one pathology class) are catalogued in Definition E.1 and Table 15 (Appendix E).

The class-level crosswalk of these pathologies against the eight (A, B, C) window classes, and the empirical window-length sensitivity sweep that grounds the CC1 admissibility corridor $\tau_{\min} \leq k \cdot \Delta t \leq \tau_{\max}$, are developed in a related engineering paper [BBI26].

4.4.4. *Formal Guarantees.* We now formalise the two key observations as propositions.

Proposition 4.4 (CC1 Violation ⇒ Premature Falsification)**.** *Let $\tau_{\min} > 0$ be the minimum causal latency of the process $P \to Q$. If $k \cdot \Delta t < \tau_{\min}$, then every trace σ produced by monitor $\mathcal{M}$ satisfies* pred_premature$(\sigma) =$ True *and has outcome* $\in \{$*Violated*, *Vacuous*$\}$.

Proof. If $k \cdot \Delta t < \tau_{\min}$, then Q cannot transition from its base state to a detectable level within the window $[x, x+k]$. Therefore, q_T is never observed: $A = 0$. The absence of q_T means pred_premature(σ) returns True (by definition: no q_T in any in-window band and $A = 0$). Since $A = 0$, the outcome is either Violated (if $B = 1$ and $H \neq 1$) or Vacuous (otherwise), which is exactly the set $\{\text{Violated}, \text{Vacuous}\}$. □

Proposition 4.5 (Inflation Risk under Independent Q)**.** *Let Q be an independent process with per-tick base rate $\rho_Q > 0$. Then for any $\varepsilon > 0$, there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$,*

$$\Pr\big[\exists\, q_T \text{ in window of length } k{+}1\big] \;>\; 1 - \varepsilon.$$

Consequently, $\Pr[OK \text{ outcome by indep. } Q] \to 1$ *as* $k \to \infty$.

Proof. The absence probability is $(1 - \rho_Q)^{k+1}$. If $\rho_Q = 1$ the absence probability is $0 < \varepsilon$ for every k and the claim is trivial, so assume $0 < \rho_Q < 1$; then $0 < 1 - \rho_Q < 1$, so $(1 - \rho_Q)^{k+1} \to 0$ as $k \to \infty$. Choosing $k_0 = \lfloor \ln \varepsilon / \ln(1 - \rho_Q) \rfloor$ gives $k_0 + 1 > \ln \varepsilon / \ln(1 - \rho_Q)$, and since $\ln(1 - \rho_Q) < 0$ this ensures the *strict* inequality $(1 - \rho_Q)^{k_0+1} < \varepsilon$ (including on the integer boundary, where the earlier $\lceil \cdot \rceil - 1$ form fails), whence $\Pr[q_T \text{ in window}] > 1 - \varepsilon$. Once q_T is observed, $A = 1$ and the outcome is OK, regardless of whether the causal link $P \to Q$ actually holds. □

Propositions 4.4 and 4.5 together delimit a *Goldilocks zone* for the window length: k must be large enough to avoid premature falsification yet small enough to prevent inflation by independent Q. This zone is precisely the interval

$$\left\lceil \frac{\tau_{\min}}{\Delta t} \right\rceil \;\leq\; k \;\leq\; \left\lfloor \frac{\tau_{\max}}{\Delta t} \right\rfloor,$$

as formulated by CC1 in Definition 4.2.

In detection-theoretic terms this window-length choice is the selection of an *operating point*: a short window is a conservative threshold that risks false negatives (premature falsification), a long window a permissive threshold that risks false positives (inflation by independent Q) — the familiar Neyman–Pearson / ROC trade-off, with the inflation term a *base-rate* effect, since a high per-tick base rate ρ_Q makes an in-window Q likely regardless of any causal link [GS66, Kay98]. The Goldilocks interval is then the band of operating points on which both error rates stay controlled. Both failure modes have field instances: the premature-falsification side (Proposition 4.4) is illustrated by early-stopping observation, as in the Hawthorne novelty effect, and independent-Q inflation (Proposition 4.5) by the threshold artifacts of collective-action models [Gra78] (Appendix G).

4.4.5. *The Guarantee Stack.* The results of this section complete the paper's guarantee stack, stated in the vocabulary of the three questions:

Question	**Guarantee**
What is observed	*Completeness*: all eight observable window classes are classified (Sections 3 and 7).
What can be read	*Precision and adaptability*: the selected monitor minimises ambiguity and its phase converges to the true offset locally (global convergence conjectured); see Section 9 and [BBI26].
What may be attributed *True*	*Physical validity and traceability*: the timing model is compatible with the process dynamics (CC1–CC4) and every causal failure mode traces to specific signatures and window classes (100% coverage, Prop. 4.4–4.5).

Together these guarantees establish the eight-tick timing model as a valid experimental instrument: its output can be trusted as evidence for or against $P \Rightarrow Q$, but only when the timing parameters are derived from, or constrained by, the governing theory of the observed process — bridging runtime verification, signal synchronisation, and the epistemology of experimental causal modelling. With the physical premises in place, the clopen grounding of Section 4.5 closes the section by deriving the coliveness of the monitored specification from the causal window skeleton, rather than imposing it.

4.5. **Clopen Grounding: Coliveness Derived.** The causal skeleton just fixed is, on the bounded window (Section 3.10), entirely *clopen*, and the coliveness of the monitored specification is what the universal quantifier *adds* to it — derived, not imposed.

Each of the four boundary events is decided by a finite prefix of the window: "event e has occurred by tick i" is both finitely confirmable *and* finitely refutable, hence clopen, in $\mathsf{S} \cap \mathsf{C}$ (PROVEN, elementary). A signature is therefore the clopen conjunction of its positional events,

$$\varphi_\sigma \;=\; P^{\uparrow} \wedge \mathbf{X}\, e_1 \wedge \mathbf{X}\,\mathbf{X}\, e_2 \wedge \mathbf{X}\,\mathbf{X}\,\mathbf{X}\, e_3,$$

with the opener $P^{\uparrow}$ at position 0 and the three window events $e_1 e_2 e_3$ at successive ticks. The four-family class marks used below — $\mathsf{S} \cap \mathsf{K}$, $\mathsf{C} \cap \mathsf{L}$, $\mathsf{L} \cap \mathsf{K}$ — are the intersection-algebra classes of the canon [Bra26a], here *specialised* to the clopen window pattern; the derivation itself is the present paper's. The coliveness mark K does *not* arise at this window level; it appears only under the quantifier lift to the infinite trace:

- quantifying the clopen window pattern with the universal $\mathbf{G}$ over *every* trigger window yields a safety property that is never finitely confirmable, $\mathbf{G}$(non-trivial clopen) $\in \mathsf{S} \cap \mathsf{K}$ (PROVEN);
- the $\mathbf{F}$-lift gives $\mathsf{C} \cap \mathsf{L}$; and
- the unbounded recurrence $\mathbf{G}\,\mathbf{F}$ gives $\mathsf{L} \cap \mathsf{K}$.

So coliveness (K, empty-interior in the trace topology) is *derived* from the clopen window skeleton by the universal $\mathbf{G}$, not an algebra imposed on the four-family classification — it is precisely what the universal $\mathbf{G}$ adds to the causal skeleton of the preceding subsections. The constructive prefix-filtration that realises this window skeleton as a deterministic clopen monitor — the trie of live prefixes and its transition function — is the operational object of the companion engineering paper [BBI26]; here it grounds only the semantic derivation.

5. LTL Formalisation

5.1. **Preliminaries.** We adopt standard LTL syntax [Pnu77] over the set of atomic propositions AP:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \vee \varphi \mid \mathbf{X}\,\varphi \mid \varphi\ \mathbf{U}\ \varphi$$

Derived operators include $\mathbf{F}\,\varphi \equiv \mathit{true}\ \mathbf{U}\ \varphi$, $\mathbf{G}\,\varphi \equiv \neg\,\mathbf{F}\,\neg\varphi$, and bounded variants $\mathbf{F}_{\leq k}\varphi$, $\mathbf{G}_{\leq k}\varphi$.

Definition 5.1 (Bounded Eventuality)**.** For $k \in \mathbb{N}$, the bounded eventuality operator is defined as:

$$\mathbf{F}_{\leq k}\varphi \;\equiv\; \bigvee_{i=0}^{k} \mathbf{X}^i\,\varphi. \tag{5.1}$$

5.2. **Core LTL Properties.**

Theorem 5.2 (Static-to-Temporal Lifting)**.** *Let $\varphi_{\text{static}} = p \to q$. Under zero observation parameters ($L = S = D = J = 0$), the LTL formula*

$$\Phi_0 \;=\; \mathbf{G}\big(w_p \to w_q\big) \tag{5.2}$$

is satisfied on a trace σ if and only if every snapshot in σ satisfies φ_{static} with respect to the monitor's observations.

Proof. Under zero parameters, $t_M^P = t_P$ and $t_M^Q = t_Q$. Hence w_p holds at tick i exactly when p holds at tick i, and similarly for w_q and q. Therefore $\mathbf{G}(w_p \to w_q)$ reduces to $\mathbf{G}(p \to q)$, which holds iff every state satisfies the material conditional. □

Theorem 5.3 (Bounded Temporal Implication — sufficiency)**.** *Under observation parameters (L, S, D, J) with cumulative worst-case offset $\Delta = L + S + D_{\max} + J$, a* sound *temporal generalisation of Φ_0 is obtained by bounding the eventuality:*

$$\Phi_k \;=\; \mathbf{G}\big(w_p \to \mathbf{F}_{\leq k} w_q\big), \qquad k \geq 2\Delta. \tag{5.3}$$

The bound is sufficient, not necessary: the emission-specific calibration bound of Theorem 7.8 is tighter. Here $D_{\max}$ denotes the maximum accumulated *drift offset over the observation horizon; under the linear-drift model of the calibration theorem (per-tick offset $D \cdot t$), $D_{\max}$ is the horizon-uniform envelope $D \cdot t_{\text{hor}}$, so the additive treatment below is the envelope of the multiplicative model used in Theorem 7.8.*

Proof. The worst case occurs when P and Q are emitted simultaneously ($t_P = t_Q$) but the monitor observes P at the earliest possible instant ($t_M^P = t_P + L + S - J$) and Q at the latest possible instant ($t_M^Q = t_Q + L + S + D_{\max} + J$). This envelope presumes the precedence guarantee established in full as Theorem 7.8 (Section 7); here we invoke only its precedence (sufficiency) direction — the observed consequent edge does not precede the observed antecedent edge (equivalently, the latched crossing reading of w_q) — so that pairing the earliest t_M^P with the latest t_M^Q measures a genuine forward gap rather than a sign-reversed one. This forward reference is not circular: the proof of Theorem 7.8 derives the tight bound directly from the observation-model offsets and does not appeal to the present theorem. The maximum forward gap between observation of P and observation of Q is therefore $2J + D_{\max} \leq 2\Delta$. Hence $k \geq 2\Delta$ suffices to guarantee that w_q appears within the bounded window after w_p. □

Theorem 5.4 (Policy Trigger Pattern)**.** *A* policy trigger *is an LTL formula that activates a diagnostic obligation upon observing w_p without subsequent w_q. It is expressed as:*

$$\Phi_{\text{trig}} = \mathbf{G}\Big(\big(w_p \wedge \neg \mathbf{F}_{\leq k} w_q\big) \rightarrow \mathbf{F}_{\leq m}\, alarm \Big), \tag{5.4}$$

where $m \geq 1$ is the diagnostic response deadline.

Proof. The nested bounded-eventuality check forces *alarm* within $\{i, \ldots, i{+}k{+}m\}$ whenever w_p holds at tick i with no w_q in $\{i, \ldots, i{+}k\}$; for finite k, m this is in the safety fragment. □

5.3. **Diagnostic Patterns.** The six forms $\mathcal{L}_1$–$\mathcal{L}_6$ are the principal LTL patterns used throughout the remainder of this paper, classified in Table 8. Pattern $\mathcal{L}_1$ is the zero-delay baseline; $\mathcal{L}_2$ is the bounded generalisation; $\mathcal{L}_3$ is its unbounded (liveness) counterpart; $\mathcal{L}_4$ captures a single counterexample (cosafety); $\mathcal{L}_5$ embeds the policy trigger from Theorem 5.4; and $\mathcal{L}_6$ is the safety dual of the bounded k-window violation pattern (the k-window generalisation of $\mathcal{L}_4$'s instantaneous violation).

The six patterns fix what the *logic* can say about a bounded implication. The first payoff of the geometric partition is to fix what the *monitor* can know about it. We first place these patterns in the four-family algebra (below); the silent window $(0, 0, 0)$ then makes the logic/observation difference concrete, and Section 6 gives it semantic content: vacuity that belongs to the logic must be separated from vacuity manufactured by the observation channel.

TABLE 8. Classification of the LTL patterns in the four-family algebra (S safety, C cosafety, L liveness, K coliveness) [Bra26a]. Each pattern's pair of marks identifies its intersection cell; the last column gives the set of definitive LTL_3 verdicts finitely reachable for the pattern (Remark 5.5).

Pattern	S	C	L	K	**Cell**	**Verdicts**
$\mathcal{L}_1$: $\mathbf{G}(w_p \rightarrow w_q)$	✓			✓	$\mathsf{S} \cap \mathsf{K}$	$\{\bot\}$
$\mathcal{L}_2$: $\mathbf{G}(w_p \rightarrow \mathbf{F}_{\leq k} w_q)$	✓			✓	$\mathsf{S} \cap \mathsf{K}$	$\{\bot\}$
$\mathcal{L}_3$: $\mathbf{G}(w_p \rightarrow \mathbf{F}\, w_q)$			✓	✓	$\mathsf{L} \cap \mathsf{K}$	$\emptyset$
$\mathcal{L}_4$: $\mathbf{F}(w_p \wedge \neg w_q)$		✓	✓		$\mathsf{C} \cap \mathsf{L}$	$\{\top\}$
$\mathcal{L}_5$: Trigger + alarm (k, m finite; Theorem 5.4)	✓			✓	$\mathsf{S} \cap \mathsf{K}$	$\{\bot\}$
$\mathcal{L}_6$: $\neg\, \mathbf{F}(w_p \wedge \mathbf{G}_{\leq k} \neg w_q)$	✓			✓	$\mathsf{S} \cap \mathsf{K}$	$\{\bot\}$

5.4. **Safety / Cosafety / Liveness Classification.** Table 8 states the classification directly in the four-family algebra of a related paper [Bra26a]; the classical tripartite reading (Alpern–Schneider [AS87], Manna–Pnueli [MP90], safety model checking [KV01]) is recovered by projecting onto the first three columns. The constructive derivation of these placements — the quantifier lift $\mathbf{G}$(non-trivial clopen) $\in \mathsf{S} \cap \mathsf{K}$ — is given in Section 4.5, and the clopen prefix-filtration that realises the window skeleton as a deterministic monitor is constructed in the engineering companion [BBI26]; the placement of $\mathcal{L}_6$ follows by its reduction to $\mathcal{L}_2$ (as shown). A violation of $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_5$, or $\mathcal{L}_6$ is witnessed by a finite prefix

(safety), yet no finite prefix ever confirms them (coliveness): all four occupy the *one-sided* cell $\mathsf{S} \cap \mathsf{K}$ — finitely refutable, never finitely confirmable. $\mathcal{L}_4$ is their mirror image in $\mathsf{C} \cap \mathsf{L}$: a single witnessing instant confirms it, and no finite prefix ever refutes it. $\mathcal{L}_3$ occupies the *zero-sided* cell $\mathsf{L} \cap \mathsf{K}$ — neither finitely refutable nor finitely confirmable — which is precisely why the unbounded regime of Definition 7.7 is inaccessible to finite monitoring. Note that $\mathcal{L}_6 \equiv \mathcal{L}_2$ semantically ($\neg\,\mathbf{F}(w_p \wedge \mathbf{G}_{\leq k}\neg w_q) = \mathbf{G}(w_p \rightarrow \mathbf{F}_{\leq k} w_q)$); we keep both because the two syntactic forms — obligation form versus forbidden-witness form — anchor different diagnostic readings in Section 7.

These four-family placements are read directly off each pattern's operator skeleton, and are exact here because the predicates are simple: w_p, w_q are atomic observation tokens and every window is a bounded $\mathbf{F}_{\leq k}$ or $\mathbf{G}_{\leq k}$. The two placements that rest on a short argument rather than a cold read-off are $\mathcal{L}_5$ (whose nested $\neg\mathbf{F}_{\leq k} w_q$ is put in the safety fragment by Theorem 5.4) and $\mathcal{L}_6$ (by its reduction to $\mathcal{L}_2$ above). Were a predicate to hide unbounded temporal structure, its placement would become provisional and require refinement; here the predicates keep the classification a direct read-off.

Remark 5.5 (Two semantic fields). Two classifications with different subjects coexist in this paper, and they live in different semantic fields. Table 8 classifies *sets of infinite traces*: its four families correspond, in the Cantor topology on Σ^ω, to the closed (S), open (C), dense (L), and empty-interior (K) cells, inherited from a related classification [Bra26a] (cf. [DL14]). For an ideal channel this topological classification fixes the *monitor-verdict* reading: $\bot$ is finitely reachable iff the property is not in L, and $\top$ iff it is not in K [Bra26a]; the converse does not hold in general (the cells $\mathsf{S} \cap \mathsf{C}$, $\{\mathsf{S}\}$ and the Quaestio all reach both verdicts), and only the forward read-off is used here, giving $\{\bot\}$, $\{\top\}$ and $\emptyset$ on $\mathsf{S} \cap \mathsf{K}$, $\mathsf{C} \cap \mathsf{L}$ and $\mathsf{L} \cap \mathsf{K}$. Table 8, Figure 4, and the validation scenarios of Section 12 all live in this one field — which definitive verdicts a sound monitor of the pair (formula, observation model) can reach on a finite prefix. The Peled–Havelund placement of Proposition 7.1 classifies a *second, different* object: not which verdict a monitor reaches, but whether the *genuine* implication can be causally confirmed or refuted from the observed signature, once consequent onsets that fall outside or after the window are taken into account. For an ideal channel the two fields correspond classically: every violation of a safety set, and every satisfaction of a cosafety set, is witnessed by a finite prefix [BLS11], while a set in $\mathsf{L} \cap \mathsf{K}$ admits neither a finite refutation nor a finite confirmation. More sharply, the monitor-verdict and causal-attribution fields *coincide* on every signature carrying an in-window settling token — a consequent denial or a retracted confirmation, whose monitor $\bot$ is also a settled causal refutation — but *diverge* on the pure-silence signatures ($(0,0,0),(0,0,1)$: $A = 0$ with no in-window denial), where a sound monitor still emits a definitive $\bot$ on the finite bad prefix of Φ_k while the causal verdict stays undetermined because an out-of-window consequent onset could still confirm. That divergence is precisely the observation-relative miscalibration. The content of Theorem 7.11 is precisely that the observation channel *breaks* this correspondence: the topological class of Φ_k is invariant ($\mathsf{S} \cap \mathsf{K}$ for every finite k), while the verdict-reachability of its monitoring problem collapses as $\Delta \rightarrow \infty$ — from one finitely reachable definitive verdict to none. This finite-k invariance is grounded, not incidental: it instantiates the *horizon-sensitivity gradient* of the related classification [Bra26a], which places $\mathsf{S} \cap \mathsf{K}$ in the *horizon-one-sided safety* regime — refutation decided at a finite horizon, confirmation available only at the unbounded (ω) horizon — and shows that L and K are both residues of the unbounded (ω) trace horizon. Here the trace horizon stays ω for every finite window bound k, so K is present at every finite k (via $\mathsf{S} \cap \mathsf{K}$), while L appears only when the window bound itself

is removed and the safety side is lost, yielding $\mathsf{L} \cap \mathsf{K}$. The degradation $\mathsf{S} \cap \mathsf{K} \to \mathsf{L} \cap \mathsf{K}$ is a trace-class effect of removing the window bound ($k \to \infty$); its monitor-side reading is the $\Delta \to \infty$ verdict collapse recorded above. These are one effect on two axes: the bounded window $\mathbf{F}_{\leq k}$ keeps refutation finite-horizon for every finite k, and only removing the bound (the unbounded $\mathbf{F}$ of $\mathcal{L}_3$) sends the refutation side to the ω-horizon, adjoining the liveness class L (WITNESSED by [Bra26a]). The landscape of Figure 4 is drawn in the monitor-verdict field. A quantitative treatment of the verdict-domain migration itself is left open.

6. Observational versus Logical Vacuity

Closure axis (crossing-closed vs. open). This section develops the departure in which a statement stays *open* — no closing edge — so a violation of $P \to Q$ is the *absence* of a $Q^{\uparrow}$ onset rather than a $Q^{\downarrow}$ event; observational vacuity is exactly this open/silent-window case.

6.1. **Background.** In classical propositional logic, the implication $P \Rightarrow Q$ is *vacuously true* whenever the antecedent P is false. This is a well-understood feature of the material conditional: rows three and four of its truth table ($P = \bot$) produce the value $\top$ regardless of Q.

From a model-checking perspective, vacuity has been studied extensively. A formula φ is said to hold *vacuously* in a model M if φ holds but some sub-formula of φ can be replaced by any formula without affecting the truth value [BBDER01, KV03]. The standard remedy is vacuity detection—a post-hoc analysis that flags properties whose satisfaction is "trivial." Classical vacuity detection asks whether the *formula* was satisfied trivially; this section asks a question it cannot ask: whether the *monitor* can know that it was. The two come apart precisely when the monitor never observes the antecedent.

Classical vacuity is a *logical* artefact—a property of the formula's structure relative to the model. In the *observational* setting of Section 3 a different phenomenon emerges: the monitor $\mathcal{M}$ may persistently observe $P{=}\bot$ (denoted p_F) not because P is logically false, but because channel delay, jitter, or miscalibration prevents $\mathcal{M}$ from ever seeing a true P within its sampling window.

Definition 6.1 (Logical Vacuity vs. Observational Miscalibration)**.** Let σ be a trace and $\mathcal{M}$ a monitor.

- **Logical vacuity**: $\forall i.\ p \notin \sigma_i$ (the antecedent is objectively false at every tick).
- **Observational miscalibration**: $\exists i.\ p \in \sigma_i$ but $\forall j.\ w_p \notin \sigma_j$ (the antecedent is true at some tick, yet the monitor never observes it as true).

This distinction is critical: logical vacuity is a property of the *system*; observational miscalibration is a property of the *monitor*. The classical Linear Temporal Logic (LTL) formula $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ cannot distinguish between these cases—both produce the same verdict (*trivially satisfied*).

6.2. **Development.** To formalise the observational vacuity scenario, we introduce an explicit "observation window" predicate *open* that indicates whether $\mathcal{M}$ has activated a monitoring window upon seeing w_p.

Definition 6.2 (Observational Vacuity Scenario)**.** An *observational vacuity scenario* on a trace σ is a configuration in which all of the following hold:

(a) $\mathcal{M}$ persistently reads p_F: $\mathbf{G}\,\neg w_p$;
(b) no observation window is ever opened: $\mathbf{G}(\neg w_p \to \neg \mathit{open})$;
(c) the implication verdict is trivially $\top$: $\sigma \models \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ holds vacuously.

The key LTL patterns that characterise this scenario are:

$$\Psi_{\text{vac},1} = \mathbf{G}(\neg w_p \to \neg \mathit{open}), \tag{6.1}$$

$$\Psi_{\text{vac},2} = \mathbf{G}\,\mathbf{F}\,\neg w_p \;\wedge\; \neg\,\mathbf{G}\,\mathbf{F}\;\mathit{open}. \tag{6.2}$$

Pattern $\Psi_{\text{vac},1}$ keeps the window closed whenever the monitor reads p_F; $\Psi_{\text{vac},2}$ strengthens this to p_F recurring infinitely often while the window opens only finitely.

Lemma 6.3 (Observational Vacuity Entails Trivial Satisfaction)**.** *If a trace σ satisfies $\mathbf{G}\,\neg w_p$, then $\sigma \models \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ for every $k \geq 0$. Moreover, the monitor's verdict is* observationally vacuous *in the sense of Definition 6.2, and no information about the ground-truth status of $P \Rightarrow Q$ can be inferred.*

Proof. If w_p never holds, the antecedent of the implication $w_p \to \mathbf{F}_{\leq k} w_q$ is false at every tick, making the entire formula trivially true. Since $\mathcal{M}$ never triggers an observation window, the consequent w_q is never tested, and hence no verdict—positive or negative—can be grounded in empirical observation. □

Proposition 6.4 (Distinguishing Observational Miscalibration)**.** *The observational vacuity scenario is distinguishable from logical vacuity* if *the monitor has access to an independent* liveness witness *ℓ_P satisfying:*

$$\mathbf{G}\,\mathbf{F}\;\ell_P \;\wedge\; \mathbf{G}(\ell_P \to \mathbf{F}_{\leq k'} w_p),$$

where ℓ_P is an auxiliary proposition that signals a genuine emission of e_P. If $\mathbf{G}\,\mathbf{F}\;\ell_P$ holds but $\mathbf{G}\,\neg w_p$ also holds, the monitor can conclude that the observation channel is miscalibrated (observation delay exceeds the window bound k'). Only this sufficient direction is established here; whether distinguishability requires *such a witness is left open. Its sole role is eliminative: it separates observational miscalibration from logical vacuity, so a persistently silent w_p under a live witness ℓ_P is attributed to a mis-set window rather than to a genuinely vacuous law (cf. Section 6.3); we do not use the converse.*

Proof. $\mathbf{G}\,\mathbf{F}\;\ell_P$ forces P-emissions to recur infinitely often, and $\mathbf{G}(\ell_P \to \mathbf{F}_{\leq k'} w_p)$ requires each to become visible within k' ticks; together they force w_p to recur, contradicting $\mathbf{G}\,\neg w_p$. Hence $\mathbf{G}\,\mathbf{F}\;\ell_P \wedge \mathbf{G}\,\neg w_p$ is satisfiable only when k' is too low—i.e. the channel is miscalibrated. □

TABLE 9. Comparison of classical vs. observational vacuity.

Aspect	**Classical (Logical)**	**Observational**
Source	P is objectively false	$\mathcal{M}$ cannot observe $P{=}\top$
Diagnosable?	Requires model-level vacuity check	Requires liveness witness ℓ_P
Verdict	Trivially $\top$	Trivially $\top$, but *uninformative*
Remedy	Coverage metrics, witness generation	Channel calibration, increase k'
LTL pattern	$\mathbf{G}(p \to q)$ vacuous	$\Psi_{\text{vac},1}$, $\Psi_{\text{vac},2}$

6.3. **Summary.** The revision of the vacuity class reveals a tripartite distinction that the classical LTL treatment collapses into a single case, as summarised in Table 9:

Observational vacuity leaves the monitor *blind* to scenarios in which P is true but unobserved: any monitor in this class gives *zero coverage* of the critical row $(P{=}\top, Q{=}\bot)$, regardless of how many ticks are observed. In the reference frame of Section 4.1, observational vacuity is the CLOSURE departure: the statement never closes, so the violation is the *absence* of a $Q^{\uparrow}$ onset rather than a $Q^{\downarrow}$ event. Having refined vacuity into its tripartite structure, we close the observational layer on its logical root: why implication—uniquely among the connectives—forces this temporal analysis at all.

6.4. **LEM, Implication, and Temporal Validity.** One logical asymmetry motivates the entire observation pipeline. The Law of Excluded Middle,

$$\varphi \ \vee \ \neg\varphi, \tag{6.3}$$

is *immanent*: at each tick t exactly one disjunct holds, with no inter-tick coupling and no observation model. As a connective, $P \Rightarrow Q$ is equally truth-functional at a single valuation; but once P and Q denote *events distributed in time*, the empirical evaluation $P(t) \Rightarrow Q(t')$ spans two instants and a theory-governed coupling assumption ($t' \in [t, t+k]$ in Φ_k) supplied not by $\Rightarrow$ but by the temporal quantifiers ($\mathbf{G}$, $\mathbf{F}_{\leq k}$) and the physics of when Q may respond to P. Hence (6.3) is checkable locally, whereas evaluating $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ additionally requires a sampling model, a detection kernel mapping noisy measurements to Boolean tokens, and a window bound k. Even the immanent reading is not free of a temporal premise: that "exactly one disjunct holds at each tick" is well-defined only if the tick is an atomic, race-free instant — the same zero-width hypothesis $\Delta t_{\text{read}} \to 0$ that the static conditional presupposes (Section 1.2). What then separates the two connectives is *memory*, not atomicity: pointwise LEM is a *memoryless* safety check — $\mathbf{G}(P \vee \neg P)$ denotes Σ^{ω}, the degenerate $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$ tautology of [Bra26a] (since Σ^{ω} is trivially cosafety) — whereas the

bounded implication Φ_k is a *finite-memory* safety check, sitting in $\mathsf{S} \cap \mathsf{K}$ for every finite k (Remark 5.5) and needing one latched bit to carry the pending antecedent across the window. Implication forces the temporal analysis because it is the one connective whose honest evaluation cannot discard that bit. This asymmetry—LEM pointwise, implication-checking temporally extended—is the root cause of every observational pathology catalogued here: the eight classes (Table 4) are the distinct outcomes of a temporally non-immanent implication evaluated through a finite-resolution pipeline. The remaining sections fix where each observed class sits in the monitorability landscape (Section 7), build the tuning machinery (Section 8), and culminate in the Bridge Proposition (Proposition 10.7) and theory ladder (Section 11). Appendix F extends this observation to the excluded middle itself.

7. Fragment Geometry and Calibration

The eight classes say what a monitor can see; this section fixes what each sighting is worth in the logic: where every class sits in the Peled–Havelund verdict landscape, and how large the window must be for those placements to be trustworthy.

7.1. **Mapping to the Peled–Havelund Verdict-Axis Partition.** Each of the eight window classes of Table 4, defined by the crossing/touch observations (A, B, C) of Definition 3.5 ($A = qT_{\leq k}$ observed, $B = qF_{\leq k}$ observed, $C = pF_{\leq k}$ observed), admits a canonical placement on the Peled–Havelund nine-cell verdict-axis partition [PH19]; the correspondence between these PH cells and the safety/cosafety/liveness/coliveness intersection algebra is established in a related paper [Bra26a]. Recall $AFR/SFR/NFR$ encode whether finite refutation is always/sometimes/never available, and symmetrically $AFS/SFS/NFS$ for finite confirmation.

Proposition 7.1 (PH-Cell Placement of Window Classes)**.** *For the bounded-eventuality implication $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$, the eight presence cells (A, B, C) map to PH verdict-axis cells as follows:*

(A, B, C)	***PH cell***
$(1, 0, 0)$	$NFR \cap AFS$
$(1, 1, 0)$	$SFR \cap SFS$ (Quaestio)
$(1, 0, 1)$	$AFR \cap AFS$ (clopen)
$(1, 1, 1)$	$SFR \cap SFS$ (Quaestio)
$(0, 0, 0)$	$SFR \cap SFS$ (Quaestio)
$(0, 1, 0)$	$AFR \cap NFS$
$(0, 0, 1)$	$SFR \cap NFS$
$(0, 1, 1)$	$AFR \cap NFS$

Field. *The cells record the* causal settleability *of the genuine implication from the signature — whether a settled causal confirmation or refutation is available once out-of-window consequent onsets are accounted for —* not *monitor bad-prefix reachability. On the monitor side every $A = 0$ signature (the silent cell $(0, 0, 0)$ included) is a Φ_k bad prefix on which a sound monitor emits $\bot$; the two readings and where they diverge are recorded in the validation scenarios and Remark 5.5.*

Proof. The FR axis records whether a settled causal refutation of the genuine implication is available from the signature. A causal refutation is *always* available (AFR) in exactly two situations: (i) the consequent is denied with no compensating in-window confirmation, $B = 1 \wedge A = 0$ — an uncontested qF-refutation $((0,1,0),(0,1,1))$; or (ii) an in-window confirmation is subsequently retracted, $A = 1 \wedge B = 0 \wedge C = 1$, so that the antecedent's retraction voids the already-latched confirmation and settles a refutation $((1,0,1))$. When a denial is *contested* by a simultaneous confirmation, $A = 1 \wedge B = 1$, refutation is only sometimes settled (SFR: $(1,1,0),(1,1,1)$); a clean uncontested confirmation with no retraction, $(A,B,C) = (1,0,0)$, admits no causal refutation (NFR); pure silence, $(0,0,0)$, and a bare retraction with no prior confirmation, $(0,0,1)$, leave refutation unsettled because an out-of-window onset could still bear on it (SFR). The FS axis records whether a settled causal confirmation is available from the signature. An in-window $A = 1$ supplies a qT-confirmation, but *only* when accompanied by $B = 0$ (no contradictory denial), giving AFS $((1,0,0),(1,0,1))$; when $A = 1 \wedge B = 1$ the confirmation is observationally contested and the property sits in SFS $((1,1,0),(1,1,1))$. For $A = 0$ the signature splits by its settling token: *pure silence* ($B = 0 \wedge C = 0$) leaves confirmation only *sometimes* available (SFS: $(0,0,0)$), since an out-of-window consequent onset can still confirm the genuine implication, whereas a denial or retraction ($B = 1$ or $C = 1$) settles *against* confirmation (NFS: $(0,1,0),(0,0,1),(0,1,1)$). These two rules reproduce each row of the table above, and match the listed PH cell of [PH19, Fig. 1]. On the monitor side, by contrast, the silent cells $(0,0,0)$ and $(0,0,1)$ ($A = 0$ with the opener already fired) are both Φ_k bad prefixes on which a sound monitor emits $\bot$; their causal cells $SFR \cap SFS$ $((0,0,0))$ and $SFR \cap NFS$ $((0,0,1))$ record that this monitor $\bot$ is not causally settled — the two-field gap of Remark 5.5. □

Remark 7.2 (Four-family reading of the PH placement). The PH verdict-axis cells of Proposition 7.1 are not a second language parallel to the four-family algebra of Table 8: the two are tied by the membership/verdict bijection of the related classification [Bra26a], under which each *corner* PH cell coincides with one intersection cell — $AFR \cap AFS = \mathsf{S} \cap \mathsf{C}$ (clopen), $AFR \cap NFS = \mathsf{S} \cap \mathsf{K}$, $NFR \cap AFS = \mathsf{C} \cap \mathsf{L}$, $NFR \cap NFS = \mathsf{L} \cap \mathsf{K}$ — while the central cell $SFR \cap SFS$ is exactly the *Quaestio*, the properties outside $\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K}$. Reading Proposition 7.1 through this bijection gives the four-family cell of each window class directly: $(1,0,0) \in \mathsf{C} \cap \mathsf{L}$; $(1,0,1) \in \mathsf{S} \cap \mathsf{C}$ (clopen); $(0,1,0),(0,1,1) \in \mathsf{S} \cap \mathsf{K}$; $(1,1,0),(1,1,1),(0,0,0)$ in the Quaestio (outside the lattice); and the mixed edge cell $(0,0,1) = SFR \cap NFS$ lies on the K side (an edge cell the bijection splits across at most two membership cells). Through that correspondence [Bra26a], Table 8 and Proposition 7.1 form a single classification rather than two parallel ones. Read on the monitor side this placement is exact and is to be kept: a sound monitor does reach $\bot$ on the finite prefixes of $(0,1,0),(0,1,1)$, so their $AFR \cap NFS = \mathsf{S} \cap \mathsf{K}$ cell correctly records that a finite refutation is available. Read causally, however, the same $(0,1,*)$ triple is by Proposition 4.4 a premature-falsification / retracted-confirmation candidate with an out-of-window onset; the gap between the monitor's $\bot$ and the absence of a settled causal violation is precisely the observation-relative mis-verdict predicted by Theorem 4.3 and Proposition 4.5, so $(0,1,0)$ and $(0,1,1)$ witness the observation-relative monitorability thesis rather than a flat genuine violation. The pure-silence classes behave oppositely. On the monitor side $(0,0,0)$ and $(0,0,1)$ ($A = 0$ with no in-window denial) are Φ_k bad prefixes reaching $\bot$, so the monitor-verdict field reads them in $AFR \cap NFS = \mathsf{S} \cap \mathsf{K}$; their causal cells $SFR \cap SFS$ $((0,0,0))$ and $SFR \cap NFS$ $((0,0,1))$ instead leave the genuine implication unsettled, since an out-of-window consequent onset

could still confirm it. Here the monitor-verdict and causal-attribution fields *diverge*, and that pure-silence divergence is exactly the observational miscalibration of Remark 7.3.

Remark 7.3 (The Observational Vacuity Paradox as Quaestio Membership). This makes the second act of the paradox (Section 1.3) formal. Definition 6.1 separates *logical vacuity* (P objectively false at every tick) from *observational miscalibration* (P true at some tick but never observed); classical LTL collapses both onto the same *trivially satisfied* verdict. Under Proposition 7.1 the distinction is formal: logical vacuity sits in $AFR \cap AFS$ (clopen), whereas observational miscalibration (the silent cell $(0,0,0)$) sits in $SFR \cap SFS$ (Quaestio), joined there by $(1,1,0), (1,1,1)$. Two independent facts make this failure *categorical*, not merely quantitative. First, the $(0,0,0) \to \{(1,0,0),(1,0,1)\}$ migration under kernel tuning crosses out of the Quaestio into a verdict-bearing corner — equivalently, across the four-family lattice from $\mathsf{S} \cap \mathsf{K}$ into $\mathsf{C} \cap \mathsf{L}$ or $\mathsf{S} \cap \mathsf{C}$ — a jump no fragment-preserving LTL transformation can effect. Second, the two fields of Remark 5.5 *diverge* on $(0,0,0)$: a sound monitor emits a definitive $\bot$ on its Φ_k bad prefix, while its causal cell $SFR \cap SFS$ leaves the genuine implication undetermined — a gap that no quantitative recalibration of the verdict can close, only a categorical change of observation regime. That the Quaestio genuinely lies *outside* the four-family lattice — the ω-regular property space decomposes as $(\mathsf{S} \cup \mathsf{C} \cup \mathsf{L} \cup \mathsf{K}) \sqcup \mathit{Quaestio}$, with the Quaestio non-empty (a witnessing property is $(p \vee GFp) \wedge q$) — is established in the related classification [Bra26a]; on our reading, this being-outside-the-lattice is what makes the failure categorical rather than quantitative.

Remark 7.4 (Sampling Bias as Logical Bias). By Proposition 7.1, the distortion introduced by the observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$ is not merely quantitative noise but a *direction-bearing* bias that propagates from the sampling level to verdict classification. Each mis-sampling of the (A, B, C) triple induces a specific PH migration: positive-evidence under-sampling ($A : 1 \to 0$) loses the q^T-confirmation and pushes an AFS cell ($(1,0,0)$) toward the Quaestio ($(0,0,0) = SFR \cap SFS$); negative-evidence over-sampling ($B : 0 \to 1$) fabricates a q^F-denial and biases the FR axis from NFR toward AFR ($(1,1,0)$ or $(0,1,0) = AFR \cap NFS$); retraction over-sampling ($C : 0 \to 1$) fabricates a p^F-retraction, crossing the $C = 0$ row of Table 4 to the $C = 1$ row and re-routing the verdict. Thus *sampling bias maps faithfully onto logical bias*: the bias direction in sampling space coincides with the shift direction in the verdict-availability scale. This justifies the calibration theorems (refined by Section 9): the bound $\Delta < \Delta_{\max}$ is exactly the condition under which sampling bias does *not* cross a PH-cell boundary, so calibration is the guarantee that sampling-level distortion stays *logical-bias-preserving*.

7.2. **Static-Like vs. Bounded Temporal Regimes.**

Definition 7.5 (Static-Like Regime)**.** The observation model is in the *static-like regime* if and only if the cumulative offset satisfies $\Delta = L + S + D_{\max} + J = 0$. In this regime, Φ_0 (Equation (5.2)) is a faithful temporal representative of the classical implication.

Definition 7.6 (Bounded Temporal Regime)**.** The observation model is in the *bounded temporal regime* if $0 < \Delta < \infty$. In this regime, Φ_k (Equation (5.3)) with $k \geq 2\Delta$ is the weakest safety formula that preserves the intended semantics of $P \Rightarrow Q$ up to temporal tolerance.

Definition 7.7 (Unbounded (Liveness) Regime)**.** The observation model is in the *unbounded regime* (also called the *liveness regime*) if the cumulative worst-case offset satisfies $\Delta = \infty$ (i.e., $D_{\max} \to \infty$ or $J \to \infty$). In this regime, no finite bound k exists such that Φ_k is a safety property equivalent to φ_{static}; the implication must be expressed as the liveness formula $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$, which is not finitely monitorable (Lemma 7.9; cf. [BLS11]).

7.3. **Calibration Theorems.**

Theorem 7.8 (Calibration Bound)**.** *Fix an emission schedule (t_P, t_Q) satisfying $\mathbf{G}(p \to q)$ in the static semantics, together with the observation-model bounds $D_{\max}$ and J. Assume the* precedence guarantee*: the observed consequent edge never precedes the observed antecedent edge, $t_M^Q \geq t_M^P$ — equivalently $(t_Q - t_P)(1+D) + (\epsilon_Q - \epsilon_P) \geq 0$ for every admissible realization, or, absent this restriction on the jitter, that w_q is read under the latched crossing modality (Remark C.3), so that a consequent raised before the antecedent remains observable at the trigger tick. Under this guarantee, the observed formula $\mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ holds for* every *trace consistent with the static implication and these bounds — that is, for every admissible drift $D \in [0, D_{\max}]$ and jitter $\epsilon_P, \epsilon_Q \in [-J, +J]$ — if and only if*

$$k \;\geq\; (t_M^Q - t_M^P)_{\max} \\ = \; (1 + D_{\max}) \cdot (\max\{t_P, t_Q\} - \min\{t_P, t_Q\}) + 2J. \quad (7.1)$$

When $t_P = t_Q$ (simultaneous emission), this simplifies to $k \geq 2J$. The equivalence is stated under $t_Q \geq t_P$; when $t_Q < t_P$ the latched crossing reading discharges the consequent at the trigger tick, so any $k \geq 0$ suffices. Without the precedence guarantee the sufficiency ($\Leftarrow$) direction fails: a negative monitored offset — the consequent edge observed strictly before the antecedent edge — escapes the forward-only window $\mathbf{F}_{\leq k}$ of Equation (5.1) *for every k (e.g. $t_P = t_Q = 0$, $D = 0$, $\epsilon_P = +J$, $\epsilon_Q = -J$ give $t_M^P = +J$, $t_M^Q = -J$, so $\mathbf{F}_{\leq k} w_q$ is false at the trigger tick for every k). The bound is thus a property of the admissible class $(D_{\max}, J)$, not of any single realized trace: a trace whose realized offset falls below the worst case satisfies Φ_k for a smaller k, but k must dominate the worst case to cover every admissible realization.*

Proof. The monitor observes P at $t_M^P = t_P + L + S + D \cdot t_P + \epsilon_P$ and Q at $t_M^Q = t_Q + L + S + D \cdot t_Q + \epsilon_Q$. The difference is:

$$t_M^Q - t_M^P = (t_Q - t_P)(1 + D) + (\epsilon_Q - \epsilon_P).$$

($\Leftarrow$) Under the precedence guarantee the monitored offset $t_M^Q - t_M^P$ is nonnegative, and this *forward* gap is maximised by extreme drift $D = D_{\max}$ and extreme jitter $\epsilon_P = -J$, $\epsilon_Q = +J$, giving the right-hand side of Equation (7.1); any k at least this large keeps every admissible realization inside the forward $\mathbf{F}_{\leq k}$ window. The opposite jitter extreme ($\epsilon_P = +J$, $\epsilon_Q = -J$) yields a negative offset, which is covered only because the precedence guarantee is in force — either it is excluded from the admissible class, or the latched reading makes a consequent raised before the trigger already observable, so the obligation is met at the trigger tick. ($\Rightarrow$) This worst-case realization is itself admissible, so if some admissible trace is to be covered for all of them, k must reach the worst-case offset; a smaller k leaves the extreme realization outside the window. The static parameters L and S cancel because they affect both observations equally. □

Lemma 7.9 (Incompatibility of Causal Interpretation)**.** *Under the adversarial delay class ($D_{\max} \to \infty$ or $J \to \infty$), no finite bound k exists such that Φ_k is a safety property equivalent to φ_{static}. Consequently, the causal interpretation "P causes Q" is* incompatible *with unbounded observation parameters.*

Proof. If $D_{\max}$ or J is unbounded, then $|t_M^Q - t_M^P|_{\max}$ diverges, and Φ_k must be replaced by the liveness formula $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$, which is not monitorable in finite time. □

Corollary 7.10 (Safe Monitoring Condition)**.** *Runtime monitoring of $P \Rightarrow Q$ is feasible (i.e., belongs to the safety fragment) if and only if L, S, $D_{\max}$, and J are all finite and known a priori.*

The regime named by Corollary 7.10 — transport bounds that both *exist* and are *known a priori* — is the *synchronous* model in the taxonomy of Dwork, Lynch, and Stockmeyer [DLS88], where fixed upper bounds on message delay and relative processor speed are known; $\Delta = 0$ is its zero-offset instance. Bounds that exist but are not known a priori, or that are known but hold only from an unknown time on, are their two versions of *partial synchrony*; there, as in the adversarial limits of Lemma 7.9 ($D_{\max} \to \infty$ or $J \to \infty$, the *asynchronous* regime), no k fixed in advance is guaranteed to suffice. Our observation parameters (L, S, D, J) are thus a physically-instantiated reading of the classical synchrony taxonomy.

7.4. **Safety–Liveness Landscape.**

7.5. **Observation-Relative Monitorability.** The results of this section consolidate into the paper's headline statement.

Theorem 7.11 (Observation-relative monitorability)**.** *Monitorability of the bounded-eventuality implication is a property of the pair* (formula, observation model)*, not of the formula alone. Precisely, let $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\le k} w_q)$ and let the observation channel carry parameters (L, S, D, J) with cumulative worst-case offset Δ:*

(i) *if $\Delta = 0$, then Φ_0 is a faithful temporal representative of the static implication (Definition 7.5);*

(ii) *if $0 < \Delta < \infty$ and $k \ge 2\Delta$ (the sufficient bound of Theorem 5.3; Theorem 7.8 gives the tighter emission-specific bound), then, under the precedence guarantee of Theorem 7.8 (the observed consequent edge does not precede the observed antecedent edge, or equivalently the latched crossing reading of w_q), Φ_k is a safety formula that preserves the intended semantics of $P \Rightarrow Q$ up to temporal tolerance, and every verdict is reached within a bounded horizon (Definition 7.6 and Theorems 5.3 and 7.8);*

(iii) *if $\Delta = \infty$, no finite k makes Φ_k a safety property equivalent to the static implication: the implication degrades to the liveness formula $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$, which is not finitely monitorable (Definition 7.7 and Lemma 7.9).*

The monitorability fragment is therefore fixed by the pair (formula, observation model)*, not by the formula alone: under a degrading observation model the faithful temporal representative and the definitive verdicts its monitoring problem can reach change with the channel parameters, while the topological cell of Table 8 remains invariant (Remark 5.5). The boundary between the regimes is crossed by parameter drift alone. In particular, a monitor deployed for Φ_k without a declared observation model has no well-defined soundness guarantee (Corollary 7.10).*

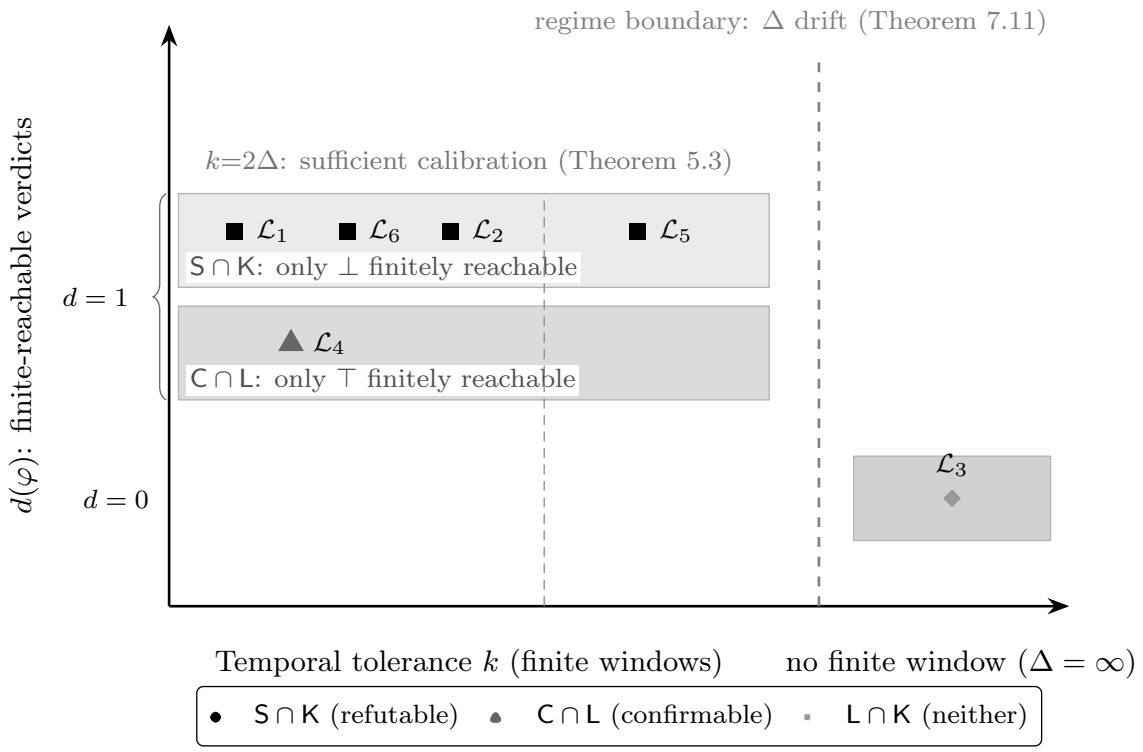


FIGURE 4. Verdict-reachability landscape of the LTL patterns $\mathcal{L}_1$–$\mathcal{L}_6$, drawn in the monitor-verdict field of Remark 5.5. The vertical axis is the number $d(\varphi) \in \{0, 1, 2\}$ of definitive LTL_3 verdicts finitely reachable [Bra26a, BLS11]: the one-sided patterns $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_5, \mathcal{L}_6$ ($\mathsf{S}\cap\mathsf{K}$, only $\bot$) and their mirror $\mathcal{L}_4$ ($\mathsf{C}\cap\mathsf{L}$, only $\top$) share the level $d = 1$; the zero-sided $\mathcal{L}_3$ sits alone at $d = 0$; the clopen level $d = 2$ is not inhabited by any implication pattern. The horizontal axis is the temporal tolerance k for finite windows, with the tick at $k = 2\Delta$ marking the *sufficient* calibration threshold of Theorem 5.3. The regime boundary (dashed) is crossed by observation-model drift $\Delta \to \infty$, *not* by enlarging k: Φ_k remains in $\mathsf{S}\cap\mathsf{K}$ for every finite k (Table 8), and only at $\Delta = \infty$ does the faithful representative of the implication degrade to $\mathcal{L}_3$, collapsing d from 1 to 0 (Theorem 7.11).

Proof. A consolidation of results already established. Case (i) is Definition 7.5 with the lifting of Equation (5.2); case (ii) is Theorem 7.8 instantiated in the bounded regime of Definition 7.6; case (iii) is Lemma 7.9 restated through Definition 7.7. The final claim is Corollary 7.10: feasibility of safety monitoring is equivalent to finiteness and a-priori knowledge of the channel parameters — which is precisely the declaration of an observation model. □

Outside formal verification, the reproducibility findings in psychological science [Ope15] — far fewer replications than originals reached significance against a fixed p-value threshold, with markedly smaller effects — can be read as an instance of this observation-relative thesis, read as non-reproducibility of the recorded signature rather than as noise (Appendix G).

Remark 7.12 (The three cases are a synchrony-taxonomy partition). Read against the classical synchrony taxonomy, the three cases of Theorem 7.11 are a possibility/impossibility partition of the pair *(formula, observation model)*. Case (i), $\Delta = 0$, is the zero-offset synchronous regime. Case (ii), $0 < \Delta < \infty$ with the bound *known* (so that k can be set to 2Δ), is the *synchronous* model of Dwork–Lynch–Stockmeyer [DLS88] (known fixed bounds); there Φ_k is monitorable as a safety property. Case (iii), $\Delta = \infty$ — the *asynchronous* regime — or, equivalently for a monitor, a finite bound that is *not known a priori*, which is their *partially synchronous* model, is the case in which no finite k fixed in advance makes Φ_k safety and the implication is not finitely monitorable; this last case is an impossibility already by Lemma 7.9. The theorem thus partitions monitorability along the synchronous / partially-synchronous / asynchronous axis on which distributed-computing possibility results are standardly stated, with possibility confined to the synchronous, known-bound side.

This closes the first question of the thesis. The eight classes, their fragment placements, and Theorem 7.11 together fix *what is observed* — and nothing so far allows the monitor to change what it sees. The next part of the paper turns to precisely that freedom.

8. Reading the Observation: Kernel, Intervals, and Tuning

The first part of the paper fixed *what is observed*: the geometric partition of the observation space into the eight window classes, their fragment assignments, and the vacuity structure of the silent window. This part turns to the second question of the thesis — *what can be read*. The object of study is no longer the observed trace but the monitor's own decision pipeline: its kernel, the interval structure it projects away, and the tuning operations that move the *same* trace between window classes without touching the specification or the signal.

8.1. **Monitor Observation Kernel and Tuning.**

The passage from the measurement layer to the observation layer (Definition 3.7) is formalised by an *observation kernel*.

Definition 8.1 (Observation Kernel)**.** An observation kernel is a parameterised operator

$$\mathcal{K}_{\boldsymbol{\theta}} = \mathcal{M}_{\theta_4} \circ \mathcal{D}_{\theta_3} \circ \mathcal{A}_{\theta_2} \circ \mathcal{F}_{\theta_1} : (\tilde{p}, \tilde{q}) \mapsto \sigma_{\text{obs}} \in \Sigma, \tag{8.1}$$

where the parameter vector $\boldsymbol{\theta} = (\hat{\varphi}, M^-, M^+, \theta, \psi)$ collects the tunable state. The four stages are a pre-filter $\mathcal{F}_{\theta_1}$ (glitch rejection / debounce), a timing alignment $\mathcal{A}_{\theta_2}$ (sampling-phase correction by the phase offset $\hat{\varphi}$), a decision stage $\mathcal{D}_{\theta_3}$ (thresholding at θ with optional majority vote), and window gating $\mathcal{M}_{\theta_4}$ (the eight-tick partition of Definition 3.11 applied with margins $[x - M^-,\ x + k + M^+]$). All four operate exclusively on the measured layer $(\tilde{p}, \tilde{q})$ and never modify the ground-truth layer (p, q), so they cannot alter the crossing/touch classification of Definition 3.5. Their signal-processing realisation — matched filter, PLL/CDR phase corrector, symbol-level decision device, gated correlator — and the runtime adaptation of $\boldsymbol{\theta}$ are developed in the companion engineering paper [BBI26].

Detection-error model and tuning objective (deferred). The decision stage $\mathcal{D}_{\theta_3}$ commits detection errors with channel false-positive and false-negative rates

$$\alpha_q(\boldsymbol{\theta}) = \Pr\big(w_q{=}\top \;\big|\; Q{=}\bot\big), \tag{8.2}$$

$$\beta_q(\boldsymbol{\theta}) = \Pr\big(w_q{=}\bot \;\big|\; Q{=}\top\big), \tag{8.3}$$

(with analogous α_p, β_p for the antecedent channel), and tuning selects the parameters that minimise the expected misclassification risk between the observed and ideal traces,

$$\boldsymbol{\theta}^\star \;=\; \arg\min_{\boldsymbol{\theta}} \; \mathbb{E}\big[\, d\big(\mathcal{K}_{\boldsymbol{\theta}}(\tilde{p},\tilde{q}),\; \mathcal{K}_{\mathrm{id}}(p,q)\big)\,\big], \tag{8.4}$$

with $\mathcal{K}_{\mathrm{id}}$ the ideal (noiseless) observation map and $d(\cdot,\cdot)$ the Hamming distance on eight-tick signatures. The objective is *not* to bias the monitor toward satisfaction of Φ_k, which appears only as the classification target: tuning adjusts the operating point on each channel's receiver operating characteristic, never the ground-truth trace, the crossing/touch classification of Definition 3.5, or the eight-class partition. The full detection-error model, its region-dependent sensitivity analysis, the minimum-risk / matched-filter development of (8.4), and the self-tuning stochastic-approximation implementation are developed in the companion engineering paper [BBI26].

Remark 8.2 (Invariance and non-circularity of tuning). The crossing/touch classification map $\mathcal{C}$ of Definition 3.5, which reads the in-window Q-validation, Q-invalidation, and P-retraction pattern of any layer-3 trace, is defined independently of $\boldsymbol{\theta}$: its codomain is the fixed set of eight window classes. Tuning $\boldsymbol{\theta}$ changes the *distribution* over observed traces—the operating point on the ROC (Equations (8.2) and (8.3))—and may route a given measured trace to a different window class, but never alters $\mathcal{C}$ and never produces an observation outside the eight classes of Table 4. Hence the count of 8 window classes, and the 50000-entry eight-tick catalogue that quotients onto them, are invariants of the classification, independent of $\boldsymbol{\theta}$.

This is why the tuning objective is not circular. Minimising (8.4) is *detection-accuracy* optimisation —choosing $\boldsymbol{\theta}$ so the observed trace matches the trace an ideal noiseless observer would produce—not *semantic* optimisation of the apparent truth of Φ_k. The monitor is a statistical detector under the causal hypothesis H_1 ("a causal link $P \to Q$ exists within k ticks"); it minimises misclassification (Equations (8.2) and (8.3)) exactly as a radar receiver tunes its matched filter without altering the target, and if the link is absent (H_0) no kernel stage can manufacture evidence for H_1. Formally, every monitor tuning is *monitor-internal*—adjusting thresholds, windows, phases, gates, or schedules while leaving the signal untouched—and so lies in the lowest layer $\mathcal{L}_1^{\mathrm{tun}}$ of the strict hierarchy $\mathcal{L}_1^{\mathrm{tun}} \subsetneq \mathcal{L}_2^{\mathrm{tun}} \subsetneq \mathcal{L}_3^{\mathrm{tun}}$ [Bra26b] (the superscript distinguishing these from the LTL patterns $\mathcal{L}_1$–$\mathcal{L}_6$); "making Φ_k true" would require synthesising new signal values, an $\mathcal{L}_3^{\mathrm{tun}}$ operation the hierarchy's strictness forbids. Equivalently, the self-tuning loop (Section 9) migrates the monitor between the point-like and interval-like *observational regimes* of Definition 8.10 by improving edge-detection reliability, never touching these combinatorial invariants.

8.2. **Interval Semantics and Refinement.** The six patterns $\mathcal{L}_1$–$\mathcal{L}_6$ are *point-event* specifications: each evaluates the Boolean signals w_p and w_q at individual ticks without reference to their rising or falling edges. This subsection formalises an *interval* (or *phase*) monitor

that tracks signal edges, establishes that the point monitor is a forgetful quotient of the interval monitor, and derives conditions under which the projection preserves the correctness of Φ_k (cf. Remark 3.14).

Discrete-time phase signals. Let $p(t), q(t) \in \{0, 1\}$ denote the ground-truth Boolean levels of the antecedent and consequent at discrete tick t. The monitor observes filtered versions $w_p(t)$ and $w_q(t)$ through the observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$ (Definition 8.1).

Definition 8.3 (Edge Operators and Interval Endpoints)**.** For a discrete-time Boolean signal s, the *rising-edge* and *falling-edge* predicates are:

$$\mathrm{rise}_s(t) \;:=\; \neg s(t{-}1) \;\wedge\; s(t), \tag{8.5}$$

$$\mathrm{fall}_s(t) \;:=\; s(t{-}1) \;\wedge\; \neg s(t). \tag{8.6}$$

The rising edge fires at the tick where s first becomes true; the falling edge fires at the tick where s first becomes false. For the antecedent and consequent, define the *interval endpoints*:

$$\begin{aligned} t_{P\uparrow} &:= \min\bigl\{t : \mathrm{rise}_{w_p}(t)\bigr\}, \\ t_{P\downarrow} &:= \min\bigl\{t > t_{P\uparrow} : \mathrm{fall}_{w_p}(t)\bigr\}, \end{aligned} \tag{8.7}$$

and analogously $t_{Q\uparrow}$, $t_{Q\downarrow}$ for w_q. The *active interval* of the antecedent is $I_P := [t_{P\uparrow},\ t_{P\downarrow})$ and of the consequent $I_Q := [t_{Q\uparrow},\ t_{Q\downarrow})$.

LTL correspondence. In LTL, $\mathrm{rise}_s(t)$ corresponds to $\neg s \wedge \mathbf{X}\, s$ evaluated at tick $t{-}1$; the interval-endpoint convention here shifts detection to the onset tick, aligning with signal-processing practice.

Definition 8.4 (Interval Satisfaction Predicates)**.** Two implication semantics arise naturally from the interval structure:

(1) *Overlap semantics.*

$$\Psi_{\mathrm{ov}} \;:=\; I_P \cap I_Q \neq \varnothing \;\iff\; t_{Q\uparrow} < t_{P\downarrow}. \tag{8.8}$$

The consequent must rise *before* the antecedent falls: both signals are simultaneously active for at least one tick.

(2) *Lag-bounded semantics.*

$$\Psi_{\mathrm{lag},\Delta} \;:=\; t_{Q\uparrow} \;\in\; \bigl[\, t_{P\uparrow},\ \ t_{P\downarrow} + \Delta \,\bigr]. \tag{8.9}$$

The consequent must rise within Δ ticks of the antecedent's falling edge. The antecedent may have ceased as a signal while the causal process it initiated continues.

Overlap semantics is a special case of lag-bounded semantics with $\Delta = 0$: $\Psi_{\mathrm{ov}} \Leftrightarrow \Psi_{\mathrm{lag},0}$ when $t_{Q\uparrow} \geq t_{P\uparrow}$.

The two mechanistic regimes—overlap-type and lag-type causation—and the token-level edge ambiguity under the monodimensional observation model are detailed in Appendix C (Remarks C.2 and C.3).

Definition 8.5 (Forgetful Projection)**.** Let $x := t_{P\uparrow}$ be the quantised opener tick and $W := [x,\ x + k]$ the observation window. The *forgetful projection*

$$\Pi_k \;:\; \mathrm{IntervalTrace} \;\longrightarrow\; \mathrm{PointTrace} \tag{8.10}$$

is defined by recording, for each tick $t \in W$, only the Boolean level $w_p(t)$ and $w_q(t)$, discarding all edge annotations, phase relationships, and interval durations. Concretely:

- The opener is set at $x = t_{P\uparrow}$.
- Q is recorded as "observed in window" iff $\exists\, t \in W : w_q(t) = 1$.
- The falling edges $t_{P\downarrow}$, $t_{Q\downarrow}$ and the active-interval durations $|I_P|$, $|I_Q|$ are forgotten.

Remark 8.6 (Quotient identity). The point monitor is the quotient of the interval monitor by the equivalence relation that identifies all interval traces sharing the same tick-level Boolean samples within W:

$$\text{PointMonitor} \;=\; \text{IntervalMonitor} \,/\, (\text{forget falling edges} + \text{forget durations}).$$

This quotient collapses information in one direction only: distinct interval traces may project to the same point trace, but every point trace has at least one interval preimage (illustrated in Figure 5).

The *concretization* (or *lifting*) γ_k of a point trace is the preimage of Π_k; its canonical representative is fixed by the conservative **latest-fall** rule (Definition C.1, deferred to Appendix C).

The round-trip identities of projection and concretization (Remark C.4) and the resulting projection coherence property (Remark C.5) are stated in Appendix C.

Definition 8.7 (Confirmation Sets)**.** For a window bound k and lag tolerance Δ, define the *confirmation sets*:

$$\mathrm{Conf}_{\mathrm{pt}}(k) := \big\{\, \tau_{\mathrm{pt}} : \tau_{\mathrm{pt}} \models \Phi_k \,\big\}, \tag{8.11}$$

$$\mathrm{Conf}^{\mathrm{ov}}_{\mathrm{iv}}(k) := \big\{\, \tau_{\mathrm{iv}} : \tau_{\mathrm{iv}} \models \Psi_{\mathrm{ov}} \wedge t_{Q\uparrow} \le x+k \,\big\}, \tag{8.12}$$

$$\mathrm{Conf}^{\mathrm{lag}}_{\mathrm{iv}}(k,\Delta) := \big\{\, \tau_{\mathrm{iv}} : \tau_{\mathrm{iv}} \models \Psi_{\mathrm{lag},\Delta} \wedge t_{Q\uparrow} \le x+k \,\big\}. \tag{8.13}$$

$\mathrm{Conf}_{\mathrm{pt}}(k)$ is the set of point traces for which the bounded response property is satisfied. $\mathrm{Conf}^{\mathrm{ov}}_{\mathrm{iv}}$ and $\mathrm{Conf}^{\mathrm{lag}}_{\mathrm{iv}}$ are the corresponding interval-level acceptance regions under overlap and lag-bounded semantics. Operationally, *confirmation* denotes set membership: a trace τ is confirmed iff $\tau \in \mathrm{Conf}_{\mathrm{pt}}(k)$ under point-like regime, or $\tau \in \mathrm{Conf}^{\mathrm{ov/lag}}_{\mathrm{iv}}$ under interval-like regime (Definition 8.10). The interval sets are evaluable only when all four edges are resolved; otherwise, only $\mathrm{Conf}_{\mathrm{pt}}(k)$ is accessible.

The inclusion relationship between confirmation sets under projection is established in Remark C.6 (Appendix C).

Proposition 8.8 (Refinement Coherence)**.** *Let τ be an interval trace satisfying $\Psi_{\mathrm{lag},\Delta}$, and let the antecedent pulse width be $d_P := t_{P\downarrow} - t_{P\uparrow}$. If the window bound satisfies*

$$k \;\ge\; d_P + \Delta + \varepsilon, \tag{8.14}$$

where $\varepsilon \ge 0$ is a safety margin absorbing quantisation and jitter, then $\Pi_k(\tau) \models \Phi_k$.

Proof sketch. Under $\Psi_{\mathrm{lag},\Delta}$, the consequent rises no later than tick $t_{P\downarrow} + \Delta = t_{P\uparrow} + d_P + \Delta$. With $x = t_{P\uparrow}$ and $k \ge d_P + \Delta + \varepsilon$, the window $W = [x,\, x+k]$ contains the tick at which w_q first becomes true. Hence $\exists\, t \in W : w_q(t) = 1$, and the point-event formula Φ_k is satisfied at the projected trace. □

Proposition 8.8 formalises the conditions under which a point-event monitor with window bound k is a *correct abstraction* of the interval semantics: the coarser monitor agrees with the finer one provided k is sufficiently large to cover the antecedent pulse width plus the lag plus jitter. This is a *refinement consistency statement*, not a new property of Φ_k per se.

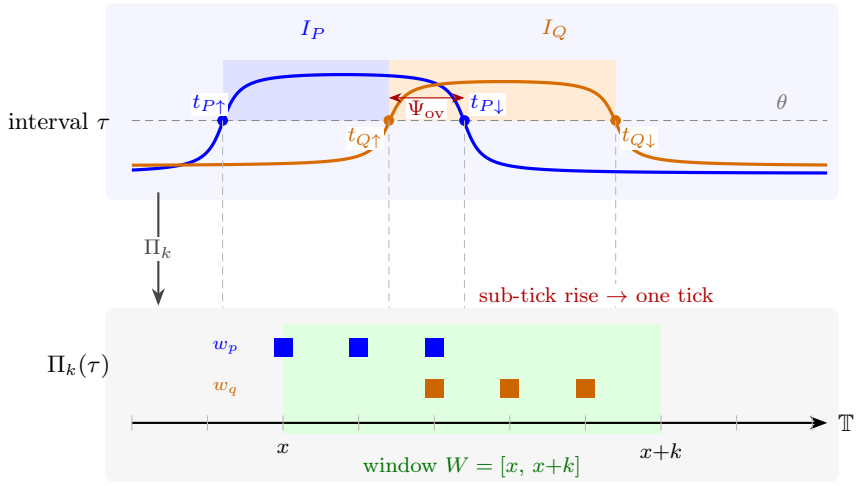


FIGURE 5. Forgetful projection Π_k as an information-losing sampling. Top: the interval trace τ rendered as two signals s_p, s_q crossing the single semantic threshold θ (dashed); their above-θ excursions are the antecedent interval I_P (blue) and consequent interval I_Q (orange), which overlap on $[t_{Q\uparrow}, t_{P\downarrow}]$, so Ψ_{ov} (hence $\Psi_{\text{lag},\Delta}$) holds. Bottom: Π_k records only the Boolean levels w_p, w_q at the integer ticks of the window $W = [x,\ x{+}k]$ (green), collapsing the sub-tick Q-rise at $t_{Q\uparrow}$ onto a single tick (grey/red projection lines). The point trace still satisfies Φ_k provided $k \geq d_P + \Delta + \varepsilon$ (Proposition 8.8); the converse does not hold, since distinct interval traces can share one projection.

Edge-pattern classification examples (Example C.7), the edge-ordering satisfaction table (Table 13), and the interval-to-point bridge table (Table 14) are collected in Appendix C.

Lemma 8.9 (Feature Collapse under Π_k)**.** *Let τ_1, τ_2 be interval traces with identical opener tick $t_{P\uparrow}$ and window bound k, but distinct edge orderings (e.g., overlap vs. lag-after-close). If $\Pi_k(\tau_1) = \Pi_k(\tau_2)$ at every tick in $W = [x,\ x + k]$, then τ_1 and τ_2 yield the same feature tuple (A, B, C) and hence the same pattern class under $\mathcal{L}_1$–$\mathcal{L}_6$.*

Proof. The features $A := \exists\, t \in W : w_q(t) = 1$, $B := \exists\, t \in W : w_q(t) = 0$, and $C := \neg w_p(x{+}k)$ depend only on the tick-level Boolean values $(w_p(t), w_q(t))_{t \in W}$. If $\Pi_k(\tau_1) = \Pi_k(\tau_2)$, these values are identical by definition, so (A, B, C) and the resulting pattern class agree. The interval predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ depend on edge orderings that Π_k discards, hence they may differ between τ_1 and τ_2. □

We use this lemma only to justify that equal projections cannot be told apart by any (A, B, C)-reading, underwriting the no-converse asymmetry of Remark 10.8.

A step-by-step calibration-to-interval trace demonstrating the $(1, 0, 0) \to (1, 0, 1)$ diagnostic upgrade under kernel tuning is developed in the supplementary material and executable in the reproducibility notebooks (Appendix A) (Appendix C).

The kernel tuning vector $\boldsymbol{\theta}$ (Definition 8.1) operates at the point-event level, shaping pulse widths and thresholds but not edge ordering; the model assumes a single decision stream with same-bin collisions resolved by the priority-max rule (Definition 3.12). The edge-detection front-end extension, the single-channel realisation, and the multi-channel analogue are detailed in Appendix C.

8.3. **Calibration Strategies for Interval Consistency.** The observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$ (Definition 8.1) can be tuned to shift the monitor between two operational regimes, depending on how reliably falling edges are detected.

Definition 8.10 (Observational Regime)**.** A monitor operates in the *point-like regime* when falling edges are unreliable (high β_p, β_q near transitions): the active interval I_P is effectively unknown and the monitor relies solely on the window assumption Φ_k, with only the opener tick x and the bound k available. It operates in the *interval-like regime* when both falling edges are reliably detected — $\Pr(\text{detect } P{\downarrow}), \Pr(\text{detect } Q{\downarrow}) \geq 1 - \varepsilon$ for a prescribed $\varepsilon > 0$ — so that the interval endpoints $t_{P\downarrow}, t_{Q\downarrow}$ resolve and the richer predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ (Definition 8.4) become directly evaluable.

Four monitor-internal calibration strategies migrate the monitor from the point-like to the interval-like regime — hysteresis (a Schmitt-trigger dead-band in $\mathcal{F}_{\theta_1}$), threshold split $(\theta_{\text{up}}, \theta_{\text{down}})$, increased sampling rate Δt, and window adjustment to the bound $k \geq d_P + \Delta + \varepsilon$ of Proposition 8.8 — each catalogued in the companion engineering paper [BBI26], which also develops the detection-error model of the threshold decision stage; the crossing/touch duality underlying the hysteresis and resolution knobs is treated in a related paper [Bra26b]. Tuning $\boldsymbol{\theta}$ thus determines *which semantics the monitor can sustain* — Φ_k alone in the point-like regime, Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ in the interval-like regime — without altering the underlying eight-class quotient (Remark 8.2).

When an observed trace $\tau_{\text{obs}} \notin \text{Conf}_{\text{pt}}(k)$ despite a genuine causal link, the failure is attributable to a measurement-layer cause — a false negative on Q, a spurious $P{\downarrow}$ setting $C{=}1$, a phase misalignment placing Q outside W, or a same-bin masking ambiguity (Remark C.3) — each repaired by a specific kernel stage (Definition 8.1), with the mechanism-by-mechanism development deferred to the companion paper [BBI26]. In every case tuning changes only the observed trace $\tau_{\text{obs}} = \mathcal{K}_{\boldsymbol{\theta}}(\tau_{\text{meas}})$, never the specification Φ_k or the combinatorial structure (8 window classes, 50000-entry eight-tick catalogue); the recovery goal is

$$\mathcal{K}_{\boldsymbol{\theta}}(\tau_{\text{meas}}) \notin \text{Conf} \ \wedge \ \mathcal{K}_{\boldsymbol{\theta}'}(\tau_{\text{meas}}) \in \text{Conf}, \tag{8.15}$$

where Conf is the regime-appropriate confirmation set (Definition 8.7) and τ_{meas} the fixed measurement-layer input. A worked recovery — lowering the consequent threshold θ_q (tuning T_1) so that a sub-threshold peak becomes an in-window validation, taking the silent window $(0,0,0)$ to $(1,0,0)$ — is given in the companion paper [BBI26].

8.4. **Algorithmic Tuning and its Justification.** The failure taxonomy above identifies *what* fails. *How* each kernel stage repairs it — the two archetypal repair algorithms (debounce / minimum-pulse suppression via the pre-filter $\mathcal{F}_{\theta_1}$, and oversampling / time-over-threshold sub-bin edge recovery), the nine elementary tuning operations they compose, and the transformation calculus — which observation class is reachable from which, in at most three tuning moves, accounting for the band delimiters — are developed in the companion engineering paper [BBI26]; the strategies rest on classical instrument principles (Δt refinement

on Nyquist–Shannon sampling [Sha49], threshold and hysteresis tuning on signal-detection theory [GS66], band gating on windowed measurement). Each repair is an instance of the *crossing* modality whose duality with the *touch* (extremum) modality is developed in a related paper [Bra26b].

9. Realising the Reading: Selection and Self-Tuning in Brief

The kernel and its tuning theory established *what can be read* in principle. Realising that reading at runtime raises two engineering problems — selecting the least ambiguous monitor, and keeping it phase-aligned against drift — whose full treatment is the subject of a related engineering paper [BBI26]. This section records only the statements and guarantees on which the core argument depends.

(i) Monitor selection. The monitor design space factors into three binary axes: a *repeat policy* (collapse repeated transitions of a proposition, or record every sample), an *edge-memory / opener policy* (edge-triggered, or level-triggered with persistent open-window state), and a *half-tick probe* (absent, or one additional sample between adjacent integer ticks). Over the exhaustively enumerated k=3 catalogue, the type $w^\star = (\rho{=}0, \omega{=}0, \pi{=}1)$ — edge-triggered, repeat-collapsing, with the half-tick probe — simultaneously minimises class ambiguity ($\mathcal{A}(w^\star) = 1$) and opener sensitivity ($\mathcal{S}(w^\star) = 0$) while maximising timing robustness ($\mathcal{R}(w^\star) = \frac{1}{2}$). This is a *witnessed* selection theorem: optimality is established by exhaustive enumeration over the catalogue, not by asymptotic argument.

(ii) Self-tuning by timing recovery. At runtime an early–late timing-error detector — the discrete analogue of the early–late gate of symbol synchronisation — compares the confirmation mass falling before and after the opener tick and drives a PLL-style update of the kernel state $\boldsymbol{\theta}$ with Robbins–Monro step sizes ($\sum_n \gamma_n = \infty$, $\sum_n \gamma_n^2 < \infty$). *Local* convergence of the phase estimate to the true offset $\varphi^\star$ is PROVEN under the standard stochastic-approximation hypotheses together with the detector's declared mean-field sign condition; *global* convergence is CONJECTURED and WITNESSED only on a finite simulation prefix (Appendix B). As the phase estimate crosses the classifier's regime thresholds, the *same* observed signal migrates through the observation classes, $(0,\cdot,\cdot) \to (1,1,\cdot) \to (1,0,0)$ — a regime-migration behaviour WITNESSED in simulation.

(iii) Non-circularity. Tuning modifies *only* the kernel parameter vector $\boldsymbol{\theta}$, never the classification rules; the non-circularity guarantee of the reading layer (Remark 8.2) is therefore inherited unchanged.

The convergence result is recorded at its proven scope (local; global conjectured and witnessed on a finite simulation prefix) in Appendix B and established in full — with the selection criteria $(\mathcal{A}, \mathcal{S}, \mathcal{R})$, the timing-error detector family, the loop architecture, the small-signal stability analysis, and the simulation evidence with its regime-band trajectory — in the companion engineering paper [BBI26].

10. Dual Semantics: the Ideal–Observed Bridge

The machinery so far operates at three description levels—interval traces, point traces, and the tuning layer—without making the boundary between the *physical* (continuous-time) world and the *observational* (discrete-time) model explicit. This section opens the third question of the thesis by introducing that distinction formally and stating a unidirectional soundness result connecting the two worlds.

Definition 10.1 (Ideal Trace)**.** An *ideal trace* is a pair of continuous-time Boolean signals

$$\tau_{\text{ideal}} \;:=\; \big(\, p : \mathbb{R}_{\geq 0} \to \{0,1\},\;\; q : \mathbb{R}_{\geq 0} \to \{0,1\} \,\big), \tag{10.1}$$

with piecewise-constant right-continuous paths and finitely many transitions in every bounded interval. The rising and falling times $t_{P\uparrow}, t_{P\downarrow}, t_{Q\uparrow}, t_{Q\downarrow} \in \mathbb{R}_{\geq 0}$ are exact, and the active intervals $I_P := [t_{P\uparrow}, t_{P\downarrow})$, $I_Q := [t_{Q\uparrow}, t_{Q\downarrow})$ have real-valued endpoints.

The ideal predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ (Definition 8.4) are naturally evaluated on τ_{ideal}: they characterise *causal regularities at the continuous level*, independent of any sampling or detection artefact.

Definition 10.2 (Full Observational Projection)**.** The *observational projection* maps an ideal trace to a discrete observed trace via three successive operations:

$$\tau_{\text{obs}} \;=\; \Pi_{\Delta t,\boldsymbol{\theta},k}(\tau_{\text{ideal}}) \;:=\; \Pi_k \circ\, \mathcal{K}_{\boldsymbol{\theta}} \circ\, S_{\Delta t}\,(\tau_{\text{ideal}}), \tag{10.2}$$

where:

(1) $S_{\Delta t}$ *(sampling)* evaluates the continuous signals at multiples of Δt, producing a discrete-time measurement trace τ_{meas} over the alphabet Σ;
(2) $\mathcal{K}_{\boldsymbol{\theta}}$ *(observation kernel, Definition 8.1)* applies the decision stages $\mathcal{F}_{\theta_1} \circ \mathcal{A}_{\theta_2} \circ \mathcal{D}_{\theta_3} \circ \mathcal{M}_{\theta_4}$;
(3) Π_k *(forgetful projection, Definition 8.5)* restricts the resulting trace to the window $W = [x,\; x{+}k]$ and discards sub-tick ordering and interval durations.

The projection is information-lossy by construction: $S_{\Delta t}$ forgets sub-bin ordering (Section 3.4), $\mathcal{K}_{\boldsymbol{\theta}}$ may mask edges (Remark C.3), and Π_k discards interval structure (Remark 8.6). The loss of $S_{\Delta t}$ is characterised exactly by the continuous-to-discrete sampling correspondence of a related paper [Bra26b]: when adequate-sampling conditions fail the omitted set — sub-tick event pairs, off-grid extrema, on-grid plateaus — is precisely the excluded stratum on which that correspondence does not hold.

All LTL formulas Φ_k are evaluated over τ_{obs}; the interval predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ are properties of τ_{ideal}.

Definition 10.3 (Evaluation Agent and Simulation Layer)**.** An *evaluation agent* $\mathcal{A}$ is a deterministic Boolean function $\mathcal{A}(\tau) := [\tau \models \Phi_k] \in \{0,1\}$ operating exclusively on τ_{obs}. The observation pipeline $\Pi_{\Delta t,\boldsymbol{\theta},k}$ (Definition 10.2) acts as a *simulation layer* between physical world and agent:

$$\underbrace{\tau_{\text{ideal}}}_{\text{physical}} \xrightarrow{S_{\Delta t}} \underbrace{\tau_{\text{meas}}}_{\text{measurement}} \xrightarrow{\mathcal{K}_{\boldsymbol{\theta}}} \tau_{\text{filt}} \xrightarrow{\Pi_k} \underbrace{\tau_{\text{obs}}}_{\text{presented to } \mathcal{A}} .$$

The pipeline of Definition 10.3 is a *sound observational simulator* (in the specification-refinement sense — no computational-indistinguishability assumption is imported) for a causal theory T and bound k when $\tau_{\text{ideal}} \models_T \Psi_{\text{lag},\Delta}$ implies $\Pi_{\Delta t,\boldsymbol{\theta},k}(\tau_{\text{ideal}}) \models \Phi_k$ under conditions (i)–(iv) of Proposition 10.7. Reducing its detection-error rates migrates the observed

class as in the tuning recovery (Equation (8.15)), adjusting the simulator's fidelity and not the truth value of Φ_k — the same non-circularity guarantee as Remark 8.2.

Definition 10.4 (Edge-Preserving Kernel)**.** Let $e \in \{P\uparrow, P\downarrow, Q\uparrow, Q\downarrow\}$ be an ideal edge at real time t_e, and let $\hat{t}_e$ be the tick at which $\mathcal{K}_{\boldsymbol{\theta}} \circ S_{\Delta t}$ reports the corresponding token transition. The observation kernel $\mathcal{K}_{\boldsymbol{\theta}}$ is ε_t*-edge-preserving* under a given regime if, for every edge e:

(a) *(Detection)* the edge is detected: $\hat{t}_e$ exists; and
(b) *(Timing accuracy)* the quantised report satisfies $|\hat{t}_e - \lfloor t_e/\Delta t \rfloor| \le \varepsilon_t$, where $\lfloor t_e/\Delta t \rfloor$ is the ideal sampling bin containing t_e.

Remark 10.5 (Probabilistic variant)*.* In noisy regimes, a $(\varepsilon_t, \varepsilon)$*-edge-preserving* kernel satisfies conditions (a) and (b) only with probability $\ge 1 - \varepsilon$ per edge; the deterministic definition (Definition 10.4) is the case $\varepsilon = 0$. All subsequent results use the deterministic variant; the probabilistic extension makes guarantees hold "with probability $\ge (1-\varepsilon)^4$" over the four edges.

Lemma 10.6 (Bin-Order Resolution)**.** *If $\mathcal{K}_{\boldsymbol{\theta}}$ is ε_t-edge-preserving with $\varepsilon_t < \Delta t/2$ (in units of bins, i.e., $\varepsilon_t < 1/2$), then for any two ideal edges e_1, e_2 with $|t_{e_1} - t_{e_2}| \ge \Delta t$, the reported ticks satisfy $\hat{t}_{e_1} \ne \hat{t}_{e_2}$. In particular, same-bin ambiguity (Section 3.4) is resolved whenever the edges are separated by at least one sampling period.*

Proof. Let $n_i = \lfloor t_{e_i}/\Delta t \rfloor$ for $i = 1, 2$. Since $|t_{e_1} - t_{e_2}| \ge \Delta t$, we have $|n_1 - n_2| \ge 1$. Edge-preservation gives $|\hat{t}_{e_i} - n_i| \le \varepsilon_t < 1/2$, so $|\hat{t}_{e_1} - \hat{t}_{e_2}| \ge |n_1 - n_2| - 2\varepsilon_t > 0$. Hence $\hat{t}_{e_1} \ne \hat{t}_{e_2}$. □

Proposition 10.7 (Observational Soundness)**.** *Let τ_{ideal} be an ideal trace. If:*

(i) $\tau_{\text{ideal}} \models \Psi_{\text{lag},\Delta}$;
(ii) *the sampling period satisfies $\Delta t \le \delta/2$, where $\delta := \min\big(|I_P|,\ t_{Q\uparrow} - t_{P\uparrow}\big)$ is the shortest relevant time scale*;
(iii) *the window bound satisfies $k \ge d_P + \Delta + \varepsilon$ (Equation (8.14)), where $d_P = |I_P|/\Delta t$ and $\varepsilon \ge 1$ absorbs quantisation jitter*;
(iv) *the detection kernel $\mathcal{K}_{\boldsymbol{\theta}}$ is ε_t-edge-preserving (Definition 10.4) with $\varepsilon_t < 1/2$;*

then

$$\Pi_{\Delta t, \boldsymbol{\theta}, k}(\tau_{\text{ideal}}) \ \models\ \Phi_k. \tag{10.3}$$

The four conditions are jointly sufficient *and not claimed minimal: the factor $1/2$ in (ii) and the margin $\varepsilon \ge 1$ in (iii) carry slack, and none of (i)–(iv) is individually necessary.*

Proof sketch. Condition (ii) ensures that $S_{\Delta t}$ places $t_{P\uparrow}$ and $t_{Q\uparrow}$ in distinct sampling bins (Section 3.4), so edge ordering is preserved modulo $\equiv_{\Delta t}$ (Section 3.4). Condition (iv) and Lemma 10.6 guarantee that $\mathcal{K}_{\boldsymbol{\theta}}$ reports these edges in distinct ticks with correct tokens in Σ, since the timing error $\varepsilon_t < 1/2$ prevents bin aliasing. Under condition (i), the consequent rises no later than $t_{P\downarrow} + \Delta$; condition (iii) then places this event within the observation window W. Hence $\exists t \in W : w_q(t) = 1$ in the projected trace, and Φ_k is satisfied. The argument composes three stages: sampling quantisation ($S_{\Delta t}$, condition ii) → kernel fidelity ($\mathcal{K}_{\boldsymbol{\theta}}$, condition iv and Definition 10.4) → window coverage (Π_k, condition iii), and is a factored restatement of Proposition 8.8. □

Interpretation. Proposition 10.7 is a one-way correctness guarantee: if the physical world satisfies the causal regularity *and* the pipeline meets conditions (i)–(iv), the discrete evaluation is sound.

Remark 10.8 (Converse does not hold). The converse of Proposition 10.7 is false in general:

$$\tau_{\text{obs}} \models \Phi_k \implies \tau_{\text{ideal}} \models \Psi_{\text{lag},\Delta}. \tag{10.4}$$

A discrete trace may satisfy Φ_k because a spurious edge (false positive) or a same-bin coincidence (Section 3.4) produced an apparent Q within the window, without any genuine causal regularity at the ideal level. This one-directionality is consistent with Remark C.5 and reflects the fundamental epistemic asymmetry: discrete verification can *confirm* a causal hypothesis under regime assumptions, but cannot *establish* causality from observational data alone.

The bridge is deliberately one-way: no finite observation forces the ideal-side conclusion. A licensed converse requires two things: the physical premises that anchor the timing model to the observed process — established with the causal-coherence conditions of the causal-validity section (Section 4.4) — and the declared observation theories that grade which window classes may be attributed *True*, developed next (Section 11).

11. The Theory Ladder of Verdict Justification

Out-of-band (Post) axis (evidence beyond the window). This section develops the departure in which a consequent onset $Q^{\uparrow}$ lands in the *Post* band: each rung *accounts for* such late evidence under a declared premise — attribution, not manufacture.

The window class records *what was observed*; whether that observation licenses evaluating the implication to *True* is a separate — and theory-laden, in Hanson's sense [Han58] — question. The same observed trace supports different verdict attributions under different declared assumptions about how emissions are transported to the monitor. We organise these assumptions as a ladder of *observation theories*, each rung licensing the True-attribution of a larger set of window classes (Figure 6) — resolving the third act of the paradox of observed implication (Section 1.3): observers who disagree are holding different rungs, not contradicting each other. The ladder is thus an epistemic stratification, in the spirit of Halpern and Moses's hierarchy of states of knowledge [HM90]: what a monitor may soundly attribute depends on its declared knowledge of the transport, exactly as what a process may soundly conclude in a distributed system depends on its knowledge of the communication environment [Lam78]. Cast in the vocabulary of distributed computing, an observation theory $\mathcal{O}$ is effectively a declared *consistency model* for the emission-to-observation channel, and the rung lattice $\mathcal{O}_{\text{sync}} \subseteq \{\mathcal{O}_{\text{rel}}, \mathcal{O}_{\text{sig}}\} \subseteq \mathcal{O}_{\text{open}}$ (Remark 11.5) is a hierarchy of progressively stronger channel guarantees. Structurally, the ladder is the OUT-OF-BAND-*Post* departure from the reference class of Section 4.1: a rung licenses a consequent onset $Q^{\uparrow}$ in the *Post* band, beyond band x and the window (HEURISTIC, attribution not manufacture).

Definition 11.1 (Observation theory)**.** An *observation theory* $\mathcal{O}$ is a declared, auditable set of physical premises about the emission-to-observation transport (clock relation, channel geometry, kinematics of emitters and monitor). Semantically, $\mathcal{O}$ determines a set $\text{Trans}(\mathcal{O})$ of *admissible transport maps*: functions ρ taking each emitter-side trace to the observed trace it produces under the premises of $\mathcal{O}$. A window class $\gamma = (A, B, C) \in \{0, 1\}^3$ is *True-attributable under* $\mathcal{O}$ if (a) some pair (τ_e, ρ) with $\rho \in \text{Trans}(\mathcal{O})$ realises class γ, i.e. $\rho(\tau_e)$ is of class γ (the quantified set is non-empty), and (b) for every emitter-side trace τ_e and every $\rho \in \text{Trans}(\mathcal{O})$ such that $\rho(\tau_e)$ is of class γ, the trace τ_e satisfies Φ_k — that is, if class γ is realisable under $\mathcal{O}$ and every $\mathcal{O}$-consistent explanation of the observation is an

emitter-side satisfaction. The non-emptiness clause (a) rules out *vacuous* True-attribution of a class that no $\mathcal{O}$-consistent transport ever produces. Entailment in this section always means this universal quantification over Trans($\mathcal{O}$).

Rung 1: synchrony and temporal linearity ($\mathcal{O}_{\text{sync}}$). Under a shared clock, linear time, and bounded classical delays — the premise set formalised by the parameters (L, S, D, J) and, in sharpened form, by the causal-coherence conditions CC1–CC4 (window plausibility, phase alignment, threshold consistency, and tick–time consistency) of Section 4.4 — the A=1 classes are True-attributable, each by its own argument: $(1, 0, 0)$ directly (in-window confirmation, stable antecedent); $(1, 1, 0)$ by glitch semantics (the validation stands, the subsequent invalidation is attributed to noise and suppressed by the debounce arguments of Section 8.4); $(1, 0, 1)$ and $(1, 1, 1)$ by retraction semantics (the antecedent's retraction *after* the consequent's confirmation does not undo a fulfilled obligation, Section 7). The A=0 classes are not True-attributable on this rung: their silence or denial is read at face value.

Rung 2: relativistic transport ($\mathcal{O}_{\text{rel}}$). Relativistic premises license True-attributions that $\mathcal{O}_{\text{sync}}$ forbids: phenomena such as time dilation make observations arrive at the monitor in a pattern *apparently* incoherent with satisfaction, while the emitter-side implication in fact holds.

Proposition 11.2 (Relativistic recovery of late evidence)**.** *Declare the premises that emitters and monitor are inertial with relative Lorentz factor bounded by a known $\gamma_{\max}$, that the relative motion is transverse (constant emitter–monitor distance), so that propagation terms remain those already modelled by (L, S, D, J) and the residual kinematic effect is pure time dilation, and that the bound k of Φ_k is calibrated in the emitters' proper time. Then an emitter-side confirmation within k proper ticks reaches the monitor within $\gamma_{\max}k$ monitor ticks (beyond the classical terms). Consequently, a trace classified $(0, 0, 0)$ on the window $[x, x+k]$ whose* `postPeri` *band (a post-window observation band $(x+k,\ x+\gamma_{\max}k]$) contains a Q-validation at some $t_q \le x+\gamma_{\max}k$ is* recoverable *under $\mathcal{O}_{\text{rel}}$: its late validation is consistent with an emitter-side confirmation within k proper ticks — although it remains $(0, 0, 0)$ (Quaestio) under $\mathcal{O}_{\text{sync}}$. The strength of this recovery depends on how γ is declared, as the epistemic note below makes precise.*

Proof. By time dilation, an interval of k ticks of emitter proper time spans at most $\gamma_{\max}k$ ticks of monitor time. An emitter-frame confirmation at proper delay $\le k$ after the opener therefore arrives at monitor delay $\le \gamma_{\max}k$; hence an observed validation at $t_q \in (x+k,\ x + \gamma_{\max}k]$ *can* be the dilated image of an in-bound emitter-frame confirmation, which is what Φ_k asserts on the emitter side. When the Lorentz factor is *pinned* to the declared $\gamma_{\max}$ (so Trans($\mathcal{O}_{\text{rel}}$) is a singleton), this dilated image is the *unique* $\mathcal{O}_{\text{rel}}$-consistent explanation, and the True-attribution of Definition 11.1 follows. □

Remark 11.3 (Epistemic status of the relativistic rung). The *universal* True-attributability of Definition 11.1 — for *every* $\rho \in$ Trans($\mathcal{O}_{\text{rel}}$) — is established (PROVEN) only under the *pinned* premise $\gamma = \gamma_{\max}$, i.e. when Trans($\mathcal{O}_{\text{rel}}$) is the singleton $\{\rho_{\gamma_{\max}}\}$. If the auditable premise declares only a *bound*, $\gamma \le \gamma_{\max}$, then Trans($\mathcal{O}_{\text{rel}}$) ranges over $\gamma \in [1, \gamma_{\max}]$, and the entailment is CONJECTURED, not proven: the $\gamma = 1$ transport reads the same late validation $t_q \in (x+k,\ x + \gamma_{\max}k]$ as an emitter-side proper delay $t_q > k$, which violates Φ_k, so the universal quantification of Definition 11.1 fails. Under the bound the rung therefore

licenses a *consistency* (plausibility) recovery — the observation is compatible with emitter-side satisfaction — rather than an entailment; the PROVEN form requires the singleton pinning.

Two remarks keep the physics honest. First, for *causally linked* (timelike-separated) events, relativity never inverts temporal order, so the relativistic rung *rescales intervals* [Ein05] (unlocking late-evidence readings of $(0,0,0)$) but does not license precedence anomalies, which still require classical channel asymmetry (expressible through L, S, D, J). Second, the premises are *quantitative and auditable* ($\gamma_{\max}$ is declared, not fitted), so the attribution is falsifiable: evidence later than $x + \gamma_{\max} k$ is not recovered. Stripped of the physics, the rung is a declared rescaling $k \to \gamma_{\max} k$ of the acceptance bound; its role in the ladder is the demonstration that an auditable premise package licenses a precisely bounded recovery — the template that every further rung must instantiate.

Rung $2'$: signal regularity ($\mathcal{O}_{\text{sig}}$). Premises need not concern transport: they may concern the *shape of the signal itself*. The rung below discharges an *unbounded* obligation from declared structural constants of s_q alone — no parameter search, no signal modification, in keeping with the non-circularity guarantee.

Proposition 11.4 (Signal-regularity discharge of the unbounded obligation)**.** *Declare the premises that, after each opener at tick x, the consequent signal increases while below threshold, with per-tick increment bounded below by a known $\delta_{\min} > 0$, and that its opening value lies in $[b_0, \theta_q)$ for a known floor b_0 (declared signal constants). Let*

$$k^* \;=\; \lceil (\theta_q - b_0)/\delta_{\min} \rceil. \tag{11.1}$$

Then, under $\mathcal{O}_{\text{sig}}(\delta_{\min}, b_0)$:

(i) *the unbounded obligation $\mathbf{G}(w_p \to \mathbf{F}\, w_q)$ collapses to the bounded Φ_{k^*}: the up-crossing of θ_q occurs within k^* ticks of every opener, so the liveness pattern $\mathcal{L}_3$ becomes finitely decidable — although it remains liveness (not finitely monitorable) under $\mathcal{O}_{\text{sync}}$;*

(ii) *a silent window of bound $k < k^*$ (class $(0,0,0)$) is True-attributable, the pending crossing being guaranteed by the declared constants.*

The attribution is falsifiable: if no validation is observed by $x + k^*$, some declared constant has failed at a finitely identifiable tick.

Proof. While $s_q < \theta_q$, the premises give $s_q(x+j) \geq b_0 + j\,\delta_{\min}$ by induction on j. For $j \geq k^*$ we have $b_0 + j\,\delta_{\min} \geq \theta_q$, so the first tick $t^* \leq x + k^*$ with $s_q(t^*) \geq \theta_q$ exists and is an up-crossing (the signal was below threshold before it); the crossing monitor latches $Q^\top$ there, giving (i), and a window with $k < k^*$ that closes before t^* is exactly case (ii). If no validation is observed by $x + k^*$, then at some tick $x+j$ either $s_q(x+j) < b_0 + j\,\delta_{\min}$ (an increment below $\delta_{\min}$, identifiable as the first such j) or the opening floor failed at $j = 0$; either way a declared premise is refuted by a finite observation. □

Thus $\mathcal{O}_{\text{sig}}$ instantiates the same template as the relativistic rung — the auditable package is now two signal constants, the bounded recovery the closed-form k^* — through signal shape rather than transport.

Rung 3: open tier ($\mathcal{O}_{\text{open}}$). Further arguments from modern physics may license further classes; we flag this rung as PLAUSIBLE and leave it as outlook. Candidate examples: measurement-contextual readings of a Q-denial (a $Q^{\perp}$ token as a context-dependent outcome rather than a refutation, potentially reopening $(0,1,0)$-type classes) and delayed-choice-style arguments for boundary events. No formal claim is made here; the point is structural — the ladder is open-ended upward, and each proposed rung must arrive with declared, auditable premises before it may license anything.

Remark 11.5 (The ladder is monotone, and each rung accounts for one previously unmodelled effect). The True-attributable sets grow monotonically with the premises: $\text{True}(\mathcal{O}) \subseteq \text{True}(\mathcal{O}')$ whenever $\mathcal{O} \subseteq \mathcal{O}'$. The declared theories form a *lattice* rather than a single chain: $\mathcal{O}_{\text{rel}}$ and $\mathcal{O}_{\text{sig}}$ extend $\mathcal{O}_{\text{sync}}$ in incomparable directions (transport versus signal shape), both below $\mathcal{O}_{\text{open}}$, and may be declared jointly. Each rung acts by *naming a previously unmodelled timing effect and accounting for it under declared premises* — the synchrony gap, the dilated proper-time interval, the residual climb ($\leq k^*$), the measurement context. This is attribution, not manufacture: the premises are hypotheses stated in advance, the observed trace is never altered, and the non-circularity guarantee of Remark 8.2 is untouched (no tuning operation is involved at all). In the terminology of Remark 10.8, the ladder supplies *graded converse premises*: each rung is a declared package under which part of the missing converse direction — from observed trace back to emitter-side satisfaction — is restored.

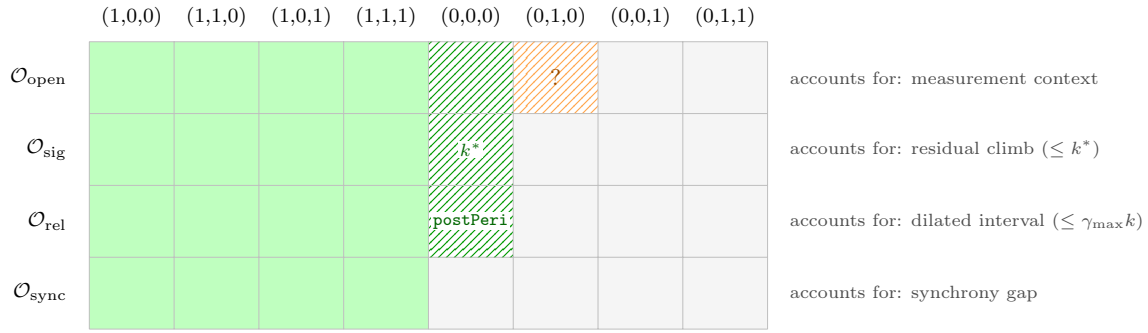


FIGURE 6. The theory ladder as a verdict-attribution map. Solid green: True-attributable on the rung; hatched green: attributable under a declared premise package — relativistic transport (a `postPeri` validation within $\gamma_{\max} k$, Proposition 11.2) or signal regularity (a pending crossing within the closed-form bound k^*, Proposition 11.4); hatched orange: PLAUSIBLE candidate on the open rung (no formal claim); gray: not attributable. Attribution grows monotonically with the premises; $\mathcal{O}_{\text{rel}}$ and $\mathcal{O}_{\text{sig}}$ extend $\mathcal{O}_{\text{sync}}$ along independent axes (Remark 11.5); each rung accounts for the timing effect named on the right.

The calibration results fix *when* temporal implication is valid and *where* each pattern sits in the safety–liveness hierarchy; the theory ladder fixes *which observations may be attributed True under which declared physics*. The three questions of the thesis are now answered in order: window classes fix *what is observed*; the kernel and its selection and self-tuning machinery fix *what can be read*; and the ladder — resting on the causal-coherence premises

of Section 4.4 — fixes *what may be attributed True*. What remains is to watch the three answers come apart on concrete traces.

12. Validation on Concrete Traces

This section validates the classification framework on three hand-constructed scenarios that exercise distinct regions of the safety–liveness–cosafety landscape (Figure 4), showing the three questions of the thesis coming apart on concrete traces: identical ground-truth processes, observed under different settings, receive different classifications, readings, and attributions. Each scenario specifies observation parameters, a finite trace prefix, the resulting eight-tick signature, the window class, and the property fragment from Table 8. The validation is *semantic*: no external datasets, tooling, or performance measurements are involved.

12.1. **Scenario 1: Bounded Confirmation (Cosafety).**

Setup. Two processes $\mathcal{P}$ and $\mathcal{Q}$ communicate through a monitor $\mathcal{M}$ with observation parameters $L = 1$, $S = 0$, $D = 0$, $J = 0$, giving cumulative worst-case offset $\Delta = 1$. The bounded-eventuality window is $k = 3$ (satisfying $k \geq 2\Delta$).

Ground process. $\mathcal{P}$ emits e_P at tick $t_P = 3$. $\mathcal{Q}$ emits e_Q at tick $t_Q = 5$. Both emitters produce a single event.

Observed trace. With no jitter ($\epsilon_P = \epsilon_Q = 0$), the monitor samples w_p at $t_M^P = 3+1+0 = 4$ and w_q at $t_M^Q = 5 + 1 + 0 = 6$. The opener tick is $x = 4$ (bandx), and the window extends to tick $x + k = 7$. The resulting observed trace prefix, restricted to $\{w_p, w_q\}$, is:

tick	0	1	2	3	4	5	6	7	8	9…
w_p					$\top$					
w_q							$\top$			

Eight-tick projection. Relative to opener $x = 4$ (closing tick at $x{+}k = 7$):
- *pre* ($t < 3$): $\varnothing$;
- *x-1* (t=3): $\varnothing$;
- *x* (t=4): $P^\top$ (opener/trigger);
- *x+1* (t=5): $\varnothing$;
- *m2* (t=6 = x+2): $Q^\top$ observed;
- *cl* (t=7, closing): $\varnothing$;
- *cl+1* (t=8): $\varnothing$;
- *post* ($t > 8$): $\varnothing$.

The eight-tick signature is $\sigma = (\varnothing, \varnothing, P^\top, \varnothing, Q^\top, \varnothing, \varnothing, \varnothing)$.

Classification. Features: $A = 1$ (Q observed as true in window), $B = 0$ (no Q-false), $C = 0$ (P does not retract). Window class: $(1, 0, 0)$ (A=1, B=0, C=0). The consequent arrives in the window interior (`m2`), not at the first-response tick `x+1`, and the out-of-band ticks are silent (no late evidence in `post`).

Property fragment. The observed pair — w_p at tick 4, w_q at tick 6, with gap $t_M^Q - t_M^P = 2 \leq k = 3$ — witnesses the bounded-reachability event $\mathbf{F}(w_p \wedge \mathbf{F}_{\leq 3} w_q)$, the *cosafety* ($\mathsf{C} \cap \mathsf{L}$) companion of the implication: a single matched pair drives it to its definitive $\top$ at tick 6. The implication itself, $\mathcal{L}_2 = \mathbf{G}(w_p \to \mathbf{F}_{\leq 3} w_q)$, is its *safety* ($\mathsf{S} \cap \mathsf{K}$) reading (Table 8): finitely refutable but never finitely confirmable, so the finite $\top$ here confirms the reachability event, not the universally-quantified implication — whose refutation face is exercised in Scenario 2.

Interpretation. Under the given parameters, the matched pair $P \Rightarrow Q$ is confirmed within the bounded window despite a 2-tick observation gap. The $(1, 0, 0)$ classification captures the *clean confirmation* semantics: exactly one Q-true signal, no ambiguity, no retraction. A monitor for the cosafety reachability property would emit a positive verdict at tick 6.

12.2. **Scenarios 2 and 3: Refutation and the Unbounded Regime.** The same ground-truth processes, observed under two further parameter settings, exercise the remaining fragments. In Scenario 2 the consequent emitter is suppressed: the window stays silent on the Q channel, the trace classifies as $(0, 0, 0)$ (the canonical silent pattern), and the observed prefix is a *bad prefix* for Φ_3 — w_p holds at the opener but w_q at no window tick — so the monitor emits a negative verdict at the closing tick: the finite-time refutation that the safety reading $\mathcal{L}_2 \in \mathsf{S} \cap \mathsf{K}$ of Table 8 predicts. Whether the silence reflects a genuinely absent consequent or observation parameters that underestimate the true delay is exactly the $(0, 0, 0)$ ambiguity separated in Section 6. In Scenario 3 the channel drift is unbounded ($D_{\max} \to \infty$), so no finite window is admissible: the implication degrades to the liveness pattern $\mathcal{L}_3$, no finite-trace verdict is sound, and the monitor can only report an *inconclusive* ("?") verdict — the zero-sided $\mathsf{L} \cap \mathsf{K}$ ($NFR \cap NFS$) cell, distinct from the central Quaestio, and the operational face of Theorem 7.11(iii). The fully worked traces, eight-tick projections, and classifications for both scenarios are in the supplementary material.

12.3. **Validation Summary.** Table 10 collects the three scenarios.

TABLE 10. Operational validation scenarios. Each scenario exercises a distinct property fragment and demonstrates a non-trivial classification.

	Sc. 1	**Sc. 2**	**Sc. 3**
Δ	2	2	∞
k	5	5	—
Q arrives?	yes	no	yes (late)
Pattern	$(1, 0, 0)$	$(0, 0, 0)$	—
Fragment	cosafety (conf.)	$\mathcal{L}_2$ (safety)	$\mathcal{L}_3$ (liveness)
Verdict	$\top$ (tick 7)	$\bot$ (tick 8)	inconclusive
Monitor action	confirm	alarm	accumulate

The three scenarios jointly demonstrate *coverage* (inputs are classified across the safety, cosafety, and liveness fragments of the Manna–Pnueli hierarchy), *non-triviality* (the same ground-truth relation $P \Rightarrow Q$ receives different patterns and verdicts as Δ and k vary, not by a priori formula choice), *diagnostic separation* (the pattern identifies the structural cause

of the verdict, the calibration theorem Theorem 7.8 the parametric cause), and *correctness* (each classification is mechanically derivable from Table 4 and Table 8, the verdicts following from the semantics of Φ_k and Φ_∞). These are *semantic* validations on controlled inputs; they do not replace empirical evaluation on instrumented systems, identified as future work (Section 16).

12.4. **Case Study: Kinetic Assay Pipeline.** Beyond the synthetic scenarios, one realistic measurement chain shows the same machinery end-to-end. A fluorescence kinetic assay instantiates every abstract parameter on real instrument knobs: the reagent-injection event is P, the fluorescence threshold crossing is Q, and the measurement chain (mixing dead-time, optical integration, ADC sampling, timestamp jitter) realises (L, S, D, J). Table 11 gives the mapping. On the numeric configuration developed in the supplementary material, the assay initially operates in the miscalibration regime ($(0, 0, 0)$: the monitor samples too early and never sees the crossing), and the Stage-C loop recovers the $(1, 0, 0)$ regime within a handful of cycles by re-aligning the sampling phase — the same migration witnessed in a related paper's simulation evidence [BBI26], now driven by instrument physics.

TABLE 11. Parameter mapping: monitor model ↔ kinetic assay instrument.

Param.	**Instrument knob**	**Range**
Δt	ADC sample period	0.1–10 ms
θ_P, θ_Q	Comparator threshold	5–500 mV
$\tau_{\min}$	Debounce width	1–5 ticks
M^-, M^+	Pre/post-trigger margin	0–3 ticks
k	Record length	5–50 ticks
L	Propagation delay	1–3 ms
J	Jitter half-width	0–2 ticks

The mapping closes the loop of the thesis on hardware: the same trace, read through instrument physics, migrates across the same classes that the geometry of Part I defined and the tuning theory of Part II predicted.

13. DISCUSSION

13.1. **Epistemic Implications.** The paradox of observed implication (Section 1.3) reveals a structural tension between the classical, atemporal notion of logical implication and a physical observer operating under the (L, S, D, J) model: the monitor $\mathcal{M}$ sees only projections of P and Q through a noisy temporal channel. This has direct consequences for empirical methodology. **Observation is theory-laden** [Han58], and hypotheses meet experience only together with auxiliary assumptions [Duh54, Qui51]: the observation parameters implicitly define which violations of $P \Rightarrow Q$ are detectable, formalised on the verdict side as the theory ladder of Section 11. **Falsifiability is temporally bounded** (Sections 4.4.2 and 5): definitive refutation of Φ_k is possible only in the safety fragment; in the liveness regime no

finite observation suffices, and under adversarial delay a counterexample may be arbitrarily deferred, making the hypothesis empirically non-falsifiable within a given window though logically falsifiable in principle. **Race conditions in reasoning** may arise when multiple implications are evaluated concurrently, the interplay of their delay parameters making the evaluation order affect the diagnostic outcome (single-formula here; the multi-formula case is future work, Section 16).

On phantom falsifiers. Even when the classical implication is vacuously satisfied, temporal observation artefacts can manufacture an apparent violation: a worked adversarial-delay example (L=3, S=1, D=0, J=2 with P=$\bot$, Q=$\top$) in which a k=3 monitor emits a false refutation of a vacuously true implication — classified under $\mathcal{L}_6$ and corrected by the calibration $k \geq 2J$ — is recorded in the supplementary material. In Popper's terms, the experimenter's apparatus has introduced a spurious falsifier.

14. Limitations

The boundary of validity is stated where each assumption is introduced; we consolidate it here by reference, adding no new claim.

(1) **Single-formula scope.** All combinatorial results (eight window classes, quotient of the 50000-entry eight-tick catalogue) are specific to the single bounded-eventuality formula $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$; multi-formula and multi-property extensions are not treated (Section 1.7, item 5; Section 16).
(2) **Monodimensional observation baseline.** The token alphabet Σ and the single crossing threshold are a deliberate baseline; micro-band spawning and interval/phase monitors are deferred (Remark 3.14, Section 3.11).
(3) **One-way soundness bridge.** The observational-soundness result relates ideal causal regularity to satisfaction of Φ_k in one direction only; the converse fails (Proposition 10.7, Remark 10.8).
(4) **Deferred convergence.** The self-tuning loop has PROVEN local and CONJECTURED global convergence, witnessed only on a finite prefix; the full treatment is deferred to the engineering companion (Section 9, Appendix B, [BBI26]).
(5) **Assumption-conditional theory-ladder rungs.** Only the simplest instances of the transport and signal-regularity rungs are proven; a systematic finite-confirmation theory grading certificates from assumption-conditional to proof-carrying remains open (Section 11, Section 16).
(6) **No causal-inference, security, or signal-theory claim.** The paper makes none of these; the corresponding non-goals are stated explicitly (Section 1.7).

15. Conclusions

The result to carry away is a single one: *monitorability is not a property of a temporal formula but of the pair (formula, observation model)* (Theorem 7.11). A bounded implication that is a safety property under one observation model degrades to a non-finitely-monitorable liveness property under another, with no change to the formula — so a monitor deployed without a declared observation model has no well-defined soundness guarantee. Everything else in the paper builds toward, or lives under, that fact.

We have shown, more fully, that the paradox of observed implication (Section 1.3) dissolves: three questions about a monitored implication separate cleanly. *What is observed*:

the monitoring class is jointly determined by the formula and the observation model, and the observation space partitions geometrically into eight classes (the quotient of the 50000-entry eight-tick catalogue, machine-verified and script-reproducible). *What can be read*: tuning migrates the observed trace between confirmation sets without altering Φ_k, up to universal readability in at most three tuning moves (engineering companion [BBI26]), guarded against truth-manufacture by the non-circularity boundary $\mathcal{L}_1^{\text{tun}}$; the accompanying self-tuning loop (Section 9) has PROVEN local convergence and CONJECTURED global convergence, with the full treatment in [BBI26]. *What may be attributed True*: a one-way observational-soundness bridge (Proposition 10.7, converse false by Remark 10.8), graded by the theory ladder (Section 11) and grounded in the causal-coherence conditions CC1–CC4 and the pathologies F1–F5 that cover the full observation-class space (Section 4.4). Underneath the headline sits its semantic groundwork: the reference-class and causal-soundness apparatus over which the headline result is stated fixes, by the declared physics (Section 4), which observed-implication signatures may be read as causally valid — the six antecedent-validated signatures are the reference on which the forward reading is well-posed, every other class a causally-graded departure, with coliveness derived from the clopen window skeleton, not imposed.
More broadly, deploying a temporal-property monitor without characterising the observation model leaves the monitor's fragment assignment — and therefore its soundness guarantee — undefined. The framework developed here makes this dependency explicit and computable.

16. Future Work

Two substantive directions and one implementation task stand out.

(1) **Monitor parametrisation.** Treating the tuning knobs of the kernel as a single parameter space over which monitor selection is *effort-optimal* — which monitor family reads a prescribed set of signatures with the least tuning effort — would extend the Stage-B criterion beyond the eight types considered here, and give the universal-readability result of the engineering companion [BBI26] a quantitative cost side.

(2) **Graded finite confirmation of liveness obligations.** The theory ladder now carries two proven rung families — transport (Proposition 11.2) and signal regularity (Proposition 11.4) — but only their simplest instances. A systematic theory of finite-confirmation premises — intrinsic regularity of the verdict alphabet, extrinsic regularity of the generating process (periodicity, stationarity, ergodicity), and actively *constructed* regularity via parametric monotonicity of the monitor — with certificates graded from assumption-conditional to proof-carrying, would populate the lattice on the unbounded side of the $k = 2\Delta$ boundary. A measure-valued refinement — grading a verdict by the measure of admissible theories that license it, rather than by a single declared rung — is the natural quantitative limit of this direction.

(3) **Interval / phase monitor implementation.** The interval semantics and refinement bridge of Section 8.2 defines the formal predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$, the forgetful projection Π_k, and the refinement coherence bound (Proposition 8.8). A full implementation—including edge-aware observation kernels, interval-specific cost functions for the self-tuning loop, and enumeration of the resulting signature refinement—remains open.

Standard extensions — multi-valued verdicts [Háj98], non-stationary (L, S, D, J) parameters [Cri89], distributed and Byzantine observers [LSP82, FPS17], a dense-time MTL reformulation [Koy90], automated monitor synthesis [BLS11], and empirical validation on instrumented systems — are also open.

Acknowledgements

The authors wish to thank the anonymous reviewers for their constructive feedback.

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Appendix A. Reproducibility Package and Notebooks

The mechanical verification of the classification is packaged as three executable Python notebooks which, together with the extended text fragments, constitute the electronic supplementary material of this paper. Each notebook opens by listing the paper claims it substantiates and asserts every reproduced number and classification programmatically, halting on any drift:

(N1) *Eight-class partition* — the eight-tick model, the eight-class quotient of the 50000-entry catalogue at $k = 3$, and the nine verification checks summarised in Table 12;
(N2) *Interval semantics* — executable versions of the worked examples and round-trip identities of Appendix C;
(N3) *Schmitt gallery and reachability* — the full eight-class worked gallery, the tuning-operation catalogue, the reachability witness behind the universal-readability result [BBI26], and the touch-probe disambiguation of the silent window [BBI26].

A-1. **The Eight-Tick Model.** The tick structure itself is defined in Definition 3.11; this appendix records the exact signature encoding used by the verification suite.

Eight-tick signature. Each triggered trace is mapped to an *eight-tick signature* — one token of Σ_{obs} per tick:

$$\sigma = \big(\sigma^{\text{pre}},\ \sigma^{x-1},\ \sigma^{x},\ \sigma^{x+1},\ \sigma^{m2},\ \sigma^{x+k},\ \sigma^{x+k+1},\ \sigma^{\text{post}}\big), \quad \sigma^{\bullet} \in \Sigma_{\text{obs}}$$

Two traces with the same eight-tick signature are observationally equivalent; the eight window classes are exactly the equivalence classes induced by the presence triple $(A, B, C) = (\exists\, Q^{\top}, \exists\, Q^{\perp}, \exists\, P^{\perp})$ over the in-band window $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ that the crossing/touch classification (Definition 3.5) assigns in Table 4.

Remark A.1 (Hand reconstruction of the machine-checked counts). The two cardinalities asserted as script-reproducible — 390625 raw tick-tuples and 50000 admissible catalogue entries, quotienting to the 8 window classes — are each reconstructible by hand from the following spec, which is the exact domain enumerated by the eight-tick model of the engineering companion [BBI26] (scripts `eightick_derive.py`, `eightick_selection.py`, `eightick_pathology.py`); the eight-tick model is owned by that companion, which the present paper cites. Fix $k = 3$; the partition has the eight ticks $\texttt{pre} \cdot \texttt{x-1} \cdot \texttt{x} \cdot \texttt{x+1} \cdot \texttt{m2} \cdot \texttt{cl} \cdot \texttt{cl+1} \cdot \texttt{post}$ of Definition 3.11, each carrying one token of Σ_{obs} ($|\Sigma_{\text{obs}}| = 5$).

(i) *Raw enumeration (*390625*).* Each of the eight ticks independently carries one of the five tokens of Σ_{obs}, so the raw space has $5^8 = 390625$ tick-tuples.

(ii) *Opener-minimal admissibility (*50000*).* Two constraints define the admissible catalogue: (V1) the opener tick `x` carries $P^\top$, and (V2) no $P^\top$ occurs at `pre` or `x-1` (the opener is the *first* $P^\top$; a later re-trigger *is* admitted). This leaves the two pre-opener ticks 4 choices each ($\Sigma_{\text{obs}} \setminus \{P^\top\}$), pins `x`, and leaves the remaining five ticks free: $4 \cdot 4 \cdot 5^5 = 50000$ admissible entries.

(iii) *Quotient to the eight window classes.* The presence triple $(A, B, C) = (\exists\, Q^\top, \exists\, Q^\perp, \exists\, P^\perp)$ over the in-band window $\{\texttt{x+1}, \texttt{m2}, \texttt{cl}\}$ partitions the 50000 admissible entries into the eight classes of Table 4; ordered $(1,0,0), (1,1,0), (1,0,1), (1,1,1), (0,0,0), (0,1,0), (0,0,1), (0,1,1)$, the per-class counts are $(7600, 7200, 7200, 2400, 3200, 7600, 7600, 7200)$, summing to 50000. Class determinism (test T8, each admissible entry lands in a single window class) is machine-checked by `eightick_derive.py`.

Example traces. Consider $k = 3$ (opener at x, closing tick at $x{+}k$):

(1) A trace with $Q^\top$ in the window interior maps to
$$(\varnothing, \varnothing, P^\top, \varnothing, Q^\top, \varnothing, \varnothing, \varnothing) \;\to\; (1,0,0).$$

(2) A trace where Q becomes true only at the closing tick maps to
$$(\varnothing, \varnothing, P^\top, \varnothing, \varnothing, Q^\top, \varnothing, \varnothing) \;\to\; (1,0,0)$$
(delayed satisfaction).

(3) A trace where Q is never observed in-band maps to the silent cell $(0,0,0)$ (potential miscalibration).

TABLE 12. Python verification results ($k = 3$, eight-tick model).

Test	**Description**	**Result**
T1	Class enumeration ($\lvert\text{classes}\rvert = 8$)	PASS
T2	Min. Hamming distance ≥ 1	PASS
T3	Class coverage (8/8, k=3)	PASS
T4	Vacuity / miscalib. separation	PASS
T5	Eight-tick catalogue enumeration (50000)	PASS
T6	Per-class distribution ($\Sigma = 50000$)	PASS
T7	Inclusion & incompatibility	PASS
T8	Tick $\to$ class determinism (0 mismatches)	PASS
T9	LTL safety + bounded eventuality	PASS

A-2. **Verification Results.** Table 12 summarises the outcomes: all nine tests pass, confirming the completeness and determinism claims of the classification and the observational/-logical vacuity separation. The eight-tick cardinalities ($5^8 = 390625$ raw, 50000 admissible, the per-class distribution summing to 50000) are machine-checked by the companion scripts `eightick_derive.py`, `eightick_selection.py`, and `eightick_pathology.py`; a re-run of this paper's own notebook (N1) against the eight-tick catalogue is pending.

Appendix B. Convergence of the Self-Tuning Loop (Deferred)

The self-tuning loop summarised in Section 9 maintains a phase estimate $\hat{\varphi}_n$ of the true observation offset $\varphi^\star$, updated after each monitoring cycle by $\hat{\varphi}_{n+1} = \hat{\varphi}_n - \mu_n e_n$, where e_n is the early–late timing error computed from the confirmation counts of the companion engineering paper's timing-error detector [BBI26] and μ_n is the step size. The full-loop construction and the loop's convergence analysis are developed there, at its proven scope, and are not re-derived here; below we prove the *local* result on which the core argument relies and record the honest status of the *global* claim.

Local almost-sure convergence of the phase estimate to $\varphi^\star$ is PROVEN under the standard Robbins–Monro stochastic-approximation hypotheses ($\sum_n \mu_n = \infty$, $\sum_n \mu_n^2 < \infty$, unbiased and bounded-variance error signal) *together with* a mean-field sign / regression (basin) condition on the band-counting timing-error detector: from any initialisation inside the basin the estimate converges almost surely ([BBI26], after [RM51]). *Global* almost-sure convergence from an arbitrary initial offset — where the fixed-span eight-tick counts saturate and the mean-field gradient collapses — is CONJECTURED, and WITNESSED only on a finite simulation prefix; it is *not* established analytically. The regime-migration behaviour this induces (miscalibration, the four A=0 cells $\to$ bounded-temporal, the contested A=1, B=1 cells $\to$ static-like $(1, 0, 0)$) is likewise a WITNESSED simulation observation, developed in full in the companion paper [BBI26].

Appendix C. Interval Semantics: Extended Remarks and Examples

This appendix collects extended remarks and worked examples from Section 8.2 that develop the interval semantics framework in full detail. All labels are preserved; body-text forward references resolve here.

Definition C.1 (Concretization / Lifting)**.** The *concretization* (or *lifting*) of a point trace is the preimage of Π_k:

$$\gamma_k(\tau_{\mathrm{pt}}) := \{\, \tau_{\mathrm{iv}} \in \mathrm{IntervalTrace} : \Pi_k(\tau_{\mathrm{iv}}) = \tau_{\mathrm{pt}} \,\}. \tag{C.1}$$

An *interval trace* $\tau_{\mathrm{iv}} = (t_{P\uparrow}, t_{P\downarrow}, t_{Q\uparrow}, t^?_{Q\downarrow})$ comprises the four edge timestamps, where $t^?_{Q\downarrow}$ may be $\perp$ (unobserved / open) if the consequent never falls within the recording horizon. The set $\gamma_k(\tau_{\mathrm{pt}})$ contains every assignment of these timestamps that is consistent with the tick-level Boolean values $(w_p(t), w_q(t))_{t\in W}$ prescribed by τ_{pt}.

A *canonical representative* $\hat{\gamma}_k(\tau_{\mathrm{pt}})$ is obtained by the **latest-fall** rule: among all interval traces in $\gamma_k(\tau_{\mathrm{pt}})$, choose the one with the *latest* $t_{P\downarrow}$ and the *latest* $t_{Q\downarrow}$ consistent with the observed Boolean samples. Formally:

(1) Set $t_{P\uparrow} := x$ (the opener tick).
(2) Set $t_{P\downarrow} := \max\{\, t \in W : w_p(t) = 1\} + 1$ (the tick immediately after the last observed $w_p = 1$). If $w_p(x + k) = 1$ (i.e., $C = 0$), set $t_{P\downarrow} := \perp$ (open).
(3) Set $t_{Q\uparrow} := \min\{\, t \in W : w_q(t) = 1\}$ if $A = 1$; otherwise $t_{Q\uparrow} := \perp$.
(4) Set $t_{Q\downarrow}$ by the same latest-fall convention applied to the w_q signal within W. If $w_q(x + k) = 1$ or $A = 0$, set $t_{Q\downarrow} := \perp$.

The latest-fall rule is *conservative*: it minimises the committed interval information, attributing the maximal pulse width consistent with the data. Any stricter reconstruction (shorter pulses) is also in γ_k but commits to more edge information than the point trace warrants.

Remark C.2 (Two mechanistic regimes). The distinction between Ψ_{ov} and $\Psi_{\text{lag},\Delta}$ captures two physically meaningful observational modes:

(1) *Overlap-type causation.* The consequent must be observed *while the enabling condition persists*—analogous to a gated detector that is active only during the antecedent pulse. The observation window is the antecedent's own active interval I_P.
(2) *Lag-type (after-effect) causation.* The antecedent acts as a one-shot trigger; the consequent may appear after the antecedent has already dropped—analogous to a stimulus–response delay or after-effect measurement. The causal process outlasts the observable signal.

In the point-event model, the patterns $\mathcal{L}_1$–$\mathcal{L}_6$ do not distinguish these two regimes: both collapse into the same tick-level check $w_p \to \mathbf{F}_{\leq k} w_q$. The interval predicates make the distinction explicit and strictly refine the diagnostic resolution.

Remark C.3 (Token-level edges under the monodimensional model). Under the monodimensional alphabet $\Sigma = \{\varnothing, P^\top, P^\perp, Q^\top, Q^\perp\}$, the Boolean predicates $w_p(t) \equiv [\sigma_{\text{obs}}(t) = P^\top]$ and $w_q(t) \equiv [\sigma_{\text{obs}}(t) = Q^\top]$ (Definition 3.8) are derived, single-valued quantities: at each tick exactly one of the five tokens is emitted. The edge operators $\text{rise}_s(t)$ and $\text{fall}_s(t)$ (Definition 8.3) are therefore computed from transitions between successive tokens:

- $\text{rise}_{w_p}(t)$: token at $t-1$ is $\neq P^\top$ and token at t is $P^\top$.
- $\text{fall}_{w_p}(t)$: token at $t-1$ is $P^\top$ and token at t is $\neq P^\top$.

The Q-priority rule (Definition 3.12) means that when both P and Q are physically present at a tick, only the Q-channel token is emitted. Consequently:

(1) A $P^\top \to Q^\top$ transition registers fall_{w_p} and rise_{w_q} simultaneously, but the *cause* of the P fall (physical de-assertion vs. Q-masking) is indistinguishable at the token level.
(2) Edges of w_p that coincide with active Q-channel tokens are unobservable: the point monitor cannot determine whether P fell before, during, or after Q activation.

This edge-masking ambiguity is exactly the information deficit that the interval monitor's explicit edge timestamps ($t_{P\downarrow}$, $t_{Q\downarrow}$) resolve, and it motivates the refinement layer of Definitions 8.5 and C.1.

Remark C.4 (Round-trip identities). The projection and concretization satisfy:

(1) $\Pi_k \circ \hat{\gamma}_k = \text{id}$ on point traces (the canonical lift reproduces the original point trace when projected back).
(2) $\hat{\gamma}_k \circ \Pi_k \neq \text{id}$ on interval traces in general: the round trip loses the edge information that Π_k discards, so $\hat{\gamma}_k(\Pi_k(\tau_{\text{iv}}))$ may assign different falling-edge timestamps than the original τ_{iv}.

This asymmetry is the operational signature of strict refinement.

Remark C.5 (Projection coherence). The point-event formula $\Phi_k = \mathbf{G}(w_p \to \mathbf{F}_{\leq k} w_q)$ is satisfied by $\Pi_k(\tau)$ whenever the interval trace τ satisfies Ψ_{ov} or $\Psi_{\text{lag},\Delta}$ (with $\Delta \leq k$) and the opener falls within the window. Conversely, Φ_k-satisfying point traces where w_q first appears only *after* w_p has fallen cannot be lifted to any Ψ_{ov}-satisfying interval trace. Hence the projection is information-lossy:

$$\Psi_{\text{ov}} \implies \Pi_k^*(\Phi_k) \not\Longrightarrow \Psi_{\text{ov}},$$

confirming that the interval monitor is a strict refinement of the point-event monitor.

Remark C.6 (Inclusion under projection). The forgetful projection preserves confirmation:

$$\Pi_k\big(\mathrm{Conf}^{\mathrm{ov}}_{\mathrm{iv}}(k)\big) \;\subseteq\; \mathrm{Conf}_{\mathrm{pt}}(k), \tag{C.2}$$

and, when $\Delta \le k$ and the bound condition of Proposition 8.8 holds, $\Pi_k(\mathrm{Conf}^{\mathrm{lag}}_{\mathrm{iv}}(k,\Delta)) \subseteq \mathrm{Conf}_{\mathrm{pt}}(k)$. Conversely, $\mathrm{Conf}_{\mathrm{pt}}(k)$ is a *collapsed* acceptance region: by Lemma 8.9, distinct interval-confirmed traces (overlap vs. lag-only) project to the same point-confirmed trace, which therefore cannot distinguish the causal mechanism that produced the confirmation.

Example C.7 (Edge-pattern classification and feature collapse). Table 13 classifies four representative edge orderings under the point-event and interval predicates, and their collapse into the feature triple (A, B, C) of Definition 3.5. Cases 1 and 3 exhibit temporal overlap ($I_P \cap I_Q \neq \varnothing$); case 2 satisfies only Ψ_{lag} (w_q rises after w_p has fallen, within the bounded lag); case 4 is a universal violation. Crucially, cases 1 and 3 are *distinct* interval traces (different falling-edge orderings) that project to the *same* point signature—the many-to-one collapse that is the hallmark of strict refinement.

TABLE 13. Edge-ordering cases and their satisfaction under Φ_k, Ψ_{ov}, and $\Psi_{\mathrm{lag},\Delta}$ (all edge gaps $\le \min(k,\Delta)$).

#	**Edge Sequence**[†]	Φ_k	Ψ_{ov}	Ψ_{lag}
1	$P{\uparrow}\,Q{\uparrow}\,P{\downarrow}\,Q{\downarrow}$	✓	✓	✓
2	$P{\uparrow}\,P{\downarrow}\,Q{\uparrow}\,Q{\downarrow}$	✓	×	✓
3	$P{\uparrow}\,Q{\uparrow}\,Q{\downarrow}\,P{\downarrow}$	✓	✓	✓
4	$P{\uparrow}\,P{\downarrow}$ (Q absent)	×	×	×

[†] $P{\uparrow} = \mathrm{rise}(w_p)$, $P{\downarrow} = \mathrm{fall}(w_p)$, etc.

TABLE 14. Interval predicates vs. point features under Π_k. Each row is an interval configuration with the indicated predicate outcomes and the resulting (A, B, C) tuple after projection. Rows 1 & 2 collapse to the same $(1, 0, 1)$ despite distinct interval semantics.

Configuration	Ψ_{ov}	Ψ_{lag}	A	B	C
Overlap ($Q{\uparrow}$ before $P{\downarrow}$)	✓	✓	1	0	1
Lag-after-close ($Q{\uparrow}$ after $P{\downarrow}$)	×	✓	1	0	1
Q rises during P (no fall in W)	✓	✓	1	0	0
Q absent	×	×	0	·	·

In Table 14, rows 1 and 2 are the central illustration: overlapping and lag-only episodes produce identical (A=1, B=0, C=1) tuples after projection, confirming that Π_k erases the edge ordering that distinguishes the two causal regimes. The concretization γ_k (Definition C.1) reconstructs the *set* of compatible interval traces, revealing the ambiguity a single point trace hides.

A fully worked calibration-to-interval trace — the $(1,0,0) \to (1,0,1)$ diagnostic upgrade under kernel tuning, step by step — and the instrumentation analogues of the tuning blocks are provided in the supplementary material; the trace is also executable in the interval-semantics reproducibility notebook (Appendix A).

Remark C.8 (Observational invariance under tuning). All tuning operations covered in Blocks A–D share a single structural property: they modify the observed trace $\tau_{\text{obs}} = \mathcal{K}_{\boldsymbol{\theta}}(\tau_{\text{meas}})$ by changing $\boldsymbol{\theta}$, but they leave the logical specification Φ_k, the interval predicates Ψ_{ov} and $\Psi_{\text{lag},\Delta}$, and the finite classification (8 window classes, backed by the 50000-entry eight-tick catalogue) untouched. Equivalently, tuning moves the point τ_{obs} within or across confirmation sets (Definition 8.7), instantiating the recovery pattern $\mathcal{K}_{\boldsymbol{\theta}}(\tau_{\text{meas}}) \notin \text{Conf} \ \wedge \ \mathcal{K}_{\boldsymbol{\theta}'}(\tau_{\text{meas}}) \in \text{Conf}$ (Equation (8.15)), without altering the definition of Conf itself. Tuning alters observational classification, not logical semantics.

Observation architectures. The model assumes a *single decision stream*: at each tick the monitor emits exactly one token from Σ (Definition 3.8). When multiple physical events fall into the same bin (Section 3.4), the priority-max rule (Definition 3.12) compresses them to one token; this is the operational source of same-bin ambiguity and the motivation for interval refinement. Two important implementation points follow:

- *Single-channel realisation.* Σ can be implemented as an interrupt-like event label or a symbol-decision output at each ADC sample—one wire, one decision per tick.
- *Multi-channel analogue (appendix only).* Some instruments provide synchronous channels (e.g., dual detectors, demodulated I/Q outputs). These can be modelled by parallel single-stream monitors, each with its own alphabet and projection; the correspondence is developed in the electronic supplementary material and is *not* the primary model of this paper.

Appendix D. Schmitt-Trigger Encoding: Reachability Witness and Touch Probe

The Schmitt-trigger encoding — the eight-class worked gallery, the elementary tuning-operation catalogue, the depth-two reachability matrix witnessing the universal-readability result [BBI26], and the touch-probe disambiguation of the silent window $(0, 0, 0)$ — is developed at full scale in the engineering companion [BBI26] (Worked Gallery and Tuning-Tree sections) and machine-verified in notebook N4 (Appendix A).

Appendix E. Trace-Level Taxonomy of Causal Pathologies

The five concrete failure modes underlying the causal-coherence conditions CC1–CC4 (Definition 4.2), summarised in the body at Section 4.4.3, are recorded here with their signature-level coverage.

Definition E.1 (Causal Pathologies)**.** Let $\mathcal{M}$ operate with window length k and threshold θ. We distinguish five trace-level failure modes:

F1. **Premature Falsification.** $k \cdot \Delta t < \tau_{\min}$: the window closes before Q can possibly arrive. Every such trace has $A = 0$ (no q_T observed); the implication is marked Violated or Vacuous even though the physical causal link is intact. This is a CC1 violation (window too short).

F2. **Causal Inflation.** $k \cdot \Delta t \gg \tau_{\max}$: the window is so long that an *independent* occurrence of Q — unrelated to P — falls inside it. The trace records $A = 1$, but the "OK" verdict is spurious. This is a CC1 violation (window too long).

F3. **Parallel (Independent) Q.** Q is already True at the opener tick x (simultaneously with the P trigger), which is impossible under a causal mechanism with non-zero latency $\tau > 0$. This violates both CC1 and CC4.

F4. **Probabilistic Causation.** The causal link $P \to Q$ has probability $p < 1$. Both q_T and q_F are observed within the window ($A = 1$ and $B = 1$), yielding a pattern with mixed evidence. This combination stresses CC1–CC3 jointly.

F5. **Noise-Induced False Positive.** Random fluctuations push the measurement of Q above θ transiently: q_T appears in one band but is surrounded by q_F in adjacent bands. This violates CC3 (threshold detectability).

Each failure mode is captured by a *signature-level predicate* that inspects only the eight-tick signature σ and its window class; no manual inspection of individual traces is required. Table 15 summarises the five predicates and their coverage.

The combined predicates cover 100% of all eight-tick signatures: under the monodimensional observation constraint, every signature belongs to at least one pathology class. This confirms that the CC conditions are not a vacuous constraint — every signature is associated with at least one causal incoherence mode.

TABLE 15. Signature-level coverage of the five causal pathologies over all 50000 eight-tick signatures. "Cov." is the number of matching signatures, machine-computed by `eightick_pathology.py` (predicate definitions documented there); "CC" is the violated condition.

Failure Mode	**CC Viol.**	**Cov.**	**Dominant Outcome**
F1 Premature Falsification	CC1 (short)	25600	Violated / Vacuous
F2 Causal Inflation	CC1 (long)	24400	OK (spurious)
F3 Parallel Q	CC1 + CC4	10000	OK (spurious)
F4 Probabilistic Causation	CC1–CC3	9600	OK (mixed)
F5 Noise-Induced FP	CC3	32000	OK (transient)
Combined coverage		50000/50000 = 100%	

Appendix F. The Excluded Middle Under Observation

This appendix locates the law of excluded middle inside the same observation-relative frame the body establishes for bounded temporal implication. The three prose leads below are illustrations, not claims of record: each instantiates a result already established above at the most elementary logical law.

S1 (Trace-level truth and deferral). On the ideal atomic trace $\mathbf{G}(P \vee \neg P)$ denotes Σ^ω, the degenerate $\mathsf{S} \cap \mathsf{C} \cap \mathsf{L}$ tautology cell of [Bra26a]; under a finite read the per-tick verdict is *deferred*, not falsified. The excluded middle is a property of the trace and it holds, whereas the deferral is a property of the monitor (Section 1.2, Section 6.4). Elementary.

S2 (Observational undefinedness under non-atomicity). Take a trace on which P's only transition falls at or outside the window edge, so the window carries neither an up-crossing (evidence for P) nor a down-crossing (evidence for $\neg P$) on P's channel. The observed value of $P \vee \neg P$ is then not the rejecting verdict but the *observationally vacuous* class $(0, 0, 0)$, the $SFR \cap SFS$ *Quaestio* cell of Proposition 7.1 (Section 6): the *absence* of in-window evidence for either literal, not a logical falsification — no proposition is at once true and false, and there is no paraconsistency. It is the same structure by which, on the closure axis, a violation is the *absence* of a consequent onset rather than a positive counter-event (Section 6). Witnessed by the construction; a uniform statement over all non-atomic models is left as an open refinement.

S3 (The excluded middle is observation-relative). Combining S1 and S2, whether $P \vee \neg P$ is observed as decided ($\top$) or as observationally undefined is a property of the pair *(formula, observation model)*, not of the formula alone: it is the observation-relative monitorability of Theorem 7.11 instantiated at the most elementary logical law. This is a heuristic bridge — an application of the theorem, not a new impossibility result; S1 keeps classical logic intact at the trace level.

Dissolve, not solve: the excluded middle is re-situated, not revised. As with bounded temporal implication, the paradox loosens once the verdict is read as joint in the observation model rather than as a property of the formula alone; the memory-versus-atomicity separation that drives this for implication is developed in Section 6.4.

Appendix G. Instantiations across domains

The observation classes of Table 4 and the confirming signatures of Table 5 are not tied to any particular domain: any process with an antecedent quantity crossing a threshold θ_P and a consequent quantity crossing a threshold θ_Q within a bounded latency window realises one of the classes. This appendix develops those realisations and, in each, records the two sensitivities the body emphasises — the window $[\tau_{\min}, \tau_{\max}]$ and the per-channel thresholds — so that one can see how the recorded signature moves under retuning.

G-1. **Physical and chemical realisations.**

Flash photolysis, $P^{\perp}\,Q^{\top}\,Q^{\perp}$, class $(1,1,1)$. A short excitation flash drives the reagent above the antecedent threshold and then decays away ($P^{\perp}$); a transient intermediate rises across θ_Q ($Q^{\top}$) and subsequently relaxes back below it ($Q^{\perp}$). Recorded over a window that contains both the rise and the decay, the signature is the full three-event $(1,1,1)$. Flash photolysis, whose intense flash creates intermediates at concentrations comparable to the reactants [NP49], is the archetype of a designed experiment that can *close* the excursion by observing the return, and hence exhibits the retraction events that most measurements miss.

Resistively-heated sample, $Q^{\top}\,P^{\perp}\,Q^{\perp}$, class $(1,1,1)$. Power on is the opener $P^{\top}$; the temperature crosses θ_Q ($Q^{\top}$); power off is the antecedent retraction $P^{\perp}$; the temperature then decays back below θ_Q ($Q^{\perp}$). The sensitivity is transparent here: if θ_Q is set too high the sample never crosses it, $Q^{\top}$ is missed, and the honest $(1,1,1)$ is recorded as the silent $(0,0,0)$; if the observation window is cut short before the sample cools, the down-crossing $Q^{\perp}$ falls out of band and the recorded class collapses to $(1,0,1)$ or, if $P^{\perp}$ also falls out, to $(1,0,0)$.

Fluorophore photobleaching, $Q^{\top}\,Q^{\perp}\,P^{\perp}$, class $(1,1,1)$. Switching the laser on ($P^{\top}$) drives the fluorescence across θ_Q ($Q^{\top}$); continued illumination bleaches the fluorophore so the emission falls back ($Q^{\perp}$); switching the laser off is the retraction ($P^{\perp}$). A window that stops at first light records only $Q^{\top}$, i.e. $(1,0,0)$, when the honest signature is $(1,1,1)$.

Sensory/receptor adaptation, $Q^{\top}\,Q^{\perp}$, class $(1,1,0)$. A sustained stimulus holds the antecedent above θ_P (no $P^{\perp}$); the response crosses θ_Q ($Q^{\top}$) and then adapts back down ($Q^{\perp}$) while the stimulus persists. The absence of C is structural — the antecedent is never retracted — so this is a two-event class regardless of window, provided the window is long enough to catch the adaptation; too short a window loses $Q^{\perp}$ and records $(1,0,0)$.

Crystallisation from a seed, $Q^{\top}\,P^{\perp}$, class $(1,0,1)$. A seed added to a supersaturated solution is the opener; crystal growth crosses θ_Q and *latches* ($Q^{\top}$, no subsequent $Q^{\perp}$); the supersaturation (antecedent) is consumed and falls back ($P^{\perp}$). The latched consequent with a retracted antecedent is the hallmark of $(1,0,1)$: the effect persists after its cause is withdrawn.

Phosphorescence, $P^{\perp}\,Q^{\top}$, class $(1,0,1)$. An excitation flash raises and then drops the antecedent ($P^{\perp}$); delayed persistent emission crosses θ_Q ($Q^{\top}$) only *after* the excitation has ceased, and persists. The consequent onset trails the antecedent retraction — the ordered $(1,0,1)$ whose event order distinguishes it from crystallisation while sharing its presence triple.

Irreversible product formation, $Q^{\top}$, class $(1,0,0)$. A reagent held above θ_P (sustained antecedent, no $P^{\perp}$) drives an irreversible reaction whose product concentration crosses θ_Q and plateaus ($Q^{\top}$, no $Q^{\perp}$). With neither down-crossing, this is the minimal confirming signature, and it is the limit to which every longer excursion above degrades once the window is narrowed enough to drop all retraction events.

Two window-and-threshold instruments. Two measurement instruments make the joint window/threshold dependence explicit. In real-time quantitative PCR, the cycle threshold C_t at which amplification is called is fixed by *both* an uncertain cycle window (where along the amplification curve one reads) *and* a per-run fluorescence threshold that must be set against the run's baseline [HFDW93]; the same amplification can be called at different cycles under different thresholds. In pulsed radar, a range gate is swept over an uncertain echo delay while a constant false-alarm-rate (CFAR) detector adaptively tunes the detection threshold to the locally sampled clutter level [FJ68]; the declared detection thus depends jointly on gate placement (the window) and the adapted threshold. Both are engineered instances of the window × threshold tuning the body treats abstractly.

G-2. **The closing-threshold selection criterion.** Most experiments measure only the confirmation $Q^{\top}$ and stop: once the effect is seen, observation ceases, and the down-crossings $P^{\perp}$ and $Q^{\perp}$ are never recorded — they either fall out of band or go undetected. The consequence is a selection effect. The confirming family (A=1) dominates real data not because retraction is rare in the world but because the closing threshold of the observation is set at first confirmation. What looks like a preponderance of $(1, 0, 0)$ "static" implications is therefore an observation artifact, not a fact about the processes observed. Good demonstrators are precisely the designs that *close* the excursion — flash photolysis, the heated sample read through cooling, the bleaching series — because they keep observing past $Q^{\top}$ and so record the $P^{\perp}$ and $Q^{\perp}$ that the default stopping rule discards.

G-3. **Sociological illustrations.** The paper's result and scope are logical-computational; the following are only qualitative, heuristic illustrations of the same window/threshold structure in social measurement — not measured witnesses, and not claims about social causation. They are included because the observation-relativity the paper formalises is unusually visible in social data, where windows and thresholds are contested.

- *Collective-action thresholds* [Gra78] *and neighbourhood tipping* [Sch71]: here the consequent threshold θ_Q (each actor's participation threshold) varies sharply across the population, so the *same* trigger yields $(1, 0, 0)$ in one setting and $(0, 0, 0)$ in another — a variable-threshold instance. This mirrors the structure of the independent-Q inflation of Proposition 4.5: many apparent confirmations are threshold artifacts, not responses to the trigger.
- *The replication crisis in psychological science* [Ope15]: replications recorded weaker and less often significant effects than the original studies; read through the present model, the recorded signature depends on the laboratory's window and threshold. Read through the present model this is a direct instance of the observation-relative thesis that Theorem 7.11 formalises — non-reproducibility as observation-relativity of the recorded signature, rather than as noise.
- *The Hawthorne / novelty effect* [RD39]: the measured effect crosses θ_Q and then decays ($Q^{\perp}$) once the novelty fades — honestly a $(1, 1, 0)$. Studies that stop at first confirmation record $(1, 0, 0)$, exactly the closing-threshold selection effect of Appendix G-2.

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